

THE POPULAR SCIENCE MONTHLY

EDITED BY
J. MCKEEN CATTELL

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MAY TO OCTOBER, 1900

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MONTHLY

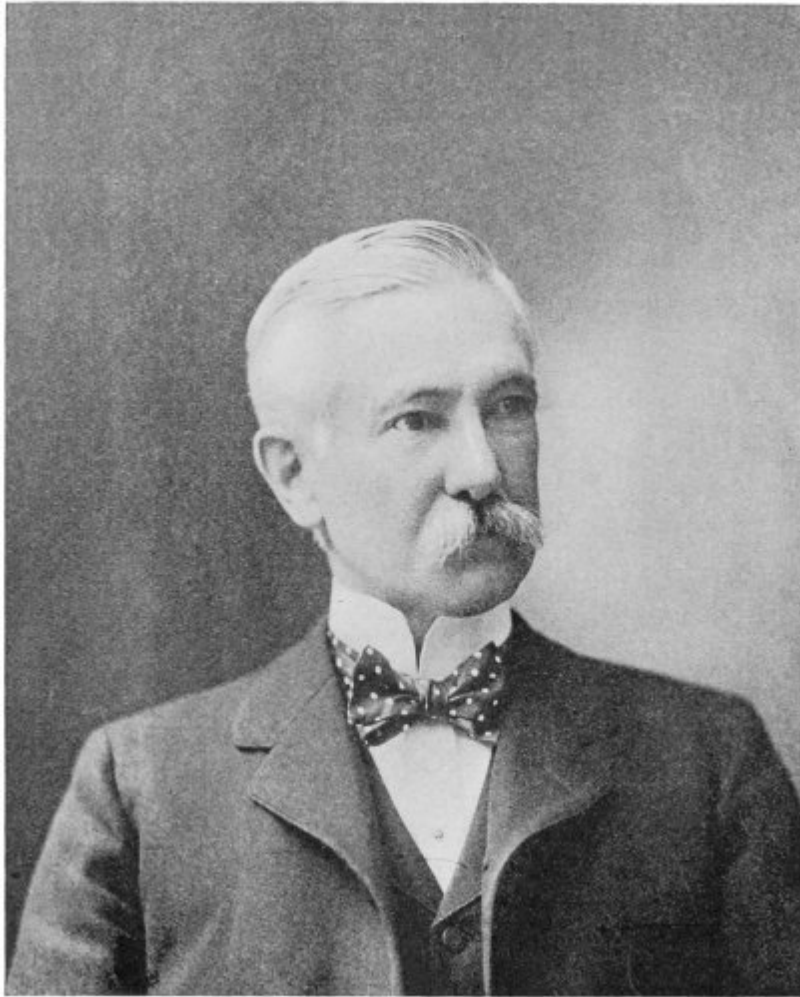
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PROFESSOR R. S. WOODWARD,
PRESIDENT OF THE AMERICAN ASSOCIATION FOR THE
ADVANCEMENT OF SCIENCE.

**THE
POPULAR SCIENCE
MONTHLY.**

AUGUST, 1900.

RHYTHMS AND GEOLOGIC TIME.^A

By G. K. GILBERT,
UNITED STATES GEOLOGICAL SURVEY.

^A Read to the American Association for the Advancement of Science, at New York, June 26, 1900, as the address of the retiring President.

CUSTOM dictates that in complying with the rule of the association I shall address you on some subject of a scientific character. But before doing so I may be permitted to pay my personal tribute to the honored and cherished leader of whose loss we are so keenly sensible on this occasion. His kindly personality, the charm which his earnestness and sincerity gave to his conversation, the range of his accomplishment, are inviting themes; but it is perhaps more fitting that I touch this evening on his character as a representative president of this body. The association holds a peculiar position among our scientific organizations of national or continental extent. Instead of narrowing its meetings by limitations of subject matter or membership, it cultivates the entire field of research and invites the interest and coöperation of all. It is thus not only the integrating body for professional investigators, but the bond of union between these and the great group of cultured men and women—the group from whose ranks the professional guild is recruited, through whom the scientific spirit is chiefly propagated, and through whose interest scientific research receives its financial support. Its aims and form of organization recognize, what pure science does not always itself recognize, that pure science is fundamentally the creature and servant of the material needs of mankind, and it thus stands for what might be called the human side of science. Edward Orton, throughout his career as teacher and investigator, was conspicuous for his

attention to the human side of science. His most abstract work was consciously for the benefit of the community, and he ever sought opportunity to make its results directly available. In promoting the interests of the people of his adopted State he incidentally accomplished much for a larger community by helping it to an appreciation of the essential beneficence of the scientific study of nature and man. As an individual he was a diligent and successful laborer in the field which the association cultivates, and when the association selected him as its standard-bearer it made choice of one who was peculiarly its representative.

The subject to which I shall invite your attention this evening is by no means novel, but might better be called perennial or recurrent; for the problem of our earth's age seems to bear repeated solution without loss of vigor or prestige. It has been a marked favorite, moreover, with presidents and vice-presidents, retiring or otherwise, when called upon to address assemblies whose fields of scientific interest are somewhat diverse—for the reason, I imagine, that while the specialist claims the problem as his peculiar theme of study, he feels that other denizens of the planet in question may not lack interest in the early lore of their estate.

The difficulty of the problem inheres in the fact that it not only transcends direct observation, but demands the extrapolation or extension of familiar physical laws and processes far beyond the ordinary range of qualifying conditions. From whatever side it is approached, the way must be paved by postulates, and the resulting views are so discrepant that impartial onlookers have come to be suspicious of these convenient and inviting stepping stones.

That vain expectation may not be aroused, I admit at the outset that I have not solved the problem and shall submit to you no estimates. My immediate interest is in the preliminary question of the available methods of approach, and it leads to the consideration of the ways, or the classes of ways, in which the measurement of time has been accomplished or attempted.

Of the artificial devices employed in practical horology there are two so venerable that their origins are lost in the obscurity of legendary myth. These are the clepsydra and the taper. In the clepsydra advantage is taken of the approximately uniform rate at which water escapes through a small

orifice, and time is measured by gaging the loss of water from a discharging vessel or the gain in a receiving vessel. The hour-glass is one of its latest forms, in which sand takes the place of water. The taper depends for its value as a timepiece on the approximate uniformity of combustion when the area of fuel exposed to the air is definitely regulated. It survives chiefly in the prayer stick and safety fuse, but the graduated candle is perhaps still used to regulate monastic vigils.

The pendulum, a comparatively modern invention, excelling the clepsydra and taper in precision, has altogether supplanted them as the servant of civilization. Its accuracy results from the remarkable property that the period in which it completes an oscillation is almost exactly the same, whatever the arc through which it swings. It regulates the movements not only of our clocks, watches and chronometers, but of barographs, thermographs and a great variety of other machines for recording events and changes in their proper order and relation in respect to time.

I must mention also a special apparatus invented by astronomers and called a chronograph. It consists ordinarily of a revolving drum about which a paper is wrapped and against which rests a pen. As the drum turns the pen draws a line on the paper. Through an electric circuit the pen is brought under the influence of a pendulum in such a way that at the middle of each swing of the pendulum the pen is deflected, making a mark at right angles to the straight line. The series of marks thus drawn constitutes a time scale. The electric arrangements are so made that the pen will also be disturbed in consequence of some independent event, such as the firing of a gun or the transit of a star; and the mark caused by such disturbance, being automatically platted on the time scale, records the time of the event.

No attempt has been made to characterize these various timepieces with fullness, because they are already well known to most of those present, and, in fact, the chief motive for giving them separate mention is that they may serve as the basis of a classification. In the use of the clepsydra and taper, time is measured in terms of a continuous movement or process; in the use of the pendulum time is measured in terms of a movement which is periodically reversed. The classification embodies the fundamental distinction between continuous motion and rhythmic motion.

Passing now from the artificial to the natural measures of time, we find that they are all rhythmic. It is true that the spinning of the earth on its axis is in itself a continuous motion, but it would yield no time measure if the earth were alone in space, and so soon as the motion is considered in relation to some other celestial body it becomes rhythmic. As viewed from, or compared with, a fixed star, the period of its rhythm is the sidereal day; compared with the sun, it is the solar day, nearly four minutes longer; and compared with the moon, it is the lunar day, still longer by 49 minutes. As the sun supplies the energy for most of the physical and all the vital processes of the earth's surface, the rhythm of the solar day is impressed in multitudinous ways on man and his environment, and he makes it his primary or standard unit of time. He has arbitrarily divided it into hours, minutes and seconds, and in terms of these units he says that the length of the sidereal day is a little more than 23 hours, 56 minutes and 4 seconds, and the average length of the lunar day is a little less than 24 hours and 49 minutes. The lunar day finds expression in the tides and is of moment to maritime folk, but the sidereal is known only to astronomers.

Next in the series of our natural time units is the month, or the rhythmic period of the moon regarded as a luminary. By our savage ancestors, who credited the moon with powers of great importance to themselves, much use was made of this unit, but as progress in knowledge has shown that the influence of the satellite had been vastly overrated, less and less attention has been paid to the returning crescent, and it is only in ecclesiastic calendars that the chronology of civilization now recognizes the natural month. Its shadow survives, without the substance, in the calendar month; and the week possibly represents an early attempt to subdivide it.

In passing to our third natural unit, the year, we again encounter solar influence, and find the rhythm of the earth's orbit echoed and reechoed in innumerable physical and vital vibrations. As the attitude of the earth's axis inclines one hemisphere toward the sun for part of the year and the other hemisphere for the remainder, the whole complex drama of climate is annually enacted, and the sequence of man's activities is made to assume an annual rhythm. The year is second only to the day as a terrestrial unit of duration; and as the day is man's standard for the minute division of time, so the year is his standard for larger divisions, and the decade, the century and the millenium are its multiples.

But the rhythms of day and night, of summer and winter, are not the only tides in the affairs of men. At birth we are small, weak and dependent, we grow larger and stronger, we become mature and independent, and then by reproducing our kind we complete the cycle, which begins again with our children. The cycle of human life is the *generation*, a time unit of somewhat indefinite length and varying in phase from family to family, but holding a place, nevertheless, in human chronology.

Still less definite is the rhythm of hereditary rulership, progressing from vigor through luxury to degeneracy, and closing its cycle in usurpation; yet it makes an epoch in the life of a nation or empire, and so the *dynasty* is one of the units of the historian.

The generation and the dynasty are of waning importance in human chronology, and they can claim no connection with the problem of geologic time; but here again I have turned aside for a moment in order to illustrate a principle of classification. The daily rhythm of waking and sleeping, of activity and rest, does not originate with man, but is imposed on him by the rhythm of light and darkness, and that in turn springs from the turning of the earth in relation to the shining sun. The yearly rhythm of sowing and harvesting, of the fan and the furnace, does not originate with man, but is imposed on him by the rhythm of the seasons, and that in turn springs from certain motions of the earth in relation to the glowing sun. But the rhythm of the generation and the rhythm of the dynasty have origin in the nature of man himself. The rhythms of human chronology may thus be grouped according to source in two classes, the *imposed* and the *original*; and the same distinction holds for other rhythms. The lunar day is an original rhythm of the earth as seen from the moon; the ground swell is an original rhythm of the ocean; but the tide is an imposed rhythm of the ocean, being derived from the lunar day. The swing of the pendulum is an original rhythm, but the regular excursion of the chronograph pen, being caused by the swing of the pendulum, is an imposed rhythm.

In giving brief consideration to each of the more important ways by which the problem of the earth's age has been approached, I shall mention first those which follow the action of some continuous process, and afterward those which depend on the recognition of rhythms.

The earliest computations of geologic time, as well as the majority of all such computations, have followed the line of the most familiar and fundamental of geologic processes. All through the ages the rains, the rivers and the waves have been eating away the land, and the product of their gnawing has been received by the sea and spread out in layers of sediment. These layers have been hardened into rocky strata, and from time to time portions have been upraised and made part of the land. The record they contain makes the chief part of geologic history, and the groups into which they are divided correspond to the ages and periods of that history. In order to make use of these old sediments as measures of time it is necessary to know either their thickness or their volume, and also the rate at which they were laid down. As the actual process of sedimentation is concealed from view, advantage is taken of the fact that the whole quantity deposited in a year is exactly equalled by the whole quantity washed from the land in the same time, and measurements and estimates are made of the amounts brought to the sea by rivers and torn from the cliffs of the shore by waves. After an estimate has been obtained of the total annual sedimentation at the present time, it is necessary to assume either that the average rate in past ages has been the same or that it has differed in some definite way.

At this point the course of procedure divides. The computer may consider the aggregate amount of the sedimentary rocks, irrespective of their subdivisions, or he may consider the thicknesses of the various groups as exhibited in different localities. If he views the rocks collectively, as a total to be divided by the annual increment, his estimate of the total is founded primarily on direct measurements made at many places on the continents, but to the result of such measurements he must add a postulated amount for the rocks concealed by the ocean, and another postulated amount for the material which has been eroded from the land and deposited in the sea more than once.

If, on the other hand, he views each group of rocks by itself, and takes account of its thickness at some locality where it is well displayed, he must acquire in some way definite conceptions of the rates at which its component layers of sand, clay and limy mud were accumulated, or else he must postulate that its average rate of accretion bore some definite ratio to the present average rate of sedimentation for the whole ocean. This course is, on the whole, more difficult than the other, but it has yielded certain

preliminary factors in which considerable confidence is felt. Whatever may have been the absolute rate of rock building in each locality, it is believed that a group of strata which exhibits great thickness in many places must represent more time than a group of similar strata which is everywhere thin, and that clays and marls, settling in quiet waters, are likely to represent, foot for foot, greater amounts of time than the coarser sediments gathered by strong currents; and studying the formations with regard to both thickness and texture, geologists have made out what are called *time ratios*—series of numbers expressing the relative lengths of the different ages, periods and epochs. Such estimates of ratios, when made by different persons, are found to vary much less than do the estimates of absolute time, and they will serve an excellent purpose whenever a satisfactory determination shall have been made of the duration of any one period.

Reade has varied the sedimentary method by restricting attention to the limestones, which have the peculiarity that their material is carried from the land in solution; and it is a point in favor of this procedure that the dissolved burdens of rivers are more easily measured than their burdens of clay and sand.

An independent system of time ratios has been founded on the principle of the evolution of life. Not all formations are equally supplied with fossils, but some of them contain voluminous records of contemporary life; and when account is taken of the amount of change from each full record to the next, the steps of the series are found to be of unequal magnitude. Though there is no method of precisely measuring the steps, even in a comparative way, it has yet been found possible to make approximate estimates, and these in the main lend support to the time ratios founded on sedimentation. They bring aid also at a point where the sedimentary data are weak, for the earliest formations are hard to classify and measure. It is true that these same formations are almost barren of fossils, but biologic inference does not therefore stop. The oldest known fauna, the Eocambrian, does not represent the beginnings of life, but a well-advanced stage, characterized by development along many divergent lines; and by comparing Eocambrian life with existing life the paleontologist is able to make an estimate of the relative progress in evolution before and after the Eocambrian epoch. The only absolute blank left by the time ratios pertains to an azoic age which

may have intervened between the development of a habitable earth crust and the actual beginning of life.

Erosion and deposition have been used also, in a variety of ways, to compute the length of very recent geologic epochs. Thus, from the accumulation of sand in beaches Andrews estimated the age of Lake Michigan, and Upham the age of the glacial lake Agassiz; and from the erosion of the Niagara gorge the age of the river flowing through it has been estimated. But while these discussions have yielded conceptions of the nature of geologic time, and have served to illustrate the extreme complexity of the conditions which affect its measurement, they have accomplished little toward the determination of the length of a geologic period; for they have pertained only to a small fraction of what geologists call a period, and that fraction was of a somewhat abnormal character.

Wholly independent avenues of approach are opened by the study of processes pertaining to the earth as a planet, and with these the name of Kelvin is prominently associated.

As the rotation of the earth causes the tides, and as the tides expend energy, the tides must act as a brake, checking the speed of rotation. Therefore the earth has in the past spun faster than now, and its rate of spinning at any remote point of time may be computed. Assuming that the whole globe is solid and rigid, and that the geologic record could not begin until that condition had been attained, there could not have been great checking of rotation since consolidation. For if there had been, it would have resulted in the gathering of the oceans about the poles and the barring of the land near the equator, a condition very different from what actually obtains. This line of reasoning yields an obscure outer limit to the age of the earth.

On the assumption that the globe lacks something of perfect rigidity, G. H. Darwin has traced back the history of the earth and the moon to an epoch when the two bodies were united, their separation having been followed by the gradual enlargement of the moon's orbit and the gradual retardation of the earth's rotation; and this line of inquiry has also yielded an obscure outer limit to the antiquity of the earth as a habitable globe.

One of the most elaborate of all the computations starts with the assumption that at an initial epoch, when the outer part of the earth was

consolidated from a liquid condition, the whole body of the planet had approximately the same temperature; and that as the surface afterward cooled by outward radiation there was a flow of heat to the surface by conduction from below. The rate of this flow has diminished from that epoch to the present time according to a definite law, and the present rate, being known from observation, affords a measure of the age of the crust. The strength of this computation lies in its definiteness and the simplicity of its data; its weakness in the fact that it postulates a knowledge of certain properties of rock—namely, its fusibility, conductivity and viscosity—when subjected to pressures and temperatures far greater than have ever been investigated experimentally.

A parallel line of discussion pertains to the sun. Great as is the quantity of heat which that incandescent globe yields to the earth, it is but a minute fraction of the whole amount with which it continually parts, for its radiation is equal in all directions, and the earth is but a speck in the solar sky. On the assumption that this immense loss of heat is accompanied by a corresponding loss of volume, the sun is shrinking at a definite rate, and a computation based on this rate has told how many millions of years ago the sun's diameter should have been equal to the present diameter of the earth's orbit. Manifestly the earth can not have been ready for habitation before the passage of that epoch, and so the computation yields a superior limit to the extent of geologic time.

Before passing to the next division of the subject—the computations based on rhythms—a few words may be given to the results which have been obtained from the study of continuous processes. Realizing that your patience may have been strained by the kaleidoscopic character of the rapid review which has seemed unavoidable, I shall spare you the recitation of numerical details and merely state in general terms that the geologists, or those who have reasoned from the rocks and fossils, have deduced values for the earth's age very much larger than have been obtained by the physicists, or those who have reasoned from earth cooling, sun cooling and tidal friction. In order to express their results in millions of years the geologists must employ from three to five digits, while the physicists need but one or two. When these enormous discrepancies were first realized it was seen that serious errors must exist in some of the observational data or else in some of the theories employed; and geologists undertook with zeal

the revision of their computations, making as earnest an effort for reconciliation as had been made a generation earlier to adjust the elements of the Hebrew cosmogony to the facts of geology. But after rediscussing the measurements and readjusting the assumptions so as to reduce the time estimates in every reasonable way—and perhaps in some that were not so reasonable—they were still unable to compress the chapters of geologic history between the narrow covers of physical limitation; and there the matter rests for the present.

The rocks which were formed as sediments show many traces of rhythm. Some are composed of layers, thin as paper, which alternate in color, so that when broken across they exhibit delicate banding. In the time of their making there was a periodic change in the character of the mud that settled from the water. Others are banded on a larger scale; and there are also bandings of texture where the color is uniform. Many formations are divided into separate strata, as though the process of accretion had been periodically interrupted. Series of hard strata are often separated by films or thin layers of softer material. Strata of two kinds are sometimes seen to alternate through many repetitions. Borings in the delta of the Mississippi show soils and remains of trees at many levels, alternating with river silts. The rock series in which coal occurs are monotonous repetitions of shale and sandstone. Belgian geologists have been so impressed by the recurrence of short sequences of strata that they have based an elaborate system of rock notation upon it.

Passing to still greater units, the large aggregates of strata sometimes called systems show in many cases a regular sequence, which Newberry called a “circle of deposition.” When complete, it comprises a sandstone or conglomerate, at base, then shale, limestone, shale and sandstone. This sequence is explained as the result of the gradual encroachment, or transgression, as it is called, of the sea over the land and its subsequent recession.

In certain bogs of Scandinavia deep accumulations of peat are traversed horizontally by layers including tree stumps in such way as to indicate that the ground has been alternately covered by forest and boggy moss. The broad glaciers of the Ice age grew alternately smaller and larger—or else were repeatedly dissipated and reformed—and their final waning was characterized by a series of halts or partial readvances, recorded in

concentric belts of ice-brought drift. Of these belts, called moraines of recession, Taylor enumerates seventeen in a single system.

In explanation of these and other repetitive series incorporated in the structure of the earth's crust, a variety of rhythmic causes have been adduced; and mention will be made of the more important, beginning with those which have the character of original rhythms.

A river flowing through its delta clogs its channel with sediment, and from time to time shifts its course to a new line, reaching the sea by a new mouth. Such changes interrupt and vary sedimentation in neighboring parts of the sea. Storms of rain make floods, and each flood may cause a separate stratum of sediment. Storms of wind give destructive force to the waves that beat the shore, and each storm may cause the deposit of an individual layer of sediment. Varying winds may drive currents this way and that, causing alternations in sedimentation.

To explain the forest beds buried in the Mississippi silts it has been suggested that the soft deposits of the delta from time to time settled and spread out under their own weight. Various alternations of strata, and especially those of the coal measures, have been ascribed to successive local subsidences of the earth's crust, caused by the addition of loads of deposit. It has been suggested also that land undergoing erosion may rise up from time to time because relieved of load, and the character of sediment might be changed by such rising. Subterranean forces, of whatever origin, seemingly slumber while strains are accumulating, and then become suddenly manifest in dislocations and eruptions, and such catastrophes affect sedimentation.

A more general rhythm has been ascribed to the tidal retardation of rotation and the resulting change of the earth's form. If the body of the earth has a rather high rigidity, we should expect that it would for a time resist the tendency to become more nearly spherical, while the water of the ocean would accommodate itself to the changing conditions of equilibrium by seeking the higher latitudes. Eventually, however, the solid earth would yield to the strain and its figure become adjusted to the slower rotation, and then the mobile water would return. Thus would be caused periodic transgressions by the sea, occurring alternately in high and low latitudes.

Another general rhythm has been recently suggested by Chamberlin in connection with the hypothesis that secular variations of climate are chiefly due to variations of the quantity of carbon dioxide in the atmosphere.^B The system of interdependent factors he works out is too complex for presentation at this time, and I must content myself with saying that his explanation of the moraines of recession involves the interaction of a peculiar atmospheric condition with a condition of glaciation, each condition tending to aggravate the other, until the cumulative results brought about a reaction and the climatic pendulum swung in the opposite direction. With each successive oscillation the momentum was less, and an equilibrium was finally reached.

^B An attempt to frame a working hypothesis of the cause of glacial periods on an atmospheric basis. *Journ. Geol.*, Vol. VII., 1899.

Few of these original rhythms have been used in computations of geologic time, and it is not believed that they have any positive value for that purpose. Nevertheless, account must be taken of them, because they compete with imposed rhythms for the explanation of many phenomena, and the imposed rhythms, wherever established, yield estimates of time.

The tidal period, or the half of the lunar day, is the shortest imposed rhythm appealed to in the explanation of the features of sedimentation. It is quite conceivable that the bottom of a quiet bay may receive at each tide a thin deposit of mud which could be distinguished in the resulting rock as a papery layer or lamina. If one could in some way identify a rock thus formed, he might learn how many half-days its making required by counting its laminæ, just as the years of a tree's age are learned by counting its rings of growth.

The next imposed rhythm of geologic importance is the year. There are rivers, like the Nile, having but one notable flood in each year, and so depositing annual layers of sediment on their alluvial plains and on the sea beds near their mouths. Where oceanic currents are annually reversed by monsoons, sedimentation may be regularly varied, or interrupted, once a

year. Streams from a glacier cease to run in winter, and this annual interruption may give a definite structure to resulting deposits. It is therefore probable that some of the laminæ or strata of rocks represent years, but the circumstances are rarely such that the investigator can bar out the possibility that part of the markings or separations were caused by original rhythms of unknown period.

The number of rhythms existing in the solar system is very large, but there are only two, in addition to the two just mentioned, which seem competent to write themselves in a legible way in the geologic record. These are the rhythms of precession and eccentricity.

Because the earth's orbit is not quite circular and the sun's position is a little out of the center, or eccentric, the two hemispheres into which the earth is divided by the equator do not receive their heat in the same way. The northern summer, or the period during which the northern hemisphere is inclined toward the sun, occurs when the earth is farthest from the sun, and the northern winter occurs when the earth is nearest to the sun, or in that part of the orbit called perihelion. These relations are exactly reversed for the southern hemisphere. The general effect of this is that the southern summer is hotter than the northern, and the southern winter is colder than the northern. In the southern part of the planet there is more contrast between summer and winter than in the northern. The sun sends to each half the same total quantity of heat in the course of a year, but the difference in distribution makes the climates different. The physics of the atmosphere is so intricate a subject that meteorologists are not fully agreed as to the theoretic consequences of these differences of solar heating, but it is generally believed that they are important, involving differences in the force of the winds, in the velocity and course of ocean currents, in vegetation, and in the extent of glaciers.

Now, the point of interest in the present connection is that the astronomic relations which occasion these peculiarities are not constant, but undergo a slow periodic change. The relation of the seasons to the orbit is gradually shifting, so that each season in turn coincides with the perihelion; and the climatic peculiarities of the two hemispheres, so far as they depend on planetary motions, are periodically reversed. The time in which the cycle

of change is completed, or the period of the rhythm, is not always the same, but averages 21,000 years. It is commonly called the precessional period.^C

^C Strictly speaking, 21,000 years is the period of the precession of the equinoxes as referred to perihelion; but the perihelion is itself in motion. As referred to a fixed star the precession of the equinoxes has an average period of about 25,700 years.

Assuming that the climates of many parts of the earth are subject to a secular cycle, with contrasted phases every 10,500 years, we should expect to find records of the cycle in the sediments. A moist climate would tend to leach the calcareous matter from the rock, leaving an earthy soil behind, and in a succeeding drier climate the soil would be carried away; and thus the adjacent ocean would receive first calcareous and then earthy sediments. The increase of glaciers in one hemisphere would not only modify adjacent sediments directly, but, by adding matter on that side, would make a small difference in the position of the earth's center of gravity. The ocean would move somewhat toward the weighted hemisphere, encroaching on some coasts and drawing down on others; and even a small change of that sort would modify the conditions of erosion and deposition to an appreciable extent in many localities.

Blytt ascribed to this astronomic cause the alternations of bog and forest in Scandinavia, as well as other sedimentary rhythms observed in Europe; and it has seemed to me competent to account for certain alternations of strata in the Cretaceous formations of Colorado. Croll used it to explain interglacial epochs, and Taylor has recently applied it to the moraines of recession.

The remaining astronomic rhythm of geologic import is the variation of eccentricity. At the present time our greatest distance from the sun exceeds our least distance by its thirtieth part, but the difference is not usually so small as this. It may increase to the seventh part of the whole distance, and it may fall to zero. Between these limits it fluctuates in a somewhat irregular way, in which the property of periodicity is not conspicuous. The effect of

its fluctuation is inseparable from the precessional effect, and is related to it as a modifying condition. When the eccentricity is large the precessional rhythm is emphasized; when it is small the precessional effect is weak.

The variation of eccentricity is connected with the most celebrated of all attempts to determine a limited portion of geologic time. In the elaboration of the theory of the Ice age which bears his name, Croll correlated two important epochs of glaciation with epochs of high eccentricity computed to have occurred about 100,000 and 210,000 years ago. As the analysis of the glacial history progresses, these correlations will eventually be established or disproved, and should they be established it is possible that similar correlations may be made between events far more remote.

The studies of these several rhythms, while they have led to the computation of various epochs and stages of geologic time, have not yet furnished an estimate either of the entire age of the earth or of any large part of it. Nevertheless, I believe that they may profitably be followed with that end in view.

The system of rock layers, great and small, constituting the record of sedimentation, may be compared to the scroll of a chronograph. The geologic scroll bears many separate lines, one for each district where rocks are well displayed, but these are not independent, for they are labeled by fossils, and by means of these labels can be arranged in proper relation. In each time line are little jogs—changes in kind of rock or breaks in continuity—and these jogs record contemporary events. A new mountain was uplifted, perhaps, on the neighboring continent, or an old uplift received a new impulse. Through what Davis calls stream piracy a river gained or lost the drainage of a tract of country. Escaping lava threw a dam across the course of a stream, or some Krakatoa strewed ashes over the land and gave the rivers a new material to work on. The jogs may be faint or strong, many or few, and for long distances the lines may run smooth and straight; but so long as the jogs are irregular they give no clue to time. Here and there, however, the even line will betray a regularly recurring indentation or undulation, reflecting a rhythm and possibly significant of a remote pendulum whose rate of vibration is known. If it can be traced to such a pendulum there will result a determination of the rate at which the chronograph scroll moved when that part of the record was made; and a

moderate number of such determinations, if well distributed, will convert the whole scroll into a definite time scale.

In other words, if a sufficient number of the rhythms embodied in strata can be identified with particular imposed rhythms, the rates of sedimentation under different circumstances and at different times will become known, and eventually so many parts of geologic time will have become subject to direct calculation that the intervals can be rationally bridged over by the aid of time ratios.

For this purpose there is only one of the imposed rhythms of practical value, namely, the precessional; but that one is, in my judgment, of high value. The tidal rhythm can not be expected to characterize any thick formation. The annual is liable to confusion with a variety of original rhythms, especially those connected with storms. The rhythm of eccentricity, being theoretically expressed only as an accentuation of the precessional, can not ordinarily be distinguished from it. But none of these qualifications apply to the precessional. It is not liable to confusion with the tidal and annual because its period is so much longer, being more than 2,000 times that of the annual. It has an eminently practical and convenient magnitude, in that its physical manifestation is well above the microscopic plane, and yet not so large as to prevent the frequent bringing of several examples into a single view. It is also practically regular in period, rarely deviating from the average length by more than the tenth part.

From the greater number of original rhythms it is distinguished, just as from the annual and tidal, by magnitude. The practical geologist would never confuse the deposit occasioned by a single storm, for example, with the sediments accumulated during an astronomic cycle of 20,000 years. But there are other original rhythms, known or surmised, which might have magnitudes of the same general order, and to discriminate the precessional from these it is necessary to employ other characters. Such characters are found in its regularity or evenness of period, and in its practical perpetuity. The diversion of the mouth of a great river such as the Hoang Ho or the Mississippi might recur only after long intervals; but from what we know of the behavior of smaller streams we may be sure that such events would be very irregular in time as well as in other ways. The intervals between volcanic eruptions at a particular vent or in a particular district may at times amount to thousands of years, but their irregularity is a characteristic

feature. The same is true of the recurrent uplifts by which mountains grow, so far as we may judge them by the related phenomena of earthquakes; and the same category would seem to hold also the theoretically recurrent collapse of the globe under the strains arising from the slowing of rotation. The carbon-dioxide rhythm, known as yet only in the field of hypothesis, is hypothetically a running-down oscillation, like the lessening sway of the cradle when the push is no longer given.

But the precessional motion pulses steadily on through the ages, like the swing of a frictionless pendulum. Its throb may or may not be caught by the geologic process which obtains in a particular province and in a particular era, but whenever the conditions are favorable and the connection is made, the record should reflect the persistence and the regularity of the inciting rhythm.

The search of the rocks for records of the ticks of the precessional clock is an out-of-door work. Pursued as a closet study it could have no satisfactory outcome, because the printed descriptions of rock sequences are not sufficiently complete for the purpose; and the closet study of geology is peculiarly exposed to the perils of hobby-riding. A student of the time problem cannot be sure of a persistent, equable sedimentary rhythm without direct observation of the characters of the repeated layers. He needs to avail himself of every opportunity to study the series in its horizontal extent, and he should view the local problem of original *versus* imposed rhythm with the aid of all the light which the field evidence can cast on the conditions of sedimentation.

Neither do I think of rhythm seeking as a pursuit to absorb the whole time and energy of an individual and be followed steadily to a conclusion; but hope rather that it may receive the incidental and occasional attention of many of my colleagues of the hammer, as other errands lead them among cliffs of bedded rocks. If my suggestion should succeed in adding a working hypothesis or point of view to the equipment of field geologists, I should feel that the search had been begun in the most promising and advantageous manner. For not only would the subject of rhythms and their interpretations be advanced by reactions from multifarious individual experiences, but the stimulus of another hypothesis would lead to the discovery of unexpected meanings in stratigraphic detail.

It is one of the fortunate qualities of scientific research that its incidental and unanticipated results are not infrequently of equal or even greater value than those directly sought. Indeed, if it were not so, there would be no utilitarian harvest from the cultivation of the field of pure science.

In advocating the adoption of a new point of view from which to peer into the mysterious past, I would not be understood to advise the abandonment of old stand-points, but rather to emulate the surveyor, who makes measurement to inaccessible points by means of bearings from different sides. Every independent bearing on the earth's beginning is a check on other bearings, and it is through the study of discrepancies that we are to discover the refractions by which our lines of sight are warped and twisted. The three principal lines we have now projected into the abyss of time miss one another altogether, so that there is no point of intersection. If any one of them is straight, both the others are hopelessly crooked. If we would succeed we should not only take new bearings from each discovered point of vantage, but strive in every way to discover the sources of error in the bearings we have already attempted.

THE PHOTOGRAPHY OF SOUND WAVES.

BY PROFESSOR R. W. WOOD,
UNIVERSITY OF WISCONSIN.

ANY one who has stood near a large naval gun during its discharge, will, I think, be prepared to admit that the sound of the explosion affects not only the ears, but the whole body as well, which experiences something not unlike a sudden blow. This blow, or concussion, as it is generally termed, is merely the impact of the wave of compressed air, spreading out in all directions around the gun. In the case of ordinary sounds, the compression of the air in the wave is so slight that only the delicate auditory nerves respond to the impact, hence we naturally conclude that sounds are perceived only by the ear. When dealing with sounds of very great intensity, this notion must be somewhat modified, for they certainly can be felt as well as heard. In some extreme cases, in fact, the sensation of feeling may be stronger than that of hearing, as in the case of which I shall speak presently. Is it also possible that we can perceive sound through the medium of any other sense organ, say the eye? 'To see a noise' certainly sounds like an absurdity; yet under certain conditions, sound waves in air can be made as distinctly visible as the ripples on a pond surrounding the splash of a stone. That they are not seen under ordinary conditions does not justify us in assuming them to be invisible. We all know that the currents of hot air rising from a stove, while not usually conspicuous, can be made visible by properly regulating the illumination, as by looking along the surface of the stove towards a window. The hot air is visible because in its optical properties it is different from the cold air surrounding it. The rays of light, passing through the unequally heated portions of the air, are bent in different directions, causing a distortion of objects seen through the heated currents. What we see, strictly speaking, is not the hot air itself, but a wavering and swimming of the objects seen through it. Yet I think we are justified in saying that the eye perceives the hot air.

Now sound waves in air, which are merely regions where the air is somewhat compressed, differ in their optical properties from the uncompressed portions, just as the hot air differs from the cold. As the pictures illustrating this article testify, they may be seen and photographed under proper conditions of illumination as readily as solid objects. We must remember, in the first place, that a sound wave travels with a velocity something greater than a thousand feet a second, rather less than the speed of a modern rifle ball, yet ten times faster than the fastest express train. The wave, even if it were stationary, could be seen only by adjusting the illumination with far greater care than was necessary in the case of the hot air, and we consequently can easily understand why we never see the waves under ordinary conditions.

While it is true that laboratory appliances are generally required to render them visible, I should like at the outset to cite an example to show that in the case of very loud sounds occurring in the open air the wave can be perceived by the eye, without the aid of any apparatus whatever. I will quote from an article by Prof. C. V. Boys, which appeared in 'Nature,' June 24, 1897. Mr. Boys first cites the following letter from Mr. E. J. Ryves: "On Tuesday, April 6th, I had occasion, while carrying out some experiments with explosives, to detonate one hundred pounds of a nitro-compound. The explosive was placed on the ground in the center of a slight depression, and in order to view the effect, I stationed myself, at a distance of about three hundred yards, on the side of a neighboring-hill. The detonation was complete, and a hole was made in the ground five feet deep and seven feet in diameter. A most interesting observation was made during the experiment. The sun was shining brightly, and at the moment of detonation the shadow of the sound wave was most distinctly seen leaving the area of disturbance. I heard the explosion as the shadow passed me, and I could follow it distinctly in its course down the valley for at least half a mile; it was so plainly visible that I believe it would photograph well with a suitable shutter."

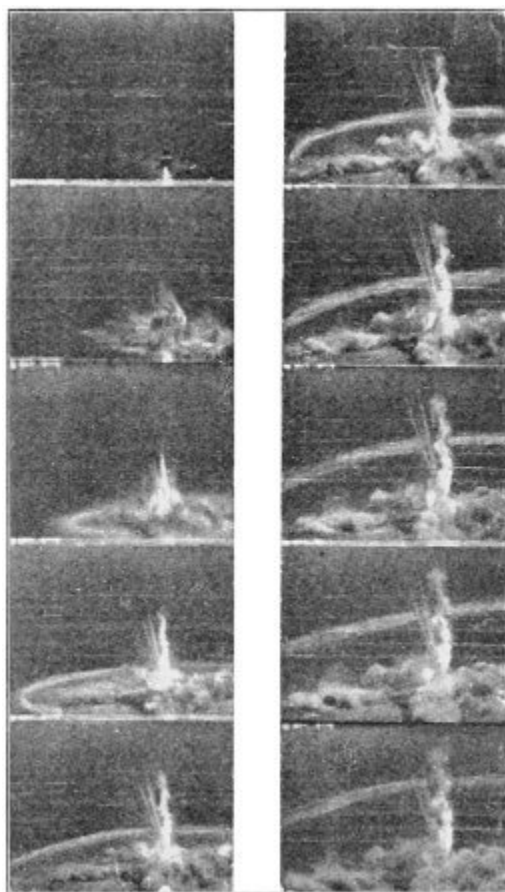
Professor Boys at once made preparations for photographing the phenomenon at the first opportunity. On May 19th the experiment was made. One hundred and twenty pounds of a nitro-compound were exploded, and an attempt made to photograph the sound shadow, both with the camera and the kinematograph, the latter instrument designed and operated by Mr.

Paul. Writing of the experiment, Professor Boys says: "On the day on which I was present, about one hundred and twenty pounds of a nitro-compound were detonated, and ten pounds of black powder were added to make sufficient smoke to show on the plate. As the growth of the smoke cloud is far less rapid than the expansion of the sound shadow, no confusion could result from this. At the time of the explosion my whole attention was concentrated upon the camera, and for the moment I had forgotten to look for the 'Ryves ring,' as I think it might be called; but it was so conspicuous that it forced itself upon my attention. I *felt*, rather than *heard*, the explosion at the moment that it passed. We stationed ourselves as near as prudence would allow, at a distance of one hundred and twenty yards, so that only about one third of a second elapsed between the detonation and the passage of the shadow. The actual appearance of the ring was that of a strong, black, circular line, opening out with terrific speed from the point of explosion as a center. It was impossible to judge of the thickness of the shadow; it may have been three feet, or it may have been more at first, and have gradually become less in thickness, or possibly in depth of shade."

Unfortunately, Professor Boys's apparatus did not work satisfactorily, but a most interesting series of pictures was secured by the kinematograph. This instrument had been constructed especially for taking pictures at a very high rate of speed, viz., eighty exposures a second, or four times the usual number. The sound wave appears in the first dozen pictures as a hazy ring of light, opening out from the center of explosion. The ring, though not very conspicuous when the pictures are viewed singly, becomes a striking object when they are projected in rapid succession on the screen. We see the rush of smoke along the ground to the box in which the explosion is confined (the smoke of the quick fuse); then comes the burst of the explosion with such startling reality that we involuntarily jump. The image of the sound wave flies out in the form of a white ring, and is gone in a moment; and there remain only the rolling clouds of smoke. It is interesting to observe the development of the explosion by running the machine quite slowly, and by thus magnifying time to follow the changes which ordinarily occur in such rapid succession that the eye is unable to perceive them.

Of this series of pictures, Professor Boys says: "The kinematograph fails to show any *black* ring; and this is not surprising, as with the exposure of about one one hundredth of a second the shadow would have to be at

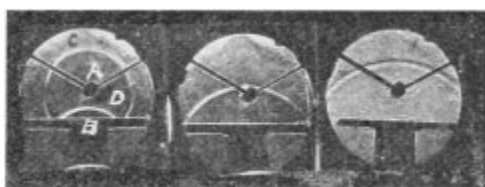
least eleven feet thick in order that some part should remain obscured during the whole exposure. As a fact, there is clearly seen a circular light shading, which does—so far as one can judge from the supposed rate of working and the known distances—expand at about the same rate as the observed shadow, but it is lighter than the ground and shaded, instead of being dark and sharp, as seen by the eye.”



**KINETOSCOPE FILM OF
EXPLOSION.**

So much for the visibility of sound under ordinary conditions. In the laboratory, by means of an optical contrivance due to the German physicist Toepler, we can secure a means of illumination so sensitive that the warm air rising from a person's hand appears like dense black smoke. Moreover, since we are working on a small scale, we can use the electric spark as the

source of light, and dispense with the photographic shutter. This is a great advantage, for the time of the exposure is, under these conditions, only about one fifty-thousandth of a second, during which time the sound wave will move scarcely a quarter of an inch. During the past year I have made a very complete series of photographs of sound waves, which illustrate in a most beautiful manner the fundamental principles of wave motion. It is not practicable to give here a full description of the apparatus used, but a brief outline may make the method intelligible. The sound photographed in each case is the crack of an electric spark, which is illuminated and photographed by the light of a second spark, occurring a brief instant later. In front of a large lens (a telescope objective, for example) two brass balls are mounted, between which the 'sound spark,' as I shall call it, passes. The instant the spark jumps across the gap, a spherical wave of condensed air starts out, which, when it reaches our ear, gives the sensation of a snap. The object is to photograph this wave before it gets beyond the limits of the lens. The camera is mounted in front of the lens and focussed on the brass balls, which appear in line in the picture, so that the sound spark is always hidden by the front one. The spark, on jumping between the balls, charges a Leyden jar, which instantly discharges itself between two wires placed behind the lens, producing the illuminating spark. This second spark can be made to lag behind the first just long enough to catch the sound wave when it is but a few inches in diameter, notwithstanding the fact that the spherical wave is expanding at the rate of eleven hundred feet a second. The photographs show in every case the circle of the lens filled up with the light of the illuminating spark, the brass balls (in line) and the rods that support them, and the sound wave, which appears in the simplest case as a circle of light and shade surrounding the balls. By placing an obstacle in the way of the wave we get the reflected wave or echo, and we shall see that the form of this echo may be very complicated.



**FIG. 1. SOUND WAVE
REFLECTED FROM A PLANE
SURFACE.**

It will be well at the outset to remind the reader of the close analogy between sound and light. A burning candle gives out spherical light waves, just as the snapping sparks give out sound waves. The form of the reflected light wave will be identical with that of a sound wave reflected under similar conditions. As we can not see the light waves themselves, we can only determine their form by calculation, and it is interesting to see that the forms photographed are identical in every case with the calculated ones. The object in view was to secure acoustical illustrations of as many of the phenomena connected with light as possible. We will begin with the very simplest case of all: the reflection of a spherical sound wave from a flat surface, corresponding to the reflection of light from a plane mirror. It can be shown by geometry that the reflected wave or echo will be a portion of a sphere, the center of which lies as far below the reflecting surface as the point at which the sound originates is above it. In the case of light, this point constitutes the image in the mirror. Referring to the photograph, we see the reflected wave in three successive positions, the interval between the sound spark and the illuminating spark having been progressively increased. The brass balls are shown at A, and beneath them the flat plate B, which acts as a reflector. In the first picture the sound wave C appears as a circle of light and shade, and has just intersected the plate. The echo appears at D. In the next two pictures the original wave has passed out of the field, and there remains only the echo.

It may, perhaps, be not out of place to remind the reader of the relation between rays of light and the wave surface. What we term light rays have no real existence, the ray being merely the path traversed by a small portion of consecutive wave surfaces. Since the wave surface always moves in a

direction perpendicular to itself, the rays are always normal to it. For instance, in the above case of a spherical wave diverging from a point, the rays radiate in all directions from the point; the same is true in the case of the echo, the rays radiating from the image point below the reflecting surface. In all subsequent cases the reader can, if interested in tracing the analogy between sound and light, draw lines perpendicular to the reflected wave surfaces representing the system of reflected waves.

We will now consider a second case of reflection. We know that if a lamp is placed in the focus of a concave mirror, the rays, instead of diverging in all directions, issue from the mirror in a narrow beam. The headlight of a locomotive and the naval searchlight are examples of the practical use made of this property. If the curvature of the mirror is parabolical, the rays leaving it are parallel; consequently mirrors of this form are employed rather than spherical ones. But what has the mirror done to the wave surface which is obviously spherical when it leaves the lamp, and what is its form after reflection? The wave surface, I have said, is always perpendicular to the rays; consequently in cases where we have parallel rays we should expect the wave to be flat or plane.



**FIG. 2. SPHERICAL SOUND
WAVE.**

Examine the second photograph, which shows a spherical, sound wave starting at the focus of a parabolic mirror. The echo appears as a *straight line*, instead of a circle as in the previous case, which shows us that the wave surface is flat.

If now our mirror is a portion of a sphere instead of a paraboloid, our reflected wave is not flat, and the reflected rays are not all parallel, the departure from parallelism increasing as we consider rays reflected from

points farther and farther away from the center of the mirror. A photograph illustrating the reflection of sound under these conditions is next shown, the echo wave being shaped like a flat-bottomed saucer. As the saucer moves upward the curved sides converge to a focus at the edge of the flat bottom, disappearing for the moment (as is shown in the fourth picture of the series), and then reappearing on the under side after passing through the focus, the saucer turning inside out.

If, instead of having a hemisphere, as in the last case, we have a complete spherical mirror, shutting the wave up inside a hollow ball, we get exceedingly curious forms; for the wave can not get out, and is bounced back and forth, becoming more and more complicated at each reflection. This is illustrated in our next photograph, the mirror being a broad strip of metal bent into a circle.^D Intricate as these wave surfaces are, they have all been verified by geometrical constructions, as I shall presently show.

^D Cylindrical mirrors have been used instead of spherical, for obvious reasons. A sectional view of the reflected wave is the same in this case as when produced by a spherical surface.

Another very interesting case of reflection is that occurring inside an elliptical mirror. When light diverges from one of the two foci of such a mirror, all the rays are brought accurately to the other focus. If rays of light come to a focus from all directions, it is evident that the wave surface must be a sphere, which, instead of expanding, is collapsing. This is very beautifully shown in the photographs. The sound wave starts in one focus and the reflected wave, of spherical form also, shrinks to a point at the other focus. (See [Fig. 5.](#))

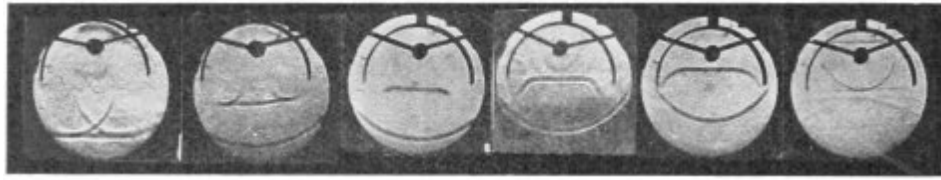


FIG. 3. A WAVE REFLECTED FROM A PORTION OF A SPHERE.

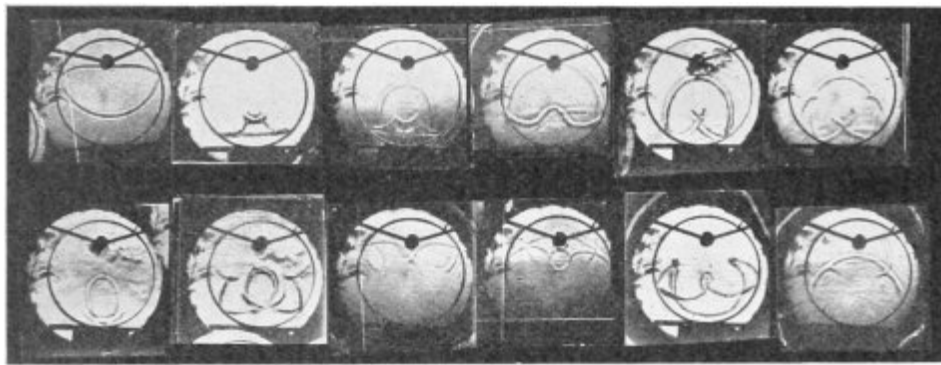


FIG. 4. A WAVE FROM A CYLINDRICAL MIRROR.

In the next series the wave starts outside of the field of the lens, and enters a hemispherical mirror. We know that a concave mirror has the power of bringing light to a focus at a point situated half-way between the surface of the mirror and its center of curvature. If the light comes from a very distant point, and the mirror is parabolic in form, the rays are brought *accurately* to a focus; which means that the reflected wave is a converging sphere,—a condition the opposite of that in which spherical waves start in the focus of such a mirror. If, however, the mirror is spherical, only a portion of the light comes to a focus. On examining the pictures we see that the reflected wave has a form resembling a volcanic cone with a bowl-shaped crater. See the third and fourth pictures of the series. The bowl of the crater shrinks to a point half-way between the surface of the mirror and its center of curvature, and represents that portion of the light which comes to a focus, while the sides of the cone run in under the collapsing bowl, and eventually cross. (No. 6 of the series.) From now on the portion which has come to a focus diverges, uniting with the sides of the cone, the whole passing out of the mirror in the form of a horseshoe.

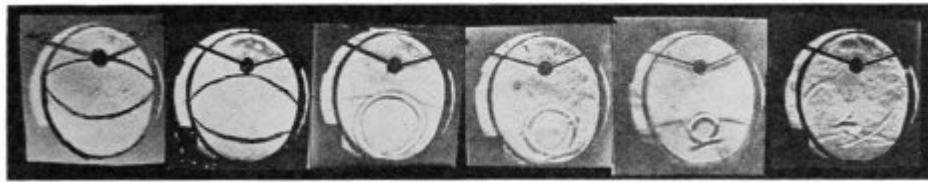


FIG. 5. A WAVE FROM AN ELLIPTICAL MIRROR.

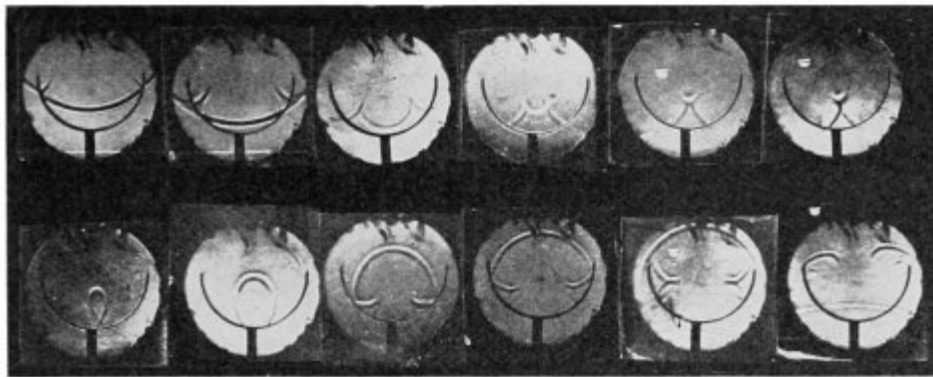


FIG. 6. A WAVE STARTING OUTSIDE THE FIELD OF THE LENS.

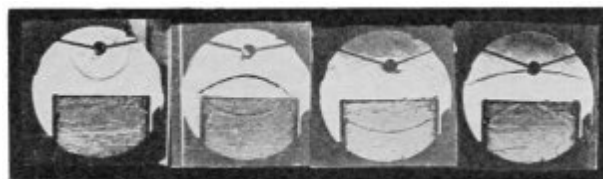


FIG. 7. A CASE OF REFRACTION.

We will now consider a case of refraction, and show the slower velocity of the sound wave in carbonic acid. A narrow glass tank, covered with an exceedingly thin film of collodion, was filled with the heavy gas and placed under the brass balls. When the sound wave strikes the collodion surface, it breaks up into two components, one reflected back into the air, the other transmitted down through the carbonic acid. An examination of the series shows that the reflected wave in air has moved farther from the collodion film than the transmitted wave, which, as a matter of fact, has been flattened

out into a hyperboloid. Exactly the same thing happens when light strikes a block of glass. We have rays reflected from the surface, and rays transmitted through the block, the waves which give rise to the latter moving slower than the ones in air.

A complete discussion of all of the cases that have been studied in this way would probably prove wearisome to the general reader. Prisms and lenses of collodion filled with carbonic acid and hydrogen gas have been made, and their action on the wave surface photographed. Diffraction, or the bending of the waves around obstacles, and the very complicated effects when the waves are reflected from corrugated surfaces, are also well shown. I shall, however, omit further mention of them and speak of but one other case, possibly the most beautiful of all.



FIG. 8. A MUSICAL TONE.

In all the cases that we have considered, it must be remembered that we have been dealing with a single wave—a pulse, as it is called. Musical tones are caused by trains of waves, the pitch of the note corresponding to the distance between the waves, or to the rate at which the separate pulses beat upon the drum of the ear. For studying the changes produced by reflection, wave trains would have been useless, owing to the confusion which would have resulted from the superposition of the different waves. Moreover, it is doubtful whether an ordinary musical tone could be photographed in this way; for the distance between the waves, even in the shrillest tones, is four or five inches, and the abrupt change in density, necessary for the perception of the wave, is not present. It is possible, however, to create a wave train or musical tone which can be photographed. The reader may perhaps have noticed that on a very still night, when walking beside a picket fence or in front of a high flight of steps, the sounds of his footsteps are echoed from the palings as metallic squeaks. Each picket, as the single wave caused by

the footfall sweeps along the fence, reflects a little wave; consequently a train of waves falls on the ear, the distance between the waves corresponding to the distance between the pickets. The closer together the pickets, the shriller the squeak. In point of fact, the distance between the waves in such a train is twice the distance between the palings, since they are not struck simultaneously by the footstep wave, but in succession.

This phenomenon, of the creation of a musical tone by the reflection of a noise, was reproduced by reflecting the crack of the spark from a little flight of steps. In the first picture the wave is seen half way between its origin and the reflecting surface. In the second it has struck the top stair, which is giving off its echo, the first wave of our artificially constructed musical tone. In the third we find the original wave at the sixth step, with a well-developed train of five waves rising from the flight. The following three pictures show the further development of the wave train. The height of each step was about a quarter of an inch; consequently the distance between the waves was half an inch. This would correspond to a note about three octaves above the highest ever used in music.

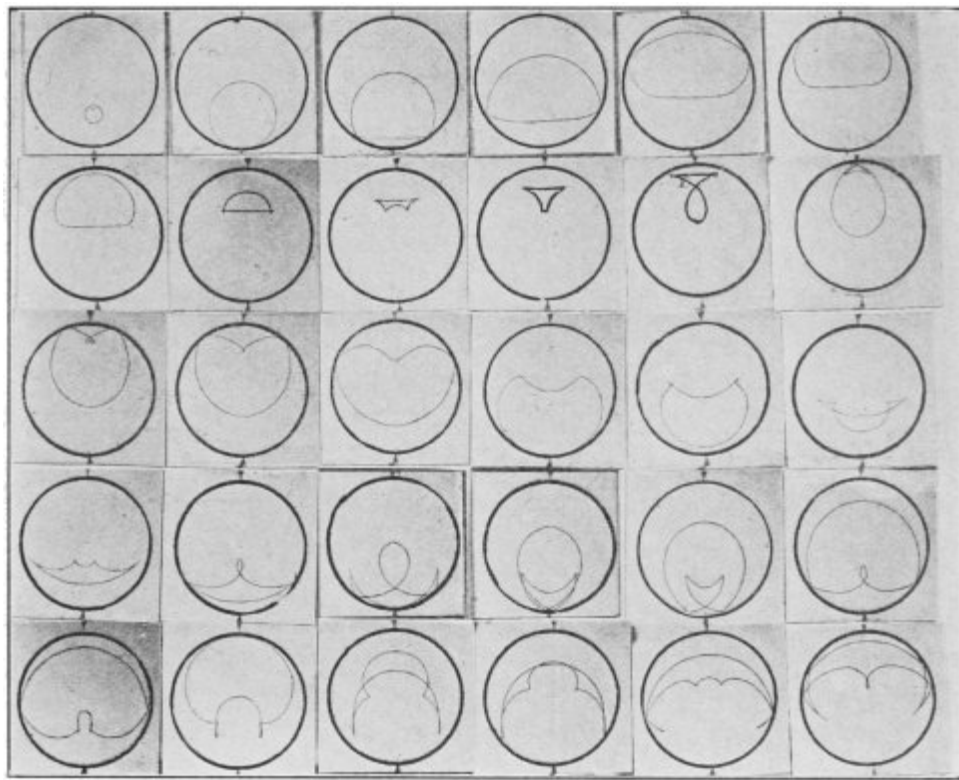


FIG 9. THE REFLECTION INSIDE THE HOLLOW SPHERE.

While experimenting with the complete circular mirror, which, it will be remembered, gave the most complicated forms, it occurred to me that a very vivid idea of how these curious wave surfaces are produced could be obtained by preparing a complete series in proper order on a kinetoscope film, and then projecting them in succession on the screen. The experimental difficulties were, however, too great to make it seem worth while to attempt to obtain a series of pictures of the actual waves, it being very difficult to accurately regulate the time interval between the two sparks. The easier method of making a large number of geometrical constructions, and then photographing them in succession on the film, was accordingly adopted. Three complete sets of drawings, to the number of about one hundred each, were prepared for three separate cases of reflection;—viz.: the entrance of a plane wave into a hemispherical mirror, the passage of a spherical wave out from the focus of a hemispherical mirror, and the multiple reflection of a spherical wave inside of a complete spherical mirror. Special methods were devised for simplifying the

constructions, and much less labor was required in the preparation of the diagrams than one would suppose. The results fully justified the labor, the evolutions of the waves being shown in a most striking manner. These films I exhibited before the Royal Society in February last, and a more complete description of the manner of preparing them may be found in the Proceedings of the Society.

A portion of one of these series is reproduced, about one in four or five of the separate diagrams being given. The series runs from left to right in horizontal rows. When projected on the screen, the spherical wave is seen gradually to expand from the focus point, like a swelling soap bubble; it strikes the surface, and the bowl-shaped echo bounces off and follows the unreflected portion across the field; these two portions are then reflected in turn, and the curiously looped wave flies back and forth across the mirror, changing continuously all the time, and becoming more complicated at each reflection. These diagrams should be compared with the photographs shown in the fourth series.

One must not suppose that these beautiful forms exist only in the laboratory. Every time we speak, spherical waves bounce off the floor, ceiling and walls of the room, while in any ordinary bowl or basin the curious crater-shaped echoes are formed. Glance once more at the wave surfaces produced within a hollow sphere, and try to imagine the complexity of the aerial vibrations caused by a fly buzzing around in an empty water-caraffe! The photographs enable us to realize what is going on around us all the time—this our perceptions are fortunately too dull to perceive. Life would be a nightmare if we were obliged to see the myriads of flying sound waves bounding and rebounding about us in every direction, and combining into grotesque and ever-changing forms. It is just as well, on the whole, that the light of the electric spark and the delicate optical device of Toepler are necessary to bring them into view.

THE PSYCHOLOGY OF RED.

BY HAVELOCK ELLIS.

AMONG all colors, the most poignantly emotional tone undoubtedly belongs to red. The ancient observation concerning the resemblance of scarlet to the notes of a trumpet has often been repeated, though it was probably unknown to the young Japanese lady who, on hearing a boy sing in a fine contralto voice, exclaimed: "That boy's voice is red." On the one hand, red is the color that idiots most easily learn to recognize; on the other hand, Kirchhoff, the chemist, called it the most aristocratic of colors; Pouchet, the zoölogist, was inclined to think that it was a color apart, not to be paralleled with any other chromatic sensation, and recalled that the retinal pigment is red; Laycock, the physician, confessed that he preferred the gorgeous red tints of an autumn sunset to either musical sounds or gustatory flavors. Artists more cautious than men of science in expressing such a preference—knowing that a color possesses its special virtue in relation to other colors, and that all are of infinite variety—yet easily reveal, one may often note, a predilection for red by introducing it into scenes where it is not naturally obvious, whether we turn to a great landscape painter like Constable or to a great figure painter like Rubens, who, with the development of his genius, displayed even greater daring in the introduction of red pigments into his work.

In all parts of the world red is symbolical of joyous emotion. Often, either alone or in association with yellow, occasionally with green, it is the fortunate or sacred color. In lands so far apart as France and Madagascar scarlet garments were at one time the exclusive privilege of the royal family. A great many different colors are symbolical of mourning in various parts of the world; white, gray, yellow, brown, blue, violet, black can be so used, but, so far as I am aware, red never. Everywhere we find, again, that red pigments and dyes, and especially red ochre, are apparently the first to be

used at the beginning of civilization, and that they usually continue to be preferred even after other colors are introduced. There is indeed one quarter of the globe where the allied color of yellow, which often elsewhere is the favorite after red, may be said to come first. In a region of which the Malay peninsula is the center and which includes a large part of China, Burmah and the lower coast of India, yellow is the sacred and preferred color, but this is the only large district which presents us with any exception to the general rule, among either higher or lower races, and since yellow falls into the same group as red, and belongs to a neighboring part of the spectrum, even this phenomenon can scarcely be said to clash seriously with the general uniformity.^E

^E A further partial exception is furnished by the tendency to prefer green which may be found in certain countries, now or formerly Mahommedan, such as North Africa and to a large extent Spain, which have an arid and more or less desert climate.

If we turn to Australia, whither the anthropologist often turns in order to explore some of the most primitive and undisturbed data of early human culture still available for study, we find the preference for red very well marked. In times of rejoicing the tribes at Port Mackay, Curr remarked, paint themselves red; in times of mourning, white. In describing the paintings and rock carvings of the Australians, Mathews states that red, white, black and occasionally yellow pigments were used, precisely the four pigments which Karl von den Steinen found in use in Central Brazil. Prof. Baldwin Spencer and Mr. Gillen, in their valuable work on the natives of Central Australia, have pointed out the significance and importance of red ochre. One of the most striking and characteristic features, they say, of Central Australians' implements and weapons is the coating of red ochre with which the native covers everything except his spear and spear-thrower. The hair is greased and red-ochred, and red ochre is the most striking feature in decoration generally. For ages past the Australian native has been

accustomed to rub this substance regularly over his most sacred objects, and then over ordinary objects.

There is, however, no need to go so far afield in order to illustrate the primitive use of red ochre. Our own European ancestors followed exactly the same methods, and the German woman of early ages used red and yellow ochre to adorn her face and body, while the finds of the ice age at Schussenquelle, described by Fraas, included a brilliant red paste (oxide of iron with reindeer fat) evidently intended for purposes of adornment. Moreover, the early artists of classic times had precisely the same predilections in color as the aboriginal Australian artists. Red, white, black and yellow are the dominant colors in the *Iliad*, and Pliny mentions that the most ancient pictures were painted in various reds, while at a later date red and yellow predominated. He also mentions that yellow was the favorite color of women for garments, and was specially used at marriages, while red being a sacred color and apt to provoke joy, was used at popular festivals, in the form of minium and cinnabar, to smear the statues of Jupiter.

This well-nigh universal recognition of the peculiarly intense emotional tone of red is reflected in language. The color words of civilized and uncivilized peoples have been investigated with interesting and on the whole remarkably harmonious results. It is only necessary here to refer to them briefly in so far as they are related to our present subject. It seems that in every country the words for the colors at the red end of the spectrum are of earlier appearance, more definite and more numerous, than for those at the violet end. On the Niger it appears that there are only three color words, red, white and black, and everything that is not white or black is called red. The careful investigation of the natives of Torres Straits and New Guinea by Dr. W. H. R. Rivers, of the Cambridge Anthropological Expedition, has shown that at Murray Island, Mabuiag and Kiwai there were definite names for red, less definite for yellow, still less so for green, while any definite name for blue could not be found. In this way as we pass from the colors of long wave-length towards those of short wave-length we find the color nomenclature becoming regularly less definite. In Kiwai and Murray Island the same word was applied to blue and black, and at Mabuiag there was a word (for sea-color) which could be applied either to blue or green, while Australian natives from Fitzroy River seemed limited to words for red,

white and black. In a neighboring region of Northern Queensland Dr. Walter Roth has reached almost identical results, the tribes having distinct names for red and yellow, as applied to ochre, while blue is confounded in nomenclature with black. In Brazil, again, while all tribes use separate words for red, yellow, white and black, only one had a word for blue and green. Even so æsthetic a people as the Japanese have no general words for either blue or green, and apply the same color word to a green tree and the unclouded sky.

Here again we may trace similar phenomena in Europe; the same greater primitiveness, precision and copiousness of the color vocabulary at the long wave end of the spectrum are found among Europeans as well as among the lowest savages. The vagueness of the Greek color vocabulary, especially at the violet end of the spectrum, has led to much controversy. Latin was especially rich in synonyms for red and yellow, very poor in synonyms for green and blue. The Latin tongue had even to borrow a word for blue from Teutonic speech; *caeruleus* originally meant dark. Even in the second century A. D. Aulus Gellius, who knew seven synonyms for red and yellow, scarcely mentions green and blue. Magnus has pointed out that a preference for the colors at the violet end of the spectrum coincided with the spread of Christianity, to which we owe it, he believes, that yellow ceased to be popular and was treated with opprobrium.^E Modern English bears witness that our ancestors, like the Homeric poets, resembled the Australian aborigines in identifying the color of the short wave end of the spectrum with entire absence of color, for 'blue' and 'black' appear to be etymologically the same word.

^E In this connection I may mention that the preference for green, which, as I have shown elsewhere ("The Color Sense in Literature," *Contemporary Review*, May, 1896), developed in English literature with the rise of Puritanism in the seventeenth century.

At this point we come across an interesting and once warmly debated question. It was maintained some twenty years ago by writers who had been

impressed by the defectiveness of the color vocabulary at the short wavelength end of the spectrum, that primitive man generally, and early Hellenic man in particular, were insensitive to the colors at that end of the spectrum, and unable to distinguish them. On investigation of individuals belonging to savage races it appeared, however, that no marked inferiority in color discrimination could be demonstrated. Hence it became clear that the vague and defective vocabulary for blue and green must be due to some other cause than vague and defective perception, and that sensation and nomenclature were not sufficiently parallel to enable us to argue from one to the other.

That, in the main, is a conclusion which still holds good. In all parts of the world it has been found that color discrimination, even amongst the lowest savages, is far more accurate than color nomenclature. Thus of an African Bantu tribe, the Mang'anja, Miss Werner states that they can discriminate all varieties of blue in beads, but call them all black. The sky is black; so is any green, brown or grey article, though a very bright grey counts as white. Violet or purple is black. Yellow is either red or white. A word supposed sometimes to mean green really means raw, unripe or even wet. Thus the Mang'anja only have three colors—black, white and red. In quite a different region, the Zulus, more advanced in color nomenclature, have not only black, white and red, but a word which may mean either green or blue, and another which means yellow, buff or grey, or some shade of brown. At the same time it now appears that the earlier scientific writers on this subject were not entirely wrong in stating that among savages there is some actual failure of perception at the short wave end of the spectrum, although they were wrong in arguing that it was necessarily involved in the defects of color vocabulary, and in imagining that it could be as extensive as that hypothesis demanded. It now appears that the conclusions reached by Hugo Magnus of Breslau, as expressed in 1883 in his study 'Ueber Ethnologische Untersuchungen des Farbensinnes,' fairly answer to the facts. In large measure relying on the examination of 300 Chukchis made by Almquist during the Nordenskiöld Expedition, Magnus concluded that although the color vision of the uncivilized has the same range from red to violet as that of the civilized and all the colors can usually be separately distinguished, there is sometimes a certain dullness, a diminished energy of sensation, as regards green and blue, the shorter and more refrangible waves of the spectrum, while the colors at the other end are perceived with much

greater vividness. Stephenson, more recently, among over one thousand Chinese, examined at various places, found only one case of color blindness, but a frequent tendency to confuse green and blue and also blue and purple, while Dr. Adele Fielde, of Swatow, China, among 1,200 Chinese of both sexes examined by Thomson's wool test, found that more than half mixed up green and blue, and many even seemed to be quite blind to violet. Ernest Krause also has argued that primitive man was most sensitive to the red end of the spectrum, hence setting about to obtain red pigments and acquiring definite names for them, an explanation which is accepted by Karl von den Steinen to account for the phenomena among the Central Brazilians. The recent investigations of Rivers at Torres Straits have confirmed the conclusions of Magnus. He found that, corresponding to the defect of color terminology, though to a much less degree, there appeared to be an actual defect of vision for colors of short wave-length; in testing with colored wools no mistake was ever made with reds, but blues and greens were constantly confused, as were blue and violet.

It may even be argued that the same defect exists to a minor degree not only among the peoples of Eastern Asia whose æsthetic sense is highly developed, but among civilized Europeans when any kind of color blindness is altogether excluded. This was noted long since by Holmgren, who remarked that some persons, though able to distinguish between blue and green wools when placed together, were liable to call the blue wool green, and the green blue, when they saw them separately. Magnus also showed that such an inability is apt to appear at a very early stage in some persons when the illumination is diminished, although the perception of red and yellow remains perfectly distinct. He further showed that blue and green at certain distances are often much more difficult to recognize than red. Most people probably are conscious of difficulty in distinguishing blue and green pigments with diminished light and find that blue easily passes into black. Violet also appears for many people to be merely a variety of blue; the word itself, we may note, is recent in our language, and plays a very small part in our poetic literature, and in fact the color itself, if we rigidly exclude purple, is extremely rare in nature. It is a noteworthy fact in this connection that in normal persons the color sense may be easily educated; this is not merely a fact of daily observation, but has been exactly demonstrated by Féré, who by means of his chromoptoscopic boxes, containing very dilute colored solutions, found that with practice it was possible to recognize solutions

which had previously seemed uncolored. It is also noteworthy that in the achromatopsia of the hysterical, as Charcot showed and as Parimand has since confirmed, the order in which the colors usually disappear is violet, green, blue, red; sometimes the paradoxical fact is found that red will give a luminous sensation in a contracted visual field when even white gives no luminous sensation. This persistence of red vision in the hysterical is only one instance of a predilection for red which has often been noted as very marked among the hysterical. Red also exerted a great fascination over the victims of the mediæval hysterical epidemics of tarantism in Italy, while the victims of the German mediæval epidemic of St. Vitus's dance imagined that they were immersed in a stream of blood which compelled them to leap up.

It may be noted that red and perhaps yellow have been stated to be the only colors visible in dreams; this is possibly due to the blood-vessels. Such an explanation is probable with regard to the various subjective visual sensations which constitute an aura in epilepsy, among which, as Gowers notes, red and reddish yellow are most frequently found. Féré has further noted that in various emotional states somewhat resembling epilepsy, and even in mystic exaltation, red may be subjectively seen. Simroth has gone so far as to argue that not only is red fundamental in human color psychology, but that in living organisms generally, even as a pigment, red is the most primitive of colors, that since the algæ at the greatest sea-depths are red it is possible that protoplasm at first only responded to rays of long wave-length, and that with increased metabolism colors became differentiated, following the order in the spectrum.

If it is really the case that in the evolution of the race familiarity with the red end of the spectrum has been earlier and more perfectly acquired than with the violet end, and that red and yellow made a more profound impression on primitive man than green and blue, we should expect to find this evolution reflected in the development of the individual, and that the child would earlier acquire a sensitiveness for red and orange and yellow than for green and blue and violet. This seems actually to be the case. The study of the color sense in children is, indeed, even more difficult than in savages; and many investigators have probably succumbed to the fallacies involved in this study. Doubtless we may thus account for some discrepancies in the attempts to ascertain the facts of color perception and

color preference in children, while doubtless also there are individual differences which discount the value of experiments made on only a single child. A few careful and elaborate investigations, however, especially that of Garbini on 600 North Italian children of various ages, have thrown much light on the matter. There is fairly general agreement that red is the first color that attracts young children and which they recognize. That is the result recorded by Uffelmann in Germany, while Preyer found yellow and red at the head; Binet in France concluded that red comes first; Wolfe in America reached the same result, and Luckey noted that his own children seemed to enjoy red, orange and yellow very much earlier than they could perceive blue, which seemed to come last. Baldwin, indeed, found in the case of his own child that blue seemed more attractive than red; his methods have, however, been criticised, and his experiments failed to include yellow. Mrs. Moore found that her baby, between the sixteenth and forty-fifth weeks, nearly always preferred a yellow ball to a red ball; this was doubtless not a matter of color, but of brightness, for there is no reason to suppose chromatic perception at so early an age. Red, orange and yellow, it may be added, are perceived by a slightly lower illumination than green, blue and violet, the last being the most difficult of all to perceive, so that it is not surprising that the colors at the violet end should be inconspicuous to young infants. Garbini, whose experiments are worth noting in more detail, found that the order of perception is red, green, yellow, orange, blue and violet, and as he experimented with a large number of children and used methods which so competent a judge as Binet regards as approaching perfection, his results may be considered a fair approach to the truth. He found that for the first few days after birth the infant shuns the light; then, about the fourteenth day, he ceases to be photophobic and begins to enjoy the light, as is shown by his being quieted when brought into a bright light and crying when taken from it; this may sometimes begin even about the fifth day. Between the fifth week and the eighteenth month children show signs of distinguishing white, black and grey objects. It is not until after the eighteenth month that their chromatic perception begins, any preference for red and yellow objects at an earlier age being due merely to their greater luminosity. Garbini considers that it is the center of the retina, or the portion most sensitive to red and yellow, which is most exercised in young infants. Between the second and third years children, both boys and girls, were found to be most successful in the recognition of red, then of green, but they

very often confused orange with red, and mixed up yellow, blue, violet and green; he thinks they tend to confuse a color with the preceding color in spectral order. Under the age of three children may be said to be color-blind, and they are liable to confuse rosy tints with green. Between the ages of three and five they are able to distinguish red in any gradation, green nearly always, with an occasional confusion with red, while yellow is sometimes confused with orange, orange sometimes replaced by rose, blue often not recognized in its gradations, and violet often selected in place of blue. At this age, also (as in hysterical anæsthesia of the retina), blue seems dark or black. In the fifth and sixth years red, green and yellow are always correctly chosen; orange gradations are not always recognized, and blue and violet come last, being sometimes confused. In the sixth year children are perfecting their knowledge of orange, blue and violet and completing their knowledge of color designations. Garbini has reached the important result that color perceptions and verbal expression of the perceptions follow exactly parallel paths, so that in studying verbal expression we are really studying perception, with the important distinction that the expression comes much later than the perception.^G These investigations of Garbini are very significant, and there can be little doubt that the evolution of the child's color sense repeats that of the race.

^G Garbini, "Evoluzione del senso cromatico nella infanzia," *Archivio per l'Antropologia*, 1894. I.

In dealing with the color perceptions of savages and children we are, of course, to some extent dealing more or less unconsciously with their color preferences. There is some interest from our present point of view in considering the conscious color preferences of young and adult civilized persons. Red, as we have seen, is the color that fascinates our attention earliest, that we see and recognize most vividly; it remains the color that attracts our attention most readily and that gives us the greatest emotional shock. It by no means necessarily follows that it is the most pleasurable color. As a matter of fact, such evidence as is available shows that very often it is not. There seems reason to think that after the first early

perception of red, and early pleasure in it, yellow or orange is frequently the favorite color, the preference often lasting during several years of childhood; Preyer's child liked and discriminated yellow best, and Miss Shinn was inclined to think that it was the favorite color of her niece, who in the twenty-eighth month showed a special fondness for daffodils and for a yellow dress. Barnes found that in children the love of yellow diminishes with age. Binet's child was specially preoccupied with orange. Aars in an elaborate and frequently varied investigation into the color preferences of eight children (four of each sex), between four and seven years of age, found that with the boys the order of preference was blue and yellow (both equal), then red, lastly green; while with the girls the order was green, blue, red and yellow; in combinations of two colors it was found that combinations of blue come first, then of yellow, then green, lastly red. It was found (as J. Cohn has found among adults and cultivated people) that the deepest and most saturated color was most pleasing; and also that the love of novelty and of variety was an important factor. It will be observed that at this age green was the girls' favorite color and that least liked by the boys, whose favorite color, in combination, was blue; the number of individuals was, however, small. This was in Germany. In America, among 1,000 children, probably somewhat older on the average (though I have not details of the inquiry), Mr. Earl Barnes found, like Dr. Aars, that more boys than girls selected blue, while the girls preferred red more frequently than the boys; Barnes considers that with growing years there is a growing tendency to select red; as is well known, girls are more precocious than boys. Among 100 students at Columbia University, the order of preference was found to be blue (34 per cent), red (22.7 per cent), and then at a more considerable distance violet, yellow, green. It is noteworthy that among 100 women students at Wellesley College the order of preference was not very different, being blue (38 per cent), red (18 per cent), yellow, green, violet; in a later investigation the order remained the same, there being only some increase in the preference for red; it was considered that association accounted for the preference for blue, while more conscious as well as more emotional elements entered into the preference for red.

By far the most extensive investigation of color preference was that carried on at Chicago by Professor Jastrow on 4,500 persons, mostly adults, of both sexes and various nationalities.^H Blue was found to be the favorite

color, less than half as many persons preferring red; of every thirty men ten voted for blue and three for red, while of every thirty women five voted for red and four for blue. The men also liked violet and on the whole confined their choice to but few colors, the women also liked pink, green (very seldom chosen by men) and yellow, and showed a tendency to choose light and dainty shades. There was on the whole a decided preference for dark shades; the least favorite colors were yellow and orange. It is evident that, as we should expect, within the elementary field of popular æsthetics, women show a more trained feeling for color than men.

^H J. Jastrow, "The Popular Æsthetics of Color," *Popular Science Monthly*, 1897.

It is not quite easy to coördinate the various phenomena of color predilection. Careful and extended observations are still required. It seems to me, however, that the facts, as at present ascertained, do suggest a certain order and harmony in the phenomena. It is difficult not to believe that there really is, both among many uncivilized peoples and also many children at an early age, even to a slight extent among civilized adults, a relative inability, by no means usually absolute, to recognize and distinguish the tones of color at the more refrangible end of the spectrum. The earliest writers on the subject were wrong when they supposed that color nomenclature at all accurately corresponded to color perception, and it is well recognized that there are no peoples who are wholly unable to distinguish between green and blue and black. But as Garbini has clearly shown, there really is a parallelism between color nomenclature and color recognition, and Garbini's wide investigation has confirmed the experiments of Preyer on a single child by showing that there is a certain hesitancy and uncertainty in recognizing the colors at the more refrangible end of the spectrum, long after children are familiar with the less refrangible end. In the same way the important investigations of Rivers have confirmed the earlier observations of Magnus and Almquist in showing that savages in many cases exhibit a certain difficulty in recognizing and distinguishing blue and green, such as they never experience with red and yellow. The vagueness of color nomenclature as regards blue and green thus indicates, though grossly

exaggerating, a real psychological fact, and in this way we have an explanation of the curious fact that in widely separated parts of the world (at Torres Straits, among the Esthonians at Rome, etc.) as civilization progressed it was found necessary to borrow a word for blue from other languages.

There is almost complete harmony among a number of observers, now very considerable, in many countries, showing that the colors children first take notice of and recognize are red and yellow, most observers putting red first. There is no true predilection for these colors at this early age because the other colors do not yet seem to have been perceived. At first, doubtless, all colors appear to the infant as light or dark, white or black. That this is so is indicated by the experience of Dr. George Harley, who at one period of his life, in order to cure an injury to the retina caused by overwork at the microscope, resolutely spent nine months in absolutely total and uninterrupted darkness. When he emerged he found that, like an infant, he was unable to appreciate distance by the eye, while he had also lost the power of recognizing colors; for the first month all light colors appeared to him perfectly white and all dark colors perfectly black. He fails to state the order in which the colors reappeared to him. It is well recognized, however, that eyes long unexposed to light become color-blind for all colors except red. Preyer's child in the fourth year was surprised that in the twilight her bright blue stockings looked grey, while for some time longer she always called dark green black. By the sixth year all colors are seen and known with fair correctness. Among young children at this age, so far as the evidence yet goes, red is rarely the preferred color, this being more often yellow, green or blue. There is doubtless room here for a great amount of individual difference, but on the whole it appears that children prefer those colors which they have most recently learnt to recognize, the colors which have all the charm of novelty and newly-won possession. It is probable, too, that (as Groos has also suggested) the stimulation of red is too painfully strong in this stage of the development of the color sense to be altogether pleasurable, in the same way that orchestral music is often only a disturbing noise to children.

One may note in this connection that hyperæsthesia to color is nearly always an undue sensibility to red and very rarely to any other color. The case has been recorded of a highly neurotic officer who, for more than thirty

years, was intolerant of red-colored objects. The dazzling produced by scarlet uniforms, especially in bright sunshine, seriously interfered with the performance of his duties, and in private life red parasols, shawls, etc., produced similar effects; he was often overcome in the streets by giddiness, sometimes almost before he realized that he was looking at a red object. Many years ago Laycock referred to the case of a lady who could not bear to look at anything red, and Elliston also had a lady patient to whom red was very obnoxious, and who, when put into a room with red curtains, drank seven quarts of fluid a day. I am not aware that any such hyperæsthesia exists in the case of other colors. It is also noteworthy that the morbid affection in which color is seen when it does not exist is most usually a condition in which red is seen (erythropsia), yellow being the color most frequently seen after red (a condition called xanthopsia); the other colors are very rarely seen, and Hilbert, in his monograph on the pathology of the color sense, considers that this is due to the fact that red and yellow make the most intense effect on the sensorium, which thus becomes liable not only to direct but to reflected irritation, in the absence of any external color stimulus. There are other facts which show that of all colors red is that which acts as the most powerful stimulus on the organism. Münsterberg, in some interesting experiments which he made to illustrate the motor power of visual impressions as measured by their arresting action on the eye-muscles, found that red and yellow have considerably more motor power in stimulating the eye than the other colors. It may be added also that, as Quantz has found, we overestimate the magnitude of colors of the less refrangible part of the spectrum and underestimate the others.

After puberty blue seems still to maintain its position, but red has now come more to the front, while yellow has definitely receded; although so favorite a color in classic antiquity, it is rarely the preferred color among ourselves. J. Cohn in Germany found that among a dozen students it was never in any degree of saturation the preferred color, while at Cornell Major found that all the subjects investigated considered yellow and orange either unpleasant or among the least pleasant colors.

While blue seems to be the color most usually preferred by men, red is more commonly preferred by women, who also show a more marked predilection for its complementary green. Whether the feminine love of red shows a fine judgment we could better decide if we knew among what

classes of the population red lovers and blue lovers respectively predominate; it may be noted, however, that the necessities of dress give the most ordinary woman an acquaintance with the elementary æsthetics of color which the average man has no occasion to acquire. In any case it might have been anticipated that, even though the typically 'cold' color should appeal most strongly to men, the most emotional of colors should appeal most strongly to women.

CHAPTERS ON THE STARS.

BY PROFESSOR SIMON NEWCOMB, U. S. N.

CONSTELLATIONS AND STAR NAMES.

IN ancient times the practice was adopted of imagining the figures of heroes and animals to be so outlined in the heavens as to include in each figure a large group of the brighter stars. In a few cases some vague resemblance may be traced between the configurations of the stars and the features of the object they are supposed to represent; in general, however, the arrangement seems quite arbitrary. One animal or man could be fitted in as well as another. There is no historic record or trace as to the time when the constellations were mapped out, or of the process by which the outlines were traced. The names of heroes, such as Perseus, Cepheus, Hercules, etc., intermingled with the names of goddesses, show that the constellations were probably mapped out during the heroic age. No maps are extant showing exactly how each figure was placed in the constellation; but in the catalogue of stars given by Ptolemy in his 'Almagest,' the positions of particular stars on the supposed body of the hero, goddess or animal are designated with such precision as he had at command, in some fairly precise position of the figure. For example, Aldebaran is said to have formed the eye of the Bull. Two other stars marked the right and left shoulders of Orion, and a small cluster marked the position of his head. A row of three stars in a horizontal line showed his belt, three stars in a vertical line below them his sword. In this way the position of the figure can be reproduced with a fair degree of certainty.

In the well-known constellation *Ursa Major*, the Great Bear, familiarly known as the Dipper, three stars form the tail of the animal, and four others a part of his body. This formation is not unnatural, yet the figure of a dipper

fits the stars much better than that of a bear. In Cassiopeia, which is on the opposite side of the pole from the Dipper, the brighter stars may easily be imagined to form a chair in which a lady may be seated without further difficulty. As a general rule, however, the resemblances of the stars to the figure are so vague that the latter might be interchanged to any extent without detracting from their appropriateness.

In any case, it was impossible so to arrange the figures that they should cover the entire heavens; blank spaces were inevitably left in which stars might be found. In order to include every star in some constellation, the figures have been nearly ignored by modern astronomers, and the heavens have been divided up, by somewhat irregular lines, into patches, each of which contains the entire figure as recognized by ancient astronomers. But all are not agreed as to the exact outlines of these extended constellations, and, accordingly, a star is sometimes placed in one constellation by one astronomer and in another constellation by another astronomer.

The confusion thus arising is especially great in the southern hemisphere, where it has been intensified by the subdivision of one of the old constellations. The ancient constellation *Argo* covered so large a region of the heavens, and included so many conspicuous stars, that it was divided into four, representing various parts of a ship—the sail, the poop, the prow and the hull.

Dr. Gould, while director of the Cordoba Observatory, during the years 1870 to 1880, constructed the ‘Uranometria Argentina,’ in which all the stars visible to the naked eye more than 10 degrees south of the celestial equator were catalogued and mapped. He made a revision of the boundaries of each constellation in such a way as to introduce greater regularity. The rule generally followed was that the boundaries should, so far as possible, run in either an east and west or a north and south direction on the celestial sphere. They were so drawn that the smallest possible change should be made in the notation of the conspicuous stars; that is, the rule was that, if possible, each bright star should be in the same constellation as before. The question whether this new division shall replace the ancient one is one on which no consensus of view has yet been reached by astronomers. Simplicity is undoubtedly introduced by Gould’s arrangement; yet, in the course of time, owing to precession, the lines on the sphere which now run north and south or east and west will no longer do so, but will deviate

almost to any extent. The only advantage then kept will be that the bounding lines will generally be arcs of great circles.

When the heavens began to be carefully studied, two or three centuries ago, new constellations were introduced by Hevelius and other astronomers to fill the vacant spaces left by the ancient ones of Ptolemy. To some of these, rather fantastic names were given; the Bull of Poniatowski, for example. Some of these new additions have been retained to the present time, but in other cases the space occupied by the proposed new constellation was filled up by extending the boundaries of the older ones.

At the present time the astronomical world, by common consent, recognizes eighty-nine constellations in the entire heavens. In this enumeration *Argo* is not counted, but its four subdivisions are taken as separate constellations.

NAMES OF THE STARS.

A glance at the heavens will make it evident that the problem of designating a star in such a way as to distinguish it from all its neighbors must be a difficult one. If such be the case with the comparatively small number of stars visible to the naked eye, how must it be with the vast number that can be seen only with the telescope? In the case of the great mass of telescopic stars we have no method of designation except by the position of the star and its magnitude; but with the brighter stars, and, indeed, with all that have been catalogued, other means of identification are available.

It is but natural to give a special name to a conspicuous star. That this was done in very early antiquity we know by the allusion to Arcturus in the Book of Job. At least two such names, Castor and Pollux, have come down to us from classical antiquity, but most of the special names given to the stars in modern times are corruptions of certain Arabic designations. As an example we may mention Aldebaran, a corruption of Al Dabaran—The Follower. There is, however, a tendency to replace these special names by a designation of the stars on a system devised by Bayer early in the seventeenth century.

This system of naming stars is quite analogous to our system of designating persons by a family name and a Christian name. The family

name of a star is that of the constellation to which it belongs. The Christian name is a letter of the Greek or Roman alphabet, or a number. As a number of men in different families may have the same Christian name, so the Greek letter or number may be given to a star in any number of constellations without confusion.

The work of Bayer was published under the title of 'Uranometria,' of which the first edition appeared in 1601. This work consists mainly of maps of the stars. In marking the stars with letters on the map, the rule followed seems to have been to give the brighter stars the earlier letters in the alphabet. Were this system followed absolutely, the brighter stars should always be called α ; the next in order β , etc. But this is not always the case. Thus in the constellation *Gemini*, the brighter star is Pollux, which is marked β , while α is the second brightest. What system, if any, Bayer adopted in detail has been a subject of discussion, but does not appear to have been satisfactorily made out. Quite likely Bayer himself did not attempt accurate observations on the brightness of the stars, but followed the indications given by Ptolemy or the Arabian astronomers. As the number of stars to be named in several constellations exceeds the number of letters in the Greek alphabet, Bayer had recourse, after the Greek alphabet was exhausted, to letters of the Roman alphabet. In this case the letter A was used as a capital, in order, doubtless, that it should not be confounded with the Greek α . In other cases smaller italics are used. In several catalogues since Bayer, new italic letters have been added by various astronomers. Sometimes these have met with general acceptance, and sometimes not.

Flamsteed was the first Astronomer Royal in England, and observed at Greenwich from 1666 to 1715. Among his principal works is a catalogue of stars in which the positions are given with greater accuracy than had been attained by his predecessors. He slightly altered the Bayer system by introducing numbers instead of Greek letters. This had the advantage that there was no limit to the number of stars which could be designated in each constellation. He assigned numbers to all the brighter stars in the order of their right ascension, irrespective of the letters used by Bayer. These numbers are extensively used to the present day, and will doubtless continue to be the principal designations of the stars to which they refer. It is very common in our modern catalogues to give both the Bayer letter and the Flamsteed number in the case of Bayer stars.

The catalogues by Flamsteed do not include quite all the stars visible to the naked eye, but various uranometries have been published which were intended to include all such stars. In such cases the designations now used frequently correspond to the numbers given in the uranometries of Bode, Argelander and Heis.

In recent times these uranometries have been supplemented by censuses of the stars, which are intended to include all the stars to the ninth or tenth magnitude. I shall speak of these in the next section; at present it will suffice to say that stars are very generally designated by their place in such a census.

There is still here and there some confusion both as to the boundaries of the constellations and as to the names of a few of the stars in them. I have already remarked that, in drawing the imaginary boundaries on a star map, as representing the celestial sphere, different astronomers have placed the lines differently. One of the regions in which this is especially true is in the neighborhood of the north pole, where some astronomers place stars in the constellation Cepheus which others place in Ursa Minor. Hence in the Bayer system the same star may have different names in different catalogues. Again, in extending the names or numbers, some astronomers use names which others do not regard as authoritative. The remapping of the southern constellation by Dr. Gould changed the boundaries of most of the southern constellations in a way already mentioned.

I have spoken of the subdivision of the great constellation *Argus* into four separate ones. Bayer having assigned to the principal stars in this constellation the Greek letters α , β , γ , etc., the general practice among astronomers since the subdivision has been to continue the designation of the stars thus marked as belonging to the constellation *Argo*. Thus, for example, we have *Argus*, which after the subdivision belonged to the constellation *Carina*. The variable star η *Argus* also belongs to the constellation *Carina*. But in the case of stars not marked by Bayer, the names were assigned according to the subdivided constellations, *Vela*, *Carina*, etc. Confusing though this proceeding may appear to be, it is not productive of serious trouble. The main point is that the same star should always have the same name in successive catalogues. Still, however, it has recently become quite common to ignore the constellation *Argus* altogether and use only the names of its subdivisions. The reader must therefore be on

his guard against any mistake arising in this way in the study of astronomical literature.

In star catalogues the position of a star in the heavens is sometimes given in connection with its name. In this case the confusion arising from the same star having different names may be avoided, since a star can always be identified by its right ascension and declination. The fact is that, so far as mere identification is concerned, nothing but the statement of a star's position is really necessary. Unfortunately, the position constantly changes through the precession of the equinoxes, so that this designation of a star is a variable quantity. Hence the special names which we have described are the most convenient to use in the case of well-known stars. In other cases a star is designated by its number in some well-known catalogue. But even here different astronomers choose different catalogues, so that there are still different designations for the same star. The case is one in which action of uniformity of practice is unattainable.

CATALOGUING AND NUMBERING THE STARS.

A catalogue or list of stars is a work giving for each star listed its magnitude and its position on the celestial sphere, with such other particulars as may be necessary to attain the object of the catalogue. If the latter includes only the more conspicuous stars, it is common to add the name of each star that has one; if none is recognized, the constellation to which the star belongs is frequently given.

The position of a star on the celestial sphere is defined by its right ascension and declination. These correspond to the longitude and latitude of places on the earth, in the following way: Imagine a plane passing through the center of the earth and coinciding with its equator, to extend out so as to intersect the celestial sphere. The line of intersection will be a great circle of the celestial sphere, called the celestial equator. The axis of the earth, being also indefinitely extended in both the north and the south directions, will meet the celestial spheres in two opposite points, known as the north and south celestial poles. The equator will then be a great circle 90° from each pole. Then as meridians are drawn from pole to pole on the earth, cutting the equator at different points, so imaginary meridians are conceived as

drawn from pole to pole on the celestial sphere. Corresponding to parallels of latitude on the earth we have parallels of declination on the celestial sphere. These are parallel to the equator, and become smaller and smaller as we approach either pole. The correspondence of the terrestrial and celestial circles is this:

To *latitude* on the earth's surface corresponds *declination* in the heavens.

To *longitude* on the earth corresponds *right ascension* in the heavens.

A little study of these facts will show that the zenith of any point on the earth's surface is always in a declination equal to the latitude of the place. For example, for an observer in Philadelphia, in 40° latitude, the parallel of 40° north declination will always pass through his zenith, and a star of that declination will, in the course of its diurnal motion, also pass through his zenith.

So also to an observer on the equator the celestial sphere always spans the visible celestial hemisphere through the east and west points.

In the case of the right ascension, the relation between the terrestrial and celestial spheres is not constant, because of the diurnal motion, which keeps the terrestrial meridians in constant revolution relative to the celestial meridians. Allowing for this motion, however, the system is the same. As we have on the earth's surface a prime meridian passing from pole to pole through the Greenwich Observatory, so in the heavens a prime meridian passes from one celestial pole to the other through the vernal equinox. Then to define the right ascension of any star we imagine a great circle passing from pole to pole through the star, as we imagine one to pass from pole to pole through a city on the earth of which we wish to designate the longitude. The actual angle which this meridian makes with the prime meridian is the right ascension of the star as it is the longitude of the place on the earth's surface.

There is, however, a difference in the unit of angular measurement commonly used for right ascensions in the heavens and longitude on the earth. In astronomical practice, right ascension is very generally expressed by hours, twenty-four of which make a complete circle, corresponding to the apparent revolution of the celestial sphere in twenty-four hours. The

reason of this is that astronomers determine right ascension by the time shown by a clock so regulated as to read 0 hrs., 0 min., 0 sec. when the vernal equinox crosses the meridian. The hour hand of this clock makes a revolution through twenty-four hours during the time that the earth makes one revolution on its axis, and thus returns to 0 hrs., 0 min., 0 sec. when the vernal equinox again crosses the meridian. A clock thus regulated is said to show sidereal time. Then the right ascension of any star is equal to the sidereal time at which it crosses the meridian of any point on the earth's surface. Right ascension thus designated in time may be changed to degrees and minutes by multiplying by 15. Thus, one hour is equal to 15° ; one minute of time is equal to $15'$ of arc, and one second of time to $1''$ of arc.

It may be remarked that in astronomical practice terrestrial longitudes are also expressed in time, the longitude of a place being designated by the number of hours it may be east or west of Greenwich. Thus, Washington is said to be 5h. 8m. 15s. west of Greenwich. This, however, is not important for our present purpose.

The first astronomer who attempted to make a catalogue of all the known stars is supposed to be Hipparchus, who flourished about 150 B.C. There is an unverified tradition to the effect that he undertook this work in consequence of the appearance of a new star in the heavens, and a desire to leave on record, for the use of posterity, such information respecting the heavens in his time that any changes which might take place in them could be detected. This catalogue has not come down to us—at least not in its original form.

Ptolemy, the celebrated author of the 'Almagest,' flourished A.D. 150. His great work contains the earliest catalogue of stars which we have. There seems to be a certain probability that this catalogue either may be that of Hipparchus adopted by Ptolemy unchanged, or may be largely derived from Hipparchus. This, however, is little more than a surmise, due to the fact that Ptolemy does not seem to have been a great observer, but based his theories very largely on the observations of his predecessors. The actual number of stars which it contains is 1,030. The positions of these are given in longitude and latitude, and are also described by their places in the figure of the constellation to which each may belong. Not unfrequently the longitude or latitude is a degree or more in error, showing that the instruments with which the position was determined were of rather rough construction.

So far as the writer is aware, no attempt to make a new catalogue of the stars is found until the tenth century. Then arose the Persian astronomer, Abd-Al-Rahman Al-Sufi, commonly known as Al-Sufi, who was born A.D. 903 and lived until 986. Nothing is known of his life except that he was a man celebrated for his learning, especially in astronomy. His only work on the latter subject which has come down to us is a description of the fixed stars, which was translated from the Arabic by Schjellerup and published in 1874 by the St. Petersburg Academy of Science. This work is based mainly on the catalogue of Ptolemy, all the stars of which he claimed to have carefully examined. But he did not add any new stars to Ptolemy's list, nor, it would seem, did he attempt to redetermine their positions. He simply used the longitudes and latitudes of Ptolemy, the former being increased by $12^{\circ} 42'$ on account of the precession during the interval between his time and that to which Ptolemy's catalogue was reduced. The translator says of his work that it gives a description of the starry heavens at the time of the author and is worthy of the highest confidence. The main body of the work consists of a detailed description of each constellation, mentioning the positions and appearances of the stars which it contains. Here we find the Arabic names of the stars, which were not, however, used as proper names, but seem rather to have been Arabic words representing some real or supposed peculiarity of the separate stars, or arbitrarily applied to them.

Four centuries later arose the celebrated Ulugh Beigh, grandson of Tamerlane, who reigned at Samarcand in the middle of the fifteenth century. Bailey says of him: "Ulugh Beigh was not only a warlike and powerful monarch, but also an eminent promoter of the sciences and of learned men. During his father's lifetime he had attracted to his capital all the most celebrated astronomers from different parts of the world; he erected there an immense college and observatory, in which above a hundred persons were constantly occupied in the pursuits of science, and caused instruments to be constructed of a better form and greater dimensions than any that had hitherto been used for making astronomical observations."

His fate was one which so enlightened a promoter of learning little deserved; he was assassinated by the order of his own son, who desired to succeed him on his throne; and in order to make his position the more secure, also put his only brother to death. A catalogue of the stars bears the name of this monarch; he is supposed to have made many or most of the

observations on which it is founded. Posterity will be likely to suppose that a sovereign used the eyes of others more than his own in making the observations. However this may be, his catalogue seems to have been the first in which the positions of the stars given by Ptolemy were carefully revised. He found that there were twenty-seven of Ptolemy's stars too far south to be visible at Samarcand, and that eight others, although diligently looked after, could not be discovered. It is curious that, like Al-Sufi, he does not seem to have added any new stars to Ptolemy's list.

Next in the order of time comes the work of Bayer, whose method of naming the stars has already been described. The main feature of this work consists of maps of all the constellations. Previous to his time, celestial globes, made especially for the use of the navigator, took the place of maps of the stars. The first edition of this book was published in 1603, and is distinguished by the fact that a list of stars in each constellation is printed on the backs of the maps. Bayer did not confine himself to the northern hemisphere, but extended his list over the whole celestial sphere, from the north to the south pole.

The catalogue of the celebrated Tycho Brahe, prepared toward the end of the sixteenth century, though of great historic value, is of no special interest to the general reader at the present time. A supplement to it, continuing its list of stars to the south pole, was published by Halley, who made the necessary observations during a journey to St. Helena in 1677.

The catalogue of Hevelius, published in 1690, offers no feature of special interest, except the addition of several new constellations which he placed between those already known. Having the aid of the telescope, he was able to include in his catalogue stars which had been invisible to his predecessors.

Modern catalogues of the stars may be divided into two classes: Those which include only stars of a special class, or stars of which the observer sought to determine the position or magnitude with all attainable precision; and catalogues intended to include all the stars in any given region of the heavens, down to some fixed order of magnitude. It may appear remarkable that no attempt of the latter sort was seriously made until more than two centuries after the telescope had been pointed at the heavens by Galileo. A reason for the absence of such an attempt will be seen in the vast number of

stars shown by the telescope, the difficulty of stopping at any given point, and the seeming impossibility of assigning positions to hundreds of thousands of stars. The latter difficulty was overcome by the improved methods of observation devised in modern times.

About the middle of the present century the celebrated Argelander commenced the work of actually cataloguing all the stars of the northern celestial hemisphere to magnitude $9\frac{1}{2}$. This work was termed a *Durchmusterung* of the northern heavens, a term which has been introduced into astronomy generally to designate a catalogue in which all the stars down to a certain magnitude are supposed to be mustered, as if a census of them were taken. The work fills three quarto volumes and contains more than 310,000 stars, of each of which the magnitude and the right ascension and declination are given. This work was extended by Schönfeld, Argelander's assistant and successor, to 22° of south declination.

In the latitudes in which most of the great observatories of the world are situated, that part of the celestial sphere within 40° or 50° of the south pole always remains below the horizon. Around this invisible region a belt of somewhat indefinite breadth, 10° or more, can be only imperfectly observed, owing to the nearness of the stars to the horizon, and the brevity of the period between their rising and setting. Up to the middle of the nineteenth century, the few observatories situated in the southern hemisphere were too ill-endowed to permit of their undertaking a complete census of this invisible region.

The first considerable work emanating from the Cordoba Observatory, under Gould, was a catalogue of all the stars from the south pole to 10° of north declination which could be seen with the naked eye. Another work, which was not issued until after Dr. Gould's death, was devoted to photographs of southern clusters of stars.

The work of the Cordoba Observatory, with which we are more especially concerned in the present connection, consists of a 'Durchmusterung' of the southern heavens, commencing at 22° of south declination, where Schönfeld's work ended, and continued to the south pole. This work is still incomplete, but two volumes have been published by Thome, extending to 41° of south declination. It is expected that the third is approaching completion. This catalogue is, in one point at least, more

complete than that of Argelander and Schönfeld, as it contains all the stars down to the tenth magnitude. The two volumes give the positions and magnitudes of no less than 340,000 stars, and therefore more than the catalogue of Argelander gives for the entire northern hemisphere. If the remaining part of the heavens, from 42° to the south pole, is equally rich, it will contain nearly half a million stars, and the entire work will comprise more than 800,000 stars.

The Royal Observatory of the Cape of Good Hope, under the able and energetic direction of Dr. David Gill, has undertaken a work of the same kind, which is remarkable for being based on photography. The history of this work is of great interest. In 1882 Gill secured the aid of photographers at the Cape of Good Hope to take pictures of the brilliant comet of that year, with a large camera. On developing the pictures the remarkable discovery was made that not only all the stars visible to the naked eye, but telescopic stars down to the ninth or tenth magnitude were also found on the negatives. This remarkable result suggested to Gill that here was a new and simple method of cataloguing the stars. It was only necessary to photograph the heavens and then measure the positions of the stars on the glass negatives, which could be done with much greater ease and certainty than measures could be made on the positions of the actual stars, which were in constant apparent motion.

As soon as the necessary arrangements could be made and the apparatus put into successful operation, Gill proceeded to the work of photographing the entire southern heavens from 18° of south declination to the celestial pole. The results of this work are found in the 'Cape Photographic Durchmusterung,' a work in three quarto volumes, in which the astronomers of all future time will find a permanent record of the southern heavens towards the end of the nineteenth century. The actual work of taking the photographs extended from 1887 to 1891. This, however, was far from being the most difficult part of the enterprise. The most arduous task of measuring the positions of a half-million of stars on the negatives, including the determining of the magnitude of each, was undertaken by Professor J. C. Kapetyn, of the University of Groningen, Holland, and brought to a successful completion in the year 1899.

What the work gives is, in the first place, the magnitude and approximate position of every star photographed. The determining of the

magnitude of a star is an important and delicate question. There is no difficulty in determining, from the diameter of the image of the star as seen in the microscope, what its photographic magnitude was at the time of the exposure, as compared with other stars on the same plate. But can we rely upon similar photographic magnitudes on a plate corresponding to similar brightnesses of the stars? In the opinion of Gill and Kapetyn we cannot. The transparency of the air varies from night to night, and on a very clear night the same star will give a stronger image than it will when the air is thick. Besides, slightly different instruments were used in the course of the work. For these reasons a scale of magnitude was determined on each plate by comparing the photographic intensity of the images of a number of stars with the magnitudes as observed with the eye by various observers. Thus on each plate the magnitude was reduced to a visual scale.

It does not follow from this that the magnitudes are visual, and not photographic. It is still true that a blue star will give a much stronger photographic image than a red star of equal visual brightness. In a general way, it may be said that the catalogue includes all the stars to very nearly the tenth magnitude, and on most of the plates stars of 10.5 were included. In fact, now and then is found a star of the eleventh magnitude.

A feature of the work which adds greatly to its value is a careful and exhaustive comparison of its results with previous catalogues of the stars. When a star is found in any other catalogue the latter is indicated. Most interesting is a complete list of catalogued stars which ought to be on the photographic negatives, but were not found there. Every such case was inexhaustibly investigated. Sometimes the star was variable, sometimes it was so red in color that it failed to impress itself on the plate, sometimes there were errors in the catalogue.

The great enterprise of making a photographic map of the heavens now being carried on as an international enterprise, having its headquarters at Paris, is yet wider in its scope than the works we have just described. One point of difference is that it is intended to include all the stars, however faint, that admit of being photographed with the instruments in use. The latter are constructed on a uniform plan, the aperture of each being 34 centimetres, or 13.4 inches, and the focal length 343 cm. Two sets of plates are taken, one to include all the stars that the instrument will photograph near poles, and the other only to take in those to the eleventh magnitude. Of

the latter it is intended to prepare a catalogue. Some portions of the German and English catalogues have already been published, and their results will be made use of in the course of the present work.

NUMBERING THE STARS.

Closely connected with the work of cataloguing the stars is that of enumerating them. In view of what may possibly be associated with any one star—planets with intellectual beings inhabiting them—the question how many stars there are in the heavens is one of perennial interest. But beyond the general statement we have already made, this question does not admit of even an approximate answer. The question which we should be able to answer is this: How many stars are there of each easily visible magnitude? How many of the first magnitude, of the second, of the third, and so on to the smallest that have been measured? Even in this form we cannot answer the question in a way which is at the same time precise and satisfactory. One magnitude merges into another by insensible gradations, so that no two observers will agree as to where the line should be drawn between them. The difficulty is enhanced by the modern system—very necessary, it is true—of regarding magnitude as a continuously varying quantity and estimating it with all possible precision. In adjusting the new system to the old one, it may be assumed that an average star of any given magnitude on the old system would be designated by the corresponding number on the new system. For example, an average star of the fourth magnitude would be called 4.0; one of the fifth, 5.0, etc. Then the brightest stars, which formerly were called of the fourth magnitude, would now be, if the estimate were carried to hundredths, 3.50, while the faintest would be 4.50. What were formerly called stars of the fifth magnitude would range from 4.50 to 5.50, and so on. But we have meet with a difficulty when we come to the sixth magnitude. On the modern system, magnitude 6.0 represents the faintest star visible to the naked eye; but the stars formerly included in this class would, on the average, be somewhat brighter than this, because none could be catalogued except those so visible.

The most complete enumeration of the lucid stars by magnitudes has been made by Pickering ('Annals of the Harvard Observatory,' Vol. XIV). The stars were classified by half magnitudes, calling

		M.		M.
Mag.	2.0 all from	1.75	to	2.25
	2.5 all from	2.25	to	2.75
	etc.,		etc.	

For the northern stars Pickering used the Harvard Photometry; for the southern, Gould's 'Uranometria Argentina.' A zone from the equator to 30° south declination is common to both; for this zone I use Gould. The number of each class in the entire sky, north and south of the celestial equator, is as follows:

	Northern Hemisphere. Pickering.	Southern Hemisphere. Gould.	Total.
1±	9	14	23
2.0	17	15	32
2.5	17	24	41
3.0	37	41	78
3.5	61	74	135
4.0	114	126	240
4.5	228	234	462
5.0	450	426	876
5.5	787	681	1,468
6.0	789	1,189	1,978
Sum.	2,509	2,824	5,333

It would seem from this that the number of lucid stars in the southern celestial hemisphere is 315 greater than in the northern. But this arises wholly from a seemingly greater number of stars of magnitude 6. In the zone 0° to 30° S., Pickering has 214 stars of this class fewer than Gould. Hence it is not likely that there is any really greater richness of the southern sky.

The total number of lucid stars is thus found to be 5,333. But it is not likely that stars of magnitudes 6.1 and 6.2 should be included in this class, though this is done in the above table. From a careful study and comparison

of the same data from Pickering and Gould, Schiaparelli enumerated the stars to magnitude 6.0. He found:

North pole to 30° S.	3,113 stars.
30° S. to south pole	<u>1,190 stars.</u>
Total lucid stars	4,303

For most purposes a classification by entire magnitudes is more instructive than one by half magnitudes. From the third magnitude downward we may assume that 40 per cent. of the stars of each half magnitude belong to the magnitude next above, and 60 per cent. to that next below. We thus find that of

		Total.
Mag. 0 and 1 there are	21 stars	21
Mag. 2 there are	52 stars	73
Mag. 3 there are	157 stars	230
Mag. 4 there are	506 stars	736
Mag. 5 there are	1,740 stars	2,476
Mag. 6 there are	5,171 stars	7,647

Here it is to be remarked that under magnitude 6 are included many other than the lucid stars, namely, all down to magnitude 6.4. The last column gives the entire number of stars down to each order of magnitude.

It will be remarked that the number of stars of each order is rather more than three times that of the order next higher. How far does this law extend? Argelander's 'Durchmusterung,' which is supposed to include all stars to magnitude 9.5, gives 315,039 stars for the northern hemisphere, from which it would be inferred that the whole sky contains 630,000 stars to the ninth magnitude. Comparing this with the number 7,647 of stars to the magnitude 6.5, we see that it is forty-fold, so that it would require a ratio of about 3.5 from each magnitude to the next lower. But it is now found that Argelander's list contains, in the greater part of the heavens, all the stars to the tenth magnitude.

On the other hand, Thome's Cordoba 'Durchmusterung' gives 340,380 stars between the parallels -22° and -42°. This is 0.14725 of the whole sky, so that, on Thome's scale of magnitude, there are about 2,311,000 stars to

the tenth magnitude in the sky. This is more than three times the Argelander number to the ninth magnitude.

It would, therefore, seem that the ratio of number for each magnitude must exceed 3, even up to the tenth. If a ratio of only 3 extends four steps farther, the whole number of stars in the sky down to magnitude 14.5 inclusive must approach 200 millions. Until the international photographic chart of the sky is subjected to a detailed examination, it is impossible to make an estimation with any approach to certainty.



COLONIES AND THE MOTHER COUNTRY (III).

BY JAMES COLLIER.

THE relations between a great state and its subject peoples will vary according to the status of these, as the relations between father and son differ according as the latter is self-supporting or still under tutelage. Roman provinces under the empire were classed as imperial when they were directly controlled by the Emperor, or senatorial when they were governed by the Senate and possessed a simulacrum of self-government. The dual status in this mother country of nearly the whole world foreshadows all subsequent relationships between a mother country and its dependencies. Spain and Portugal governed their colonies imperially, appointing all officers, immediately or through their representative mediatingly enacting all laws, and leaving almost as little freedom to their own countrymen as to the down-trodden indigenes. More humanely, indeed, but in spite of conceded French citizenship and theoretical equality, the French have ruled their scattered dependencies with as little of the reality of public life. The Dutch colonies are similarly controlled. The British Empire presents a variegated picture where every color is blended and every form of policy known among men is displayed. From it alone an Aristotle might delineate the metaphysics of government or a Spencer construct its physics. In Egypt and Crete, with practical possession, imperial England is vassal to the Sultan, and she now holds the conquered Soudan jointly with Egypt, but acknowledges no suzerainty. She is herself suzerain of the two South African Boer republics and regent of Zanzibar. In her magnificent dependency of India, 692 sovereignties and chiefships form a 'protected' girdle around her own possessions, or interlace or approach them. Between these beneficent despotisms and the free states of Australia, South Africa or North America there seems to be every possible variety of mingled

absolutism and self-government. Certain territories are governed by chartered companies; one (Rhodesia) by a chartered company under the control of the Crown. Three native territories are governed by officers under the High Commissioner of South Africa; four others by the officers of Cape Colony. The status of Crown colonies administered more or less directly by the Imperial Government is almost as various. One colony may be dependent on another, as Natal was for years on Cape Colony. Others exhibit in an ascending scale the acquisition of the attributes of self-government. The governor rules at first alone despotically, then with an executive council, next with a nominated legislative council, further with the latter partly elected, and finally with it wholly elective. At these successive stages the colony is in a decreasing degree under the control of the Imperial Government, and a scale might be drawn showing groups of colonies indefinitely arrested at one or another of them. Only colonies destined for complete freedom victoriously pass through them all and emerge into full political manhood.

The duration of their infancy and youth is determined by internal and external circumstances: (1) When a colony is systematically founded and quickly peopled it may rapidly traverse the period of dependence, and (like New Zealand or South Australia) be granted responsible government in about fifteen years. (2) Convict colonies, like Tasmania and New South Wales, may have fifty or sixty years of pupillage. (3) A colony of retarded growth, like West Australia, may be nearly as long a minor. (4) Colonies that have long to struggle with an overwhelming mass of indigenes, like Cape Colony, may take half a century to ripen, and even then, like Natal, may retain traces of the earlier state. (5) When the mother country is herself despotically governed, as England was under the Stuarts, the Commonwealth and the early Hanoverians, colonies that possess every attribute qualifying them for freedom, like many of the North American colonies, may be forcibly retained in partial dependence. (6) The New England colonies, free from the start, were connected with Britain by a shadowy tie of nominal allegiance, tightened at times into real subjection. Lastly, a colony may revert, like Jamaica, after years of Parliamentary institutions, to the dependent position of a Crown colony.

So various and so intricate, so weak here, so strong there, and withal so marvelously compacted, is the network of relations forming the anatomy of

the wonderful new type of social organism constituted by a mother country, its free and its subject colonies, its protected states and its dependencies.

The brain sometimes inhibits natural movements and enforces injurious actions, as a morbid conscience often prescribes irksome duties and forbids innocent pleasures. Fathers have misdirected the career of their sons, and the unwisdom of mothers (Lady Ashton, in 'The Bride of Lammermoor,' is a tragic, but far from a rare example) has destroyed the happiness of their daughters. So governments inevitably hinder and blunder, worry colonies by vexatious interferences or goad them into insurrection. For more than thirty years Bishop Fonseca, the president of the Council of the Indies, lay like an incubus on the Spanish colonies in South America. His main object seemed to be to throw impediments in the way of the great discoverers and rulers—Columbus and Cortez. When Cortez planned the conquest of Mexico he experienced protracted opposition from Fonseca, who "discouraged recruits, stopped supplies and sequestered the property" Cortez sent to Spain. The conqueror bitterly complained that he "had found it harder to contend against his own countrymen than against the Aztecs." The story of Spain's South American colonies is one of injustice, oppression and downright robbery. The natives naturally suffered most. They were condemned to forced labor in the mines under circumstances of extreme barbarity, in order that large sums of money might be sent annually to Spain. This insatiable demand neutralized all the efforts of the best-intentioned viceroys and rendered all attempts at good government nugatory. The Indians had further to submit to grinding oppression by the local officials and to the exactions and tyranny of the priests. The Spanish colonists had their own grievances. Articles of commerce were excluded, or had their prices heightened by the monopoly of the Cadiz merchants. They were oppressed by the military despotism of the government. The political development of the colonies was made impossible by the continued use of them for the purposes of the mother country. What Spain was for three centuries, that was she till the other day in her few remaining colonies. Lord Brassey writes of Cuba: "The casual visitor can not fail to be impressed with the evidences of inefficient administration. The fiscal policy is intensely exclusive. The taxation is heavy, and the government absolutely despotic. The police maintain a system of intolerable espionage. Every salaried servant of the local government is a Spaniard, who regards Cuba as a vassal state, over which Spain has unlimited rights, without reciprocal

duties or obligations. The system has already severed all her noble settlements in South America from the mother country. In time it must involve the loss of Cuba.”

If it were the case that the genesis and growth of the myriad buds formed round a prolific hydroid were accelerated by magnetic shoots (so to speak) from the parent zoöphyte, and ‘persons’ were thus differentiated, we should have a true analogue to a kind of action exercised by the mother country on its colonies. For it long supplies them with the greater part of their brain power, governing force, culture, science and experience of all sorts, and when these have done their work a new political, intellectual and moral center is created, which is henceforth self-subsistent; the colony has received a soul, a mind, a heart. First, the governor is usually sent out by the metropolis. Of six hundred and seventy-two rulers of South America, from its conquest to its independence, only eighteen were Americans. In French and Dutch colonies there are possibly no exceptions. Many of the charter and proprietary colonies of North America elected their own governors, and the insurrectionary governor of a Crown colony, New York, was popularly elected. The lieutenant-governors of the provinces of the Canadian Dominion are locally appointed. With these and one or two other exceptions, the governor may be considered as symbolizing (in so far as he has the capacity) the entire civilization of the mother country. He brings much or little to the colony he comes to govern. Sometimes, as in the case of Sir George Gray, he brings intellectual superiority, and he may thus stimulate its literary development, but that is rare. He oftener imparts an aroma of gentility that is much appreciated by a certain class. He may be of practical utility by applying the experience of a military engineer, as did Sir William Jervois. He may have had large colonial experience, like Sir Hercules Robinson, and use that to solve the intricate political problems of his colony. If he is a collector, like Sir George Gray, he may enrich it by bequests of libraries and museums. If he possesses literary gifts and has passed through an eventful time, he may enrich colonial history by dictating his biography, like one colonial governor, or writing his reminiscences, like so many. And lastly, after returning to the mother-land, he may continue to watch over the interests of the colony or colonies he ruled; he may become president or member of the Council of the Indies, like three viceroys of Peru, or Parliamentary under-secretary for the colonies, like Sir James Fergusson, or even his former colony’s agent-general, like Sir W. Robinson.

In Crown colonies the chief legal and administrative officials are imperial appointees, and are only superseded by local ministers when the colony is granted responsible government. In a unique case, that of Queensland, after a constitution had been conceded, the first governor took out with him the first premier; and he too was afterwards able to safeguard its interests as permanent under-secretary in London.

The Greek metropolis sometimes sent priests to its colonies, and bishops are long appointed by the mother church. During the three centuries of Peruvian dependence fully one in seven bishops—one hundred and five against seven hundred and six—were native Americans. Canada seems to have at length arrived at complete independence and appoints Canadians. In Australasia and South Africa the metropolitans and most of the suffragans are still nominated in England; a dean may be transferred from one colony to another as a bishop; or a small and poor diocese may elect one of its incumbents. Local jealousies and possibly the absence of a commanding spirit combine with the desire to have the best the home church can afford to give or the colonial church procure to dictate the extraneous selection. The stream of ecclesiastical culture flows likewise through the immigration or importation of ministers of all denominations. It means, among Catholics as among Protestants, the periodical addition to the spiritual wealth of the colonies of an amount of talent and high character which they would have been slow to acquire by natural growth. University or collegiate professors are for quite as long appointed by a committee of selection in the mother country. Such men—some of them brilliant, laborious, enthusiastic—are a real acquisition to communities immersed in material pursuits and cut off from the movement of science in Europe, and their position is deservedly high and well remunerated. Doctors, lawyers, artists, teachers, experts in many departments, place the colonies in the same position relatively to less-favored communities, as the sons of a squire relatively to the sons of an artisan. In this respect, as in most others, a colony follows the example of the mother country. The introduction of literature, the sciences and the arts into the mother-land was to a large extent at all stages in its history the work of aliens. It is so still; the names of Bunsen, Rosen, Max Müller, Goldstücker, Aufrecht, and a score of others, are proofs that men as well as things that are ‘made in Germany’ are still imported into England. To descend to the mechanical arts, “the ranks of skilled workmen in America were and are renewed from the more fertile soil of Europe”; even the

workmen in the Portland stone-quarries are imported from England. The second mode in which foreign culture was introduced into the mother-land in common with all others—visits made abroad for discipleship or instruction—has all along been, and is now increasingly, maintained. Colonial students go to Europe to be trained in medicine and law. Experts go to become acquainted with advances in science and medicine, or with recent improvements in mechanical processes. The wealthier colonists who spend occasional seasons in Europe bring back new (or antiquated) social or political notions, and Americans who thus try to import into the United States an aristocratic style of living have to be ridiculed out of it. The third method by which an infusion of foreign civilization may pass into another community is by books, works of plastic art, music, tools, implements and instruments, and into this vast inheritance of the mother country the daughter colonies have entered. They participate in the advances made by other countries as well. The Canadian colonies owe only less to the United States than to England, and American railway cars, agricultural implements and household utensils are in use in Australasia. In New Zealand a French Masonic lodge has struck root.

The new colonial centers thus formed react on the father-land, as we may conceive the daughter buds to react on the parent hydroid. The discovery of the New World and the successive entrance of the five great maritime powers upon a long and fierce rivalry for its possession transformed the politics of Europe. Great wars were undertaken solely with this object. The political center of gravity was shifted from the Mediterranean to the Atlantic. New industrial interests were created. Insular and stagnant powers, isolated Continental powers, received a fresh lease of life, and, along with warlike Continental powers, were expanded to the measure of the globe. New sympathies were generated. Wider horizons were opened out. The heart and brain of all were in a manner enlarged. The policy of the mother country is even now being modified by its colonies. “The paramount object in legislating for colonies should be the welfare of the parent state,” frankly avows the law officer of the Dutch East India Company at the Cape of Good Hope, in 1779. The Ashburton Treaty and the Oregon Agreement were entered into as if England and the United States were alone interested in their provisions. Treaties are now concluded in the interests of the colonies. Treaties are ‘denounced’ in order to allow them freedom to tax foreign commodities. They are represented by

commissioners, on an equal footing with those of Britain, at conferences preparatory to the conclusion of treaties, and colonial conferences are summoned in order that the general views of the colonies may be ascertained.

There is a more direct reaction, resembling the adoption by an admiring father of the sentiments and opinions of a son who is rising in the world. The Greek cities that had planted colonies imitated the republican institutions of these and deposed their kings. “The American colonists,” says Bancroft, “founded their institutions on popular freedom and ‘set an example to the nations.’ Already the ... Anglo-Saxon emigrants were the hope of the world.” The filial free colonies of Britain are exerting an influence on the domestic policy of the father-land. An aged colonial ruler used to console himself for exclusion from the English Parliament by cherishing the belief that ideas and measures of his had passed into the public life of England. Much of this is mere hallucination; some of it is reality. The testimony of a sagacious and experienced statesman on this subject is decisive:

“To the influence of the American Union must be added that of the British colonies. The success of popular self-government in these thriving communities is reacting on political opinion at home with a force that no statesman neglects, and that is every day increasing. There is even a danger that the influence may go too far. They are solving some of our problems, but not under our conditions, and not in presence of the same difficulties. Still, the effect of colonial prosperity—a prosperity alike of admirable achievement and boundless promise—is irresistible. It imparts a freedom, an elasticity, an expansiveness, to English political notions, and gives our people a confidence in free institutions and popular government, which they would never have drawn from the most eloquent assumptions of speculative system-mongers, nor from any other source whatever, save practical experience carefully observed and rationally interpreted.”^I

^I Morley, ‘Studies in Literature,’ pp. 126–7.

The New Zealand system of local government is a model which Great Britain, at one time famous in that line, has not been ashamed to imitate; the English county councils have been molded on those of her colony. From the same colony the mother country borrowed her First Offenders' act. The restriction of electors to the exercise of a single vote—unimportant excepting in principle in populous England, but important in young countries where property is widely held—was perseveringly proposed, and at length carried, by the aristocratic leader of the democratic party in New Zealand, whence it is spreading to the adjacent colonies; it has been for some years adopted by the British Liberals as an article in their programme, and it is also a plank in the European socialist platform. The general adhesion to an eight hours' day in the Australasian colonies is having an effect in England and is probably the measure to which Mr. Morley refers as likely to be dangerous; his opposition to it cost him his seat at Newcastle. The adoption of female suffrage in two of these colonies and the certainty of its adoption in others are habitually cited by the advocates of the cause in England as an argument for its adoption in England. The nationalization of the land has been a popular notion in these same colonies ever since Henry George's famous book was published, and the large extent of private lands bought back by the governments of New Zealand and Queensland has strengthened the hands of the land-nationalizers in Europe. The advanced government socialism of most of these colonies, made inevitable by the lack of private capital, and its apparent success, furnish socialists of the German type with weapons and encourage them to prophesy 'the dawn of a revolutionary epoch.'

The spiritual reaction of the colonies on the mother-land is much less considerable, yet is not nil. One or two instances stand out prominently. Jonathan Edwards is one of the giants of British as well as of American theology, and his treatise on the freedom of the will has counted for as much as Butler's Analogy in the development of English theological thought. Sam Slick has been the father or foster-father of the portentous overgrowth of humor by which the United States balances the devouring activity of its public and the overstrain of its private life, but he has been practically inoperative on the very different quality of English humor. From South Africa have come influences of a sterner sort. "Who could have foreseen," asks Mr. Stead, "that the new, and in many respects the most distinctive, note of the literature of the last decade of the nineteenth century

would be sounded by a little chit of a girl reared in the solemn stillness of the Karoo, in the solitude of the African bush? The Cape has indeed done yeoman's service to the English-speaking world. To that pivot of the empire we owe our most pronounced types of the imperial man and the emancipated woman"—Cecil Rhodes and Olive Schreiner.

CAUSES OF DEGENERATION IN BLIND FISHES.

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IT may now be profitable to take up the causes leading to the small degree of degeneration found in *Chologaster*, the degenerations of the eye in *Amblyopsis*, *Typhlichthys* and *Troglichthys* to a mere vestige, together with the total disappearance of some of the accessory structures of the eye, as the muscles.

In the outset of this consideration we must guard against the almost universal supposition that animals depending on their eyes for food are or have been colonizing caves, or that the blind forms are the results of catastrophes that have happened to eyed forms depending on their eyesight for their existence. This idea, so prevalent, vitiates nearly everything that has been written on the degeneration of the eyes of cave animals.

Another word of warning ought perhaps to be added. The process of degeneration found in the *Amblyopsidæ* need not necessarily be expected to be identical with the degeneration of the same organs in another group of animals, and, however much the conditions in one group may illuminate the conditions in another, cross-country conclusions must be guarded against.

The degeneration of organs ontogenetically and phylogenetically has received a variety of explanations:

1. The organ diminishes with disuse (ontogenetic degeneration—Lamarck, Roux, Packard), and the effect of this disuse appears to some extent in the next generation (phylogenetic degeneration—Lamarck, Roux, Packard, Kohl).

2. Through a condition of panmixia the general average maintained by selection is reduced to the birth mean in one generation (ontogenetic—Romanes, Lankester, Lloyd Morgan, Weismann) to the greatest possible degeneration in succeeding generations (phylogenetic—Weismann), or but little below the birth average of the first generation (Weismann's later view, Romanes, Morgan, Lankester).

3. Through natural selection (reversed), the struggle of persons, the organ may be caused to degenerate either (A) by the migration of persons with highly developed eyes from the colony living in the dark (Lankester), or (B) through economy of weight and nutriment or liability to injury (phylogenetic purely—Darwin, Romanes).

4. Through the struggle of parts for room or for food an unused organ in the individual may be crowded (ontogenetic—Roux). This may lead to the development of the used organ as against the disused through a compensation of growth (Goethe, Saint-Hilaire, Roux); this ontogenetic result becomes phylogenetic through transmission of the acquired character (Roux), or is in its very nature phyloblastic (Kohl).

5. Through the struggle between soma and germ to produce the maximum of efficiency of the former with the minimum expenditure to the latter (ontogenetic and phylogenetic—Lendenfeld).

6. Through germinal selection, the struggle of the representatives of organs in the germ (ontogenetic and phylogenetic—Weismann).

The idea of ontogenetic degeneration is intimately bound up with the idea of phylogenetic degeneration. Logically we ought to consider first the causes of individual degeneration, and then the processes or causes that led to the transmission of this. Practically it is impossible to do so, because many of the explanations are general. Only No. 4 of the above may be taken in the ontogenetic sense purely, though it was certainly also meant to explain phylogenetic degeneration. In many of the explanations of particular cases of degeneration more than one of the above principles are invoked, though only one was meant to be used. In most cases, however, the discussions of degeneration have been in general terms, without direct bearing on any specific instance of degeneration in all its details. It must be evident that such discussions can only by accident lead to right results.

By the Lamarckian ontogenetic degeneration is considered the result of lack of use and consequent diminished blood supply. The results of the diminution caused by the lack of use during one generation are transmitted in some degree to the next generation, which thus starts at a lower level. A continuation of the same conditions leads finally to the great reduction and ultimate disappearance of an organ.

No one, so far as I am aware, has succeeded in accounting for the degeneration of the eye by means of this view. Packard's^J explanations are evidently a mixture of Lamarckism and Darwinism.

^J *American Naturalist*, September, 1894, vol. xxviii, p. 727.

Packard says: "When a number, few or many, of normal-seeing animals enter a totally dark cave or stream, some may become blind sooner than others," some having the eye slightly modified by disuse, while others may have in addition physical or functional defects, especially in the optic nerves and ganglia. "The result of the union of such individuals and adaptation to their Stygian life would be broods of young, some with vision unimpaired, others with a tendency to blindness, while in others there would be noticed the first steps in degeneration of nervous power and nervous tissue." Packard evidently had invertebrates in mind. He clearly admits the cessation of selection or panmixia in that those born with defects may breed with the others. He supposes that the blind fauna may have arisen in but few or several generations, a supposition that may be applicable to invertebrates, but certainly is not to vertebrates. At first those becoming so modified that they can do without the use of their eyes would greatly preponderate over those 'congenitally blind.' "So all the while the process of adaptation was going on, the antennæ and other tactile organs increasing in length and in the delicacy of structures, while the eyes were meanwhile diminishing in strength of vision and their nervous force giving out, after a few generations—perhaps only two or three—the number of congenitally blind would increase, and eventually they would, in their turn, preponderate in numbers." Packard seems here to admit the principle of

degeneration as the result of compensation of growth, the nervous force of the eye giving out with the increase of the tactile and olfactory organs. It is somewhat doubtful in what sense the term 'congenitally blind' is used, but it probably means born blind as the result of transmitted disuse, rather than blind as the result of fortuitous variation. The effects of disuse are thus supposed, through their transmission, to have given rise to generations of blind animals. The continued degeneration is not discussed.

Romanes maintained that the beginning of degeneration was due to cessation of selection, and continued degeneration to the reversal of selection and final failing of the power of heredity. Selection he supposed to be reversed because the organ no longer of use "is absorbing nutriment, causing weight, occupying space and so on, uselessly. Hence, even if it be not also a source of actual danger, economy of growth will determine a reversal of selection against an organ which is now not only useless, but deleterious." This process will continue until the organ becomes rudimentary and finally disappears.

Roux^K attempted chiefly to explain degeneration in the individual. Degeneration is looked upon as the result of a struggle among the parts for (a) room and (b) food. Without doubting that both these principles are active agents in degeneration, it may be seriously doubted whether they are effective in the degeneration of the eyes in question. Certainly there can be no question of a struggle for room, for the position and room formerly occupied by the eye is now filled with fat, which can not have been operative against the eye. The presence of this large fat mass in the former location of the eye, the large reserve fat mass in the body, the uniformly good condition of the fish and the low vitality, which enables them to live for months without visible food, all argue against the possibility that the struggle for food between parts was an active agent in the degeneration of the eyes.

^K Gesammelte Abhandlungen, 1895.

Kohl^L considers that “*Der Grund und direkter oder indirekter Anlass zum Eintreten der Entwicklungshemmung ist Lichtmangel.*” The method of operation of the lack of light he conceived to be as follows:

^L Rudimentäre Wirbelthieraugen, 1893.

Other organs were developed to compensate for the disuse of the eye; and as the developmental force was used in the formation of these organs, each succeeding generation developed its eye less. The degeneration is thus explained as the result of a struggle of parts, although this term is nowhere used, acting through the principle of compensation. The same objections may be offered to this explanation of Kohl as to all his theoretical discussions—they are based on the assumption of conditions and processes that have no existence. The high development of ‘compensating’ organs is not primarily the result of the loss of the eye, but the high development of the former organs permitted the disuse and later degeneration of the latter. His whole process is a phylogenetic one, without a preceding ontogenetic one, though on this point he does not seem to be very clear himself, for on one page we are told that degeneration leads to retardation, and on another that degeneration is a consequence of retardation.

Ledenfeld^M endeavors to apply Roux’s *Kampf der Theile*, with reversed selection, to explain the conclusions reached by Kohl on the processes and causes of degeneration. The struggle is represented as taking place between the germ and soma, the former endeavoring to keep the latter at the lowest efficient point as weapon for the germ. If a series of individuals get into the dark the organs of vision are of no advantage and reversed selection will bring about their degeneration. The saving in ontogeny appears first as a retardation and then as a cessation of development.

^M Zoölogischer Centralblatt, 1896.

Weismann^N more recently accepts the view of Romanes, Morgan and Lankester on the inadequacy of panmixia to explain the whole phenomena of degeneration, and in his ‘Germinal Selection’ rejects the idea of reversed selection, and suggests a new explanation for what Romanes attributed to the failure of heredity and the Lamarckians to transmission of the effects of disuse. The struggle of the parts of Roux has been crowded by him back to the representatives of these parts in the germ.

^N The Monist, 1896, pp. 250–274.

“The phenomena observed in the stunting, or degeneration, of parts rendered useless ... show distinctly that ordinary selection, which operates by the removal of entire persons—personal selection, as I prefer to call it—can not be the only cause of degeneration, for in most cases of degeneration it can not be assumed that slight individual vacillations in the size of the organ in question have possessed selective value. On the contrary, we see such retrogressions effected apparently in the shape of a continuous evolutionary process determined by internal causes, in the case of which there can be no question whatever of selection of persons or of a survival of the fittest—that is, of individuals with the smallest rudiments. The gradual diminution continuing for thousands and thousands of years and culminating in its final and absolute effacement” can only be accomplished by germinal selection. Germinal selection as applied to degeneration is the formal explanation of Romanes’ failure of heredity through the struggle of parts for food. “Powerful determinants will absorb nutriment more rapidly than weaker determinants. The latter, accordingly, will grow more slowly and will produce weaker determinants than the former.” If an organ is rendered useless, the size of this organ is no longer an element in personal selection. This alone would result in a slight degeneration. Minus variations are, however, supposed to rest “on the weaker determinants of the germ, such as absorb nutriment less powerfully than the rest. This will enable the stronger determinants to deprive them even of the full quantum of food corresponding to their weakened capacity of assimilation, and their descendants will be weakened still more. Inasmuch, now, as no weeding out

of the weaker determinants of the hind leg [or eye] by personal selection takes place on our hypothesis, inevitably the average strength of this determinant must slowly but constantly diminish—that is, the hind leg [or eye] must grow smaller and smaller until it finally disappears altogether.... Panmixia is the indispensable precondition of the whole process; for, owing to the fact that persons with weak determinants are just as capable of life as those with strong, ... solely by this means is a further weakening effected in the following generations.”

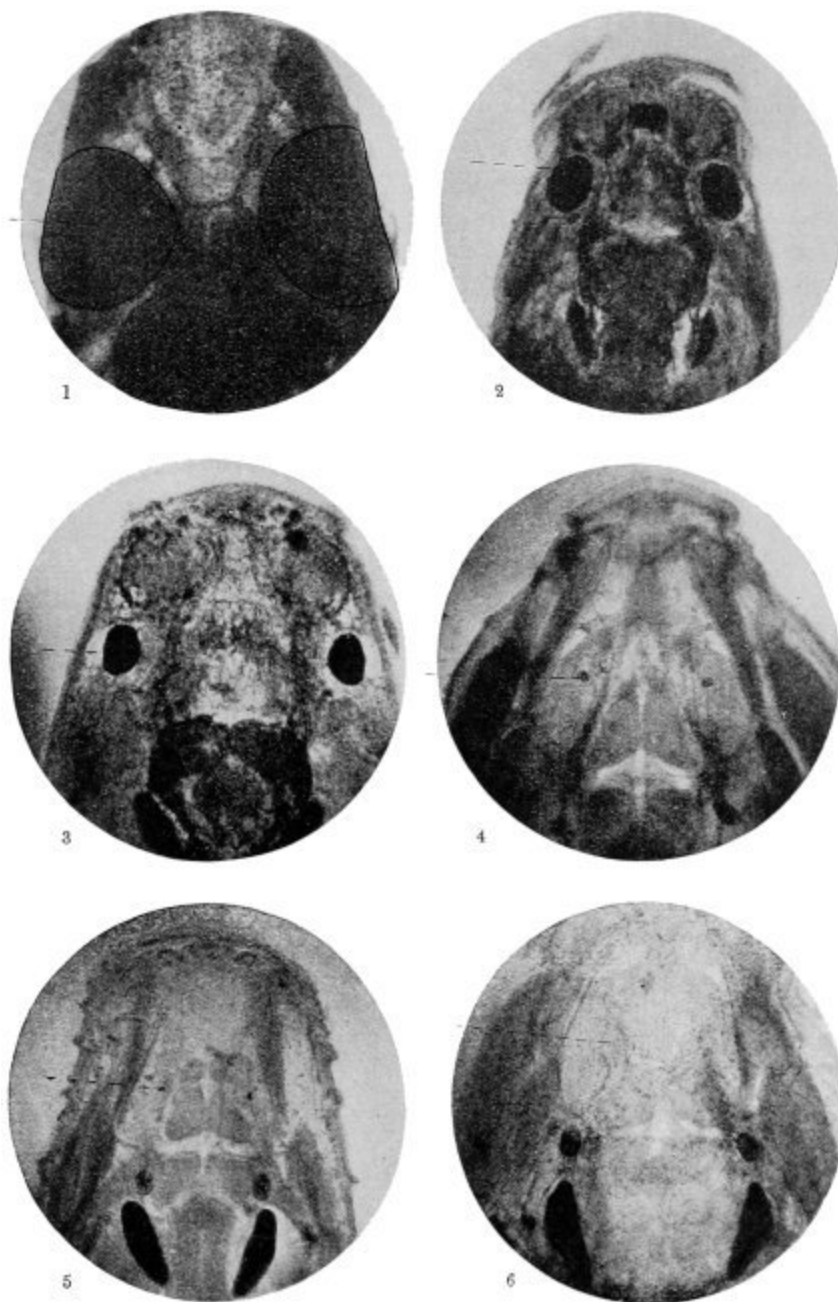
This theory presupposes the complex structure of the germ plasm formulated by Weismann and rejected by various persons for various reasons. But granting Weismann the necessary structure of the germ plasm, can germinal selection accomplish what is claimed for it? I think not. Granting that variations occur about a mean, would not all the effects claimed for minus variations be counteracted by positive variations? Eye determinants, which, on account of their strength, secure more than their fair share of food, and thereby produce eyes that are as far above the mean as the others are below, and leave descendent determinants that are still stronger than their ancestry would balance the effect produced by weak-eye determinants. It is evident that a large, really extravagant development of the eye in such a fish as *Chologaster* would not effect the removal of the individual by personal selection; still less so in *Amblyopsis*, which not only lives in comparative abundance, but has lived for twenty months in confinement without visible food, and in which the eye is minute. It seems that all the admitted objections to degeneration by panmixia apply with equal force to germinal selection. This, however, would be changed were the effect of disuse admitted to affect the determinants, and this it seems Weismann has unconsciously admitted. So far we have considered germinal selection in the abstract only. All its suppositions are found to be but a house of cards when the actual conditions of degeneration are considered. We find that degeneration is not a horizontal process affecting all the parts of an organ alike, as Weismann presupposes, not even a process in the reverse order of phyletic development, but the more vital, most worked parts degenerate first with disuse and panmixia; the passive structures remain longest. The rate of degeneration is proportional to the past activity of the parts, and the statement that “passively functioning parts—that is, parts which are not alterable during the individual life by function—by the same laws also degenerate when they become useless” finds no basis in

fact, and is an example of the inexact utterances abundant in the discussion of degeneration on which it is entirely unsafe to build lofty theoretical structures. As one example of the unequal degeneration we need only call attention to the scleral cartilages and the rest of the eye of *Troglichthys rosæ*.

All are agreed that natural selection alone is insufficient to explain all, if any, of the processes of degeneration. All either consciously or not admit the principle of panmixia, and all are now agreed that this process alone can not produce extensive degeneration. All are agreed that the important point is degeneration beyond the point reached by panmixia, the establishment of the degenerating process, whatever it may be, in the germ, or, in other words, the breaking of the power of heredity. It is in the explanation of the latter that important differences of opinion exist.

Weismann attempts to explain the degeneration beyond the point which panmixia can reach by a process which not only is insufficient, even if all his premises are granted, to produce the desired result without the help of use transmission, but has as its result a horizontal degeneration which has no existence in fact.

Romanes supposed degeneration, beyond the point which may be reached by panmixia, to be the result of personal selection and the failure of the hereditary force. The former is not applicable to the species in question, and is denied by such an ardent Darwinist as Weismann to be applicable at all in accounting for degeneration. Moreover, the process as explained by Romanes would result in a horizontal degeneration which has no existence in fact. The second assumption, the failure of hereditary force, is not distinguishable, as Morgan has pointed out, from the effect of use transmission.



FIGS. 1-6.—Photographs of the upper halves of the heads of specimens of nearly the same length under the same magnification, to show the gradual decrease of the eye. The dotted line leads to the eye in all cases.

FIG. 1.—*Zygonectes notatus*.

FIG. 2.—*Chologaster papilliferus*.

FIG. 3.—*Chologaster Agassizii*.

FIG. 4.—*Amblyopsis spelæus*.

FIG. 5.—*Troglichthys rosæ*.

FIG. 6.—*Typhlichthys subterraneus*.

The struggle of parts in the organism has not affected the eye through the lack of room, since the space formerly occupied by the eye is now filled by fat and not by an actively functioning organ. It is not affected by the struggle for food, for stored food occupies the former eye space. It could only be affected by the more active selection of specific parts of food by some actively functioning organ. It is possible that this has in fact affected the degeneration of the eye. The theory explains degeneration in the individual, and implies that the effect in the individual should be transmitted to the next generation. This second part seems but the explanation of the workings of the Lamarckian factor.

The Lamarckian view—that through disuse the organ is diminished during the life of the individual, in part, at least, on account of the diminution of the amount of blood going to a resting organ, and that this effect is transmitted to succeeding generations—not only would theoretically account for unlimited progressive degeneration, but is the only view so far examined that does not on the face of it present serious objections. Is this theory applicable in detail to the conditions found in the Amblyopsidæ? Before going further, objections may again be raised against the universal assumption that the cessation of use and the consequent panmixia was a sudden process. This assumes that the caves were peopled by a catastrophe. But it is absolutely certain that the caves were not so peopled, that the cessation of use was gradual, and the cessation of selection must also have been a gradual process. There must have been ever-widening bounds within which the variation of the eye would not subject the possessor to elimination.

Chologaster is in a stage of panmixia as far as the eye is concerned. It is true the eye is still functional, but that the fish can do without its use is evident by its general habit and by the fact that it sometimes lives in caves. The present conditions have apparently existed for countless generations—as long as the present habits have existed—and yet the eye still maintains a

higher degree of structure than reverse selection, if operative, would lead us to expect, and a lower than the birth mean of fishes depending on their eyes, the condition that the state of panmixia alone would lead us to expect. There is a staying quality about the eye with the degeneration, and this can only be explained by the degree of use to which the eye is subjected.

The results in *Chologaster* are due to panmixia and the limited degree of use to which the eye is put. *Chologaster Agassizii* shows the rapid diminution with total disuse.

The difference in the conditions between *Chologaster* and *Amblyopsis*, *Typhlichthys* and *Troglichthys*, is that in the former the eyes are still in use, except when living in caves; in the latter they have not been in a position to be used for hundreds of generations. The transition between conditions of possible use and absolute disuse may have been rapid with each individual after permanently entering a cave. Panmixia, as regards the minute eye, continued. Reversed selection, for economy, can not have affected the eye for reasons already stated. The mere loss of the force of heredity, unless this was caused by disuse, or the process of germinal selection, can not have brought about the conditions, because some parts have been affected more than others.

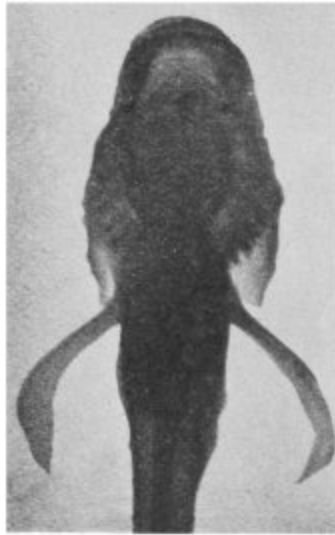


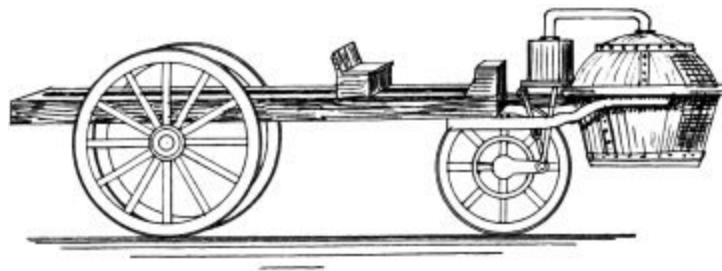
FIG. 7.—Head of a very young *Amblyopsis* before all the yolk is absorbed.

Considering the parts most affected and the parts least affected, the degree of use is the only cause capable of explaining the conditions. Those parts most active during use are the ones reduced most—viz., the muscles, the retina, optic nerve and dioptric appliances, the lens and vitreous parts. Those organs occupying a more passive position, e. g., the scleral cartilages, have been much less affected. The lens is one of the latest organs affected, not at all during use, possibly because during use it would continuously be in use. It disappears most rapidly after the beginning of absolute disuse both ontogenetically and phylogenetically. All indications point to use and disuse as the effective agents in molding the eye. The process, however, does not give results with mathematical precision. In *Typhlichthys subterraneus* the pigmented layer is affected differently from that of *Amblyopsis*. The variable development of the eye muscles in different species would offer another objection if we did not know of the variable condition of these structures in different individuals. Chilton has objected to the application of the Lamarckian factor to explain degeneration, on account of the variable effects of degeneration in various invertebrates. But such differences in the reaction are still less explainable by any of the other theories.

THE EVOLUTION AND PRESENT STATUS OF THE AUTOMOBILE.

BY WILLIAM BAXTER, JR.

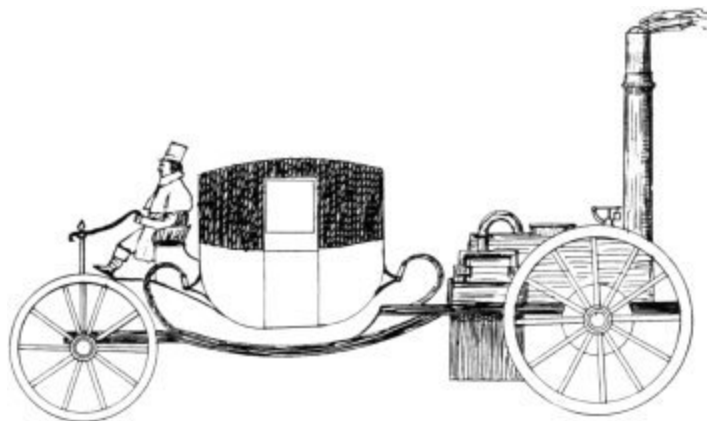
IN this closing year of a century which is marked by unparalleled advances in science and its applications to the industrial arts, we are very much inclined to take it for granted that none of the inventions that are regarded by us as indicative of the highest order of progressive tendency, could by any possibility have been thought of by our forefathers; and as the automobile is looked upon as an ultra-progressive idea, no one who has not investigated the subject would believe for a moment that its conception could antedate the present generation, much less the present century. The records, however, show that the subject engrossed the attention of inventive minds many hundreds of years ago. In fact, as far back as the beginning of the thirteenth century a Franciscan monk named Roger Bacon prophesied that, the day would come when boats and carriages would be propelled by machinery.



**FIG. 1. CUGNOT'S STEAM GUN CARRIAGE,
MADE IN 1763.**

The first authentic record of a self-propelled carriage dates back to the middle of the sixteenth century. The inventor was Johann Haustach, of Nuremburg. The device is described as a chariot propelled by the force of springs, and it is said that it attained a speed of two thousand paces per hour, about one mile and a quarter. Springs have been tried by many inventors since that time, but always without success from the simple fact that the amount of energy that can be stored in a spring is practically insignificant.

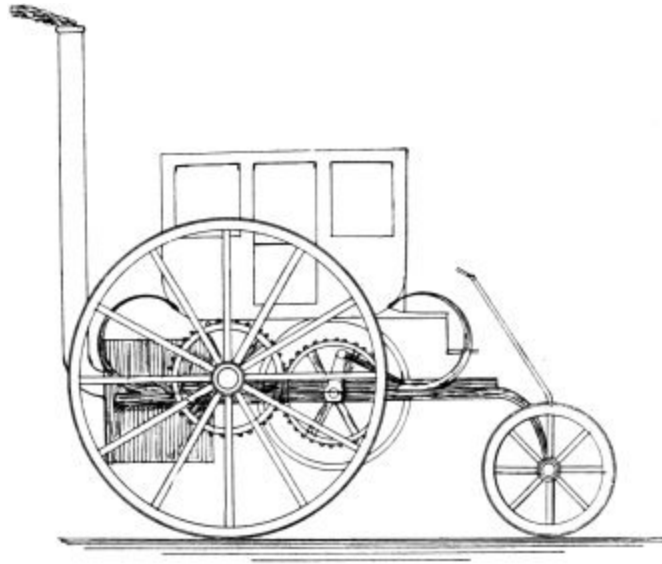
In 1763 a Frenchman by the name of Cugnot devised a vehicle that was propelled by steam, and a few years after the date of his first experiment, constructed for the French Government a gun carriage which is shown in [Fig. 1](#). As will be seen, the design was of the tricycle type, and it was intended to mount the gun between the rear wheels. The boiler, which resembles a huge kettle, hung over the front end and was apparently devoid of a smoke stack. Motion was imparted to the front wheel by means of a ratchet. Although this invention is very crude, it must be regarded as meritorious if we consider that it was made before the steam engine had been developed in a successful form for stationary purposes.



**FIG. 2. SYMINGTON'S STEAM COACH, MADE
IN 1784.**

The next effort to solve the problem was made by W. Symington in the year 1784, the carriage devised by him being illustrated in [Fig. 2](#). This

coach, although pretentious in appearance, was crude mechanically, but it actually ran. The service, however, was not what could be called satisfactory.



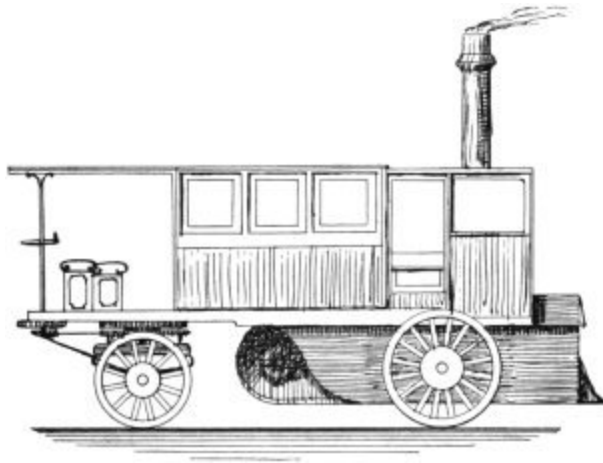
**FIG. 3. TREVITHICK'S STEAM CARRIAGE,
MADE IN 1803.**

In 1803, Richard Trevithick brought out the carriage shown in [Fig. 3](#), which could run, but was artistically a failure. Moreover, the machinery was such as would soon give out, even if well designed, on account of its exposed position.



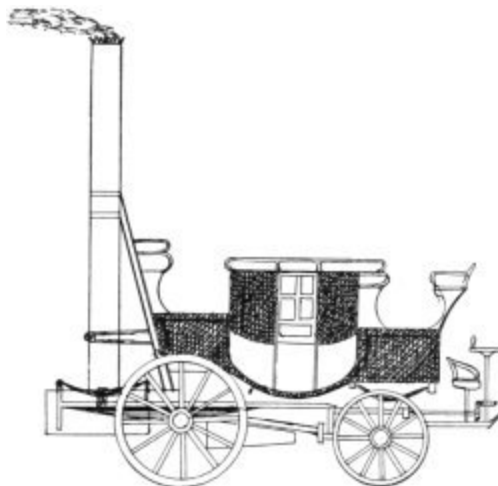
FIG. 4. STEAM COACH, MADE BY JAMES AND ANDERSON, ABOUT 1810.

Between 1805 and 1830, quite a number of steam vehicles were invented and put into practical operation. [Fig. 4](#) shows a very elaborate coach of this period, which was invented by W. H. James, and constructed with the assistance of Sir James Anderson, Bart. The machinery used in this design consisted of two powerful steam engines, one being connected with each one of the hind wheels in a manner similar to that employed in locomotives at the present time. The wheels were not fast upon the axle, hence they could revolve at different velocities in rounding curves. In this respect this invention embodied one of the features commonly used by automobiles of the latest design. Two boilers were provided, one for each engine, and the record says that with one boiler the speed was six to seven miles per hour.



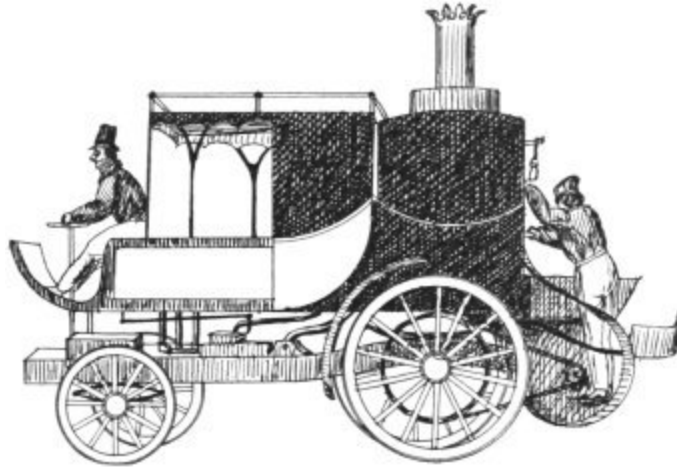
**FIG. 5. STEAM OMNIBUS, MADE BY
HANCOCK.**

[Fig. 5](#) shows an omnibus invented by Hancock. This vehicle ran on a regular route, carrying passengers from Pentonville to Finsbury Square, London. [Fig. 6](#) shows a carriage invented by Burstall and Hiel, which attracted a great deal of attention. It was probably the most complete and perfect mechanically of any invention that had been made up to that time.



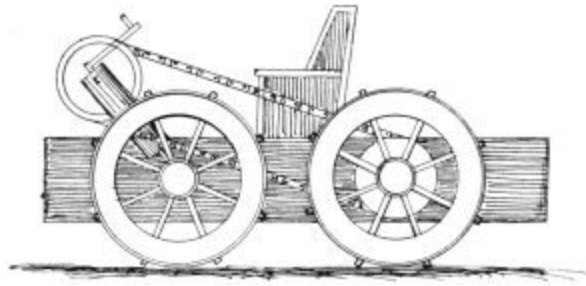
**FIG. 6. BURSTALL AND HIEL'S
STEAM CARRIAGE, MADE PRIOR
TO 1825.**

[Fig. 7](#) shows a carriage invented by Squire and Maceroni, who had been for a long time in the service of Goldsworth Gurney, one of the most noted experimenters of his day in steam propulsion. A number of carriages were made by these workers, on designs similar to [Fig. 7](#), and it is said that they ran at a high rate of speed, probably ten miles per hour.



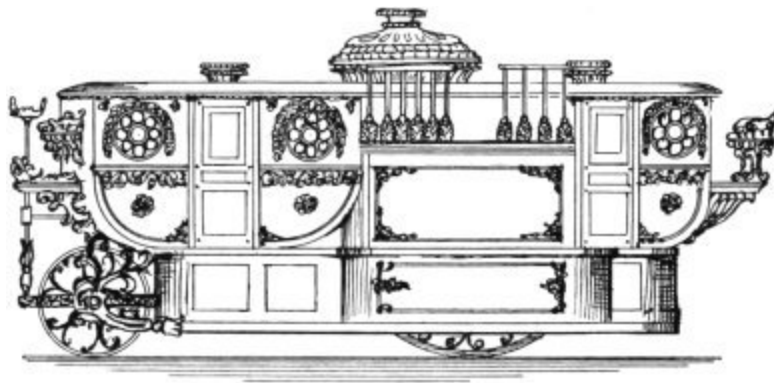
**FIG. 7. STEAM CARRIAGE, MADE BY
SQUIRE AND MACERONI.**

[Fig. 8](#) illustrates an invention that is interesting from the fact that it was to be operated by compressed air, and perhaps was the first effort to utilize this form of stored energy for the propulsion of vehicles. It was not a success, but its failure was due to the fact that the inventor labored under the delusion that the laws of nature could be circumvented by skillfully contrived mechanical devices so as to obtain something from nothing. The body of the carriage was used as a reservoir for the compressed air, and within the wheels were placed a number of pumps, the short bars projecting from the peripheries being the ends of the plungers. The expectation was that as the wheels revolved, the plungers would be depressed, and thus air would be pumped into the reservoirs and this air would operate the engine that propelled the vehicle; hence the apparatus would supply its own power, and realize perpetual motion. If this attempt to controvert the laws of nature had not been relied upon, better results might have been obtained.



**FIG. 8. COMPRESSED AIR WAGON,
MADE ABOUT 1810.**

The highly ornamental coach shown in [Fig. 9](#) was invented by Dr. Church about 1832. In addition to being ornamental, it was of massive construction and large capacity, being able to accommodate fifty passengers. Its operation is said to have been very satisfactory, a high rate of speed being attained and all grades on ordinary roads being easily mounted. The inventor swamped himself in endeavoring to compete with railroads.



**FIG. 9. SIDE VIEW OF DR. CHURCH'S STEAM COACH,
MADE IN 1832.**

Perhaps the most perfect of all the early automobiles was the one devised by Scott Russell, the celebrated designer of the Great Eastern. This carriage is shown in [Fig. 10](#). It was operated successfully, and was able to

mount the steepest hills and to attain a high rate of speed, but as coal was used for fuel and the engines were of large capacity, it is probable that the smoke, exhaust steam and noise of the machinery were decidedly objectionable features. A line of these coaches was put in commission in Glasgow in 1846, each one having a seating capacity of twenty-six, six inside and twenty on the top. After several months of successful operation, the line was withdrawn on account of the opposition of the authorities and of the general public.

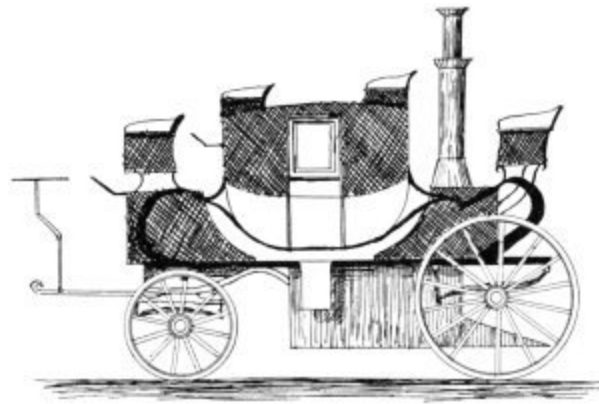


FIG. 9A. DR. CHURCH'S STEAM COACH ON THE ROAD.

These few examples of the early attempts to solve the problem of mechanical propulsion of vehicles are sufficient to show that the automobile is not entirely a creation of the progressive mind of the latter part of the nineteenth century, but that it engrossed the attention of inventors more than one hundred and thirty years ago. The success attained by the workers in this field at different periods was directly in proportion to the degree to which the form of power used had been perfected at the time. The first inventors attained but slight success, owing to the fact that, in their time, the

steam engine was in a crude form, but as the construction of the latter improved, so did that of the vehicles operated by it.

Before the days of steam, the power of wind mills was utilized to propel vehicles, and with such success that in the sixteenth and seventeenth centuries wind-propelled wagons or 'Charvolants,' as they were called, were very numerous upon the flat plains of the Netherlands.

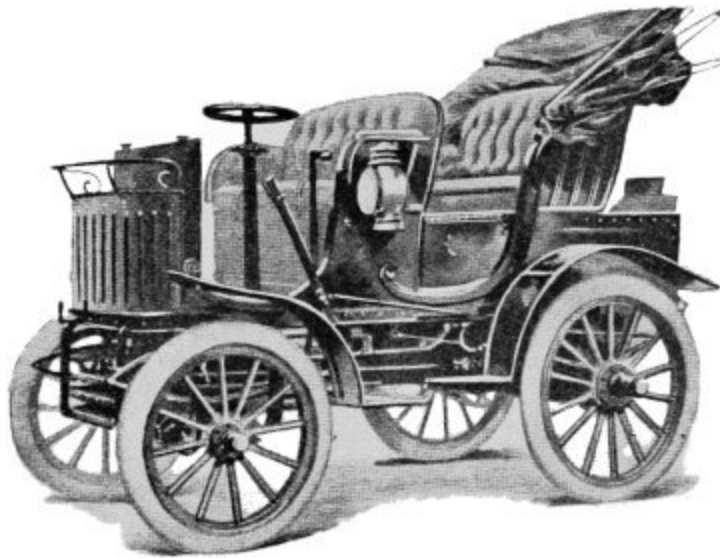


**FIG. 10. SCOTT RUSSELL'S STEAM
CARRIAGE, MADE IN 1845.**

From 1845 up to the early nineties, a period of nearly half a century, very little was done in the way of developing the automobile. From time to time inventors in various parts of the world devoted themselves to the subject, but they were generally looked upon as visionary cranks, and their work attracted little attention. During this period there was an almost universal prejudice against the use of any kind of mechanical power upon the streets or public highways, and it is even possible that if during these years any one had invented a horseless carriage, perfect in every way, he would have failed to obtain proper recognition. Prejudice against mechanically-propelled vehicles has gradually worn away, probably because of the introduction of cable and trolley cars, and at the present time the majority of people desire to see the substitution of mechanical for animal power. As a result of this change in public opinion, self-propelled vehicles are accepted as entirely satisfactory, which a few years ago would have been regarded as failures. Notwithstanding this tolerant feeling,

however, it is very doubtful whether the cumbersome coaches of the early part of the century would be received with favor at the present time when taste and requirements are entirely different. What is now desired is a light, fast-running and attractive vehicle, which could not be constructed along the lines followed by the inventors of former days. The automobile of to-day is a far more perfect device than its predecessors, although it can not be said to have reached a state of perfection. As motive power, steam, gasoline and electricity are used. Which of the three is the best, taking all things into consideration, it would be difficult to say, as each one has its defects as well as its advantages, and the evident superiority of each one in a certain direction is offset by deficiencies in other directions.

In every civilized country, where the mechanic arts are far enough advanced, automobiles are now being manufactured, but France is the country where modern development first began, and up to the present time it has maintained its leading position, although in quality of product, other nations, if not on a par with it, are certainly not very far behind.



**FIG. 11. SERPOLLET CARRIAGE, A MODERN
STEAM AUTOMOBILE OF FRENCH DESIGN.**

The perfection to which the steam automobile has been developed in these latter days is due mainly to the efforts of L. Serpollet, a distinguished

French engineer. Other highly successful steam carriages are now manufactured in England and in this country, as well as in several European nations, but Serpollet was the first to bring forth a successful fast-running and attractive vehicle, and the others have profited by his work.

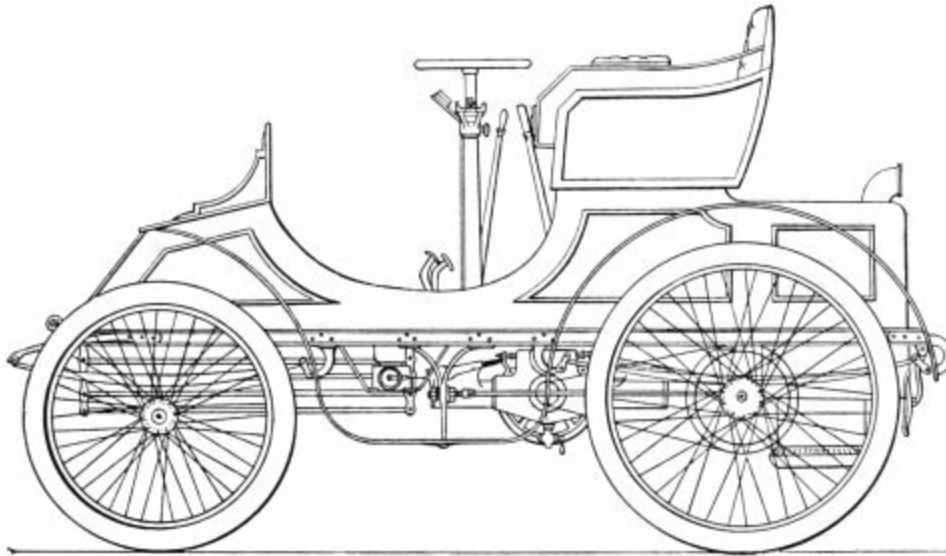
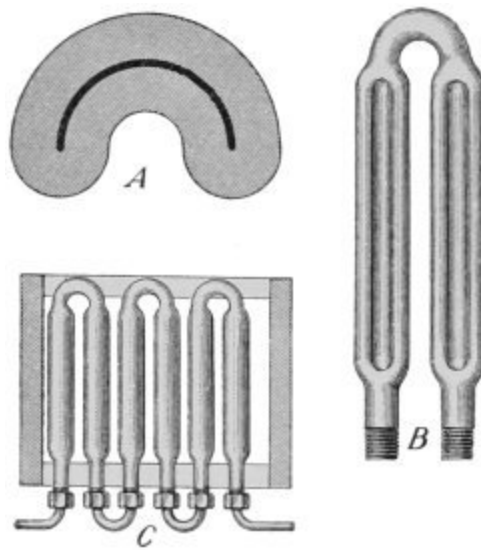


FIG. 12. SIDE VIEW OF SERPOLLET CARRIAGE, SHOWING LOCATION OF ENGINE, BOILER, CONDENSER, ETC.

One of the many designs of Serpollet carriages is shown in [Fig. 11](#); [Fig. 12](#) shows more fully the arrangement and location of the machinery. The engine used in these vehicles is made with four cylinders of the single action type; that is, they take steam at one end only. By using this construction, while the number of cylinders is increased, the other parts are greatly simplified, as the piston rods, crossheads and guides can be dispensed with. In addition, the whole engine can be made very compact.



**FIG. 13. SHOWING DETAILS OF
THE BOILER OF THE
SERPOLLET CARRIAGE.**

The boiler is of the flash type; that is, it carries no water ordinarily, but when the engine is in operation, a pump injects into the boiler at each stroke of the engine as much water as may be required to generate the steam necessary to propel the vehicle; the instant the water enters the boiler it is converted into steam. As the amount of steam is proportional to the amount of water, it can be seen that by regulating the water supply, the power of the engine and thereby the speed of the carriage, can be controlled. This is the method actually employed to control the speed. In starting, a handle is moved which connects the engine, the boiler and the pump in the proper relation; and while under way the velocity is varied by the manipulation of a lever which controls the amount of water injected into the boiler. The fuel used is kerosene, which is vaporized and then fed into a properly constructed burner. The amount of oil supplied to the burner is regulated by the same lever that regulates the supply of water, so that both are increased or reduced in the proper proportion. The boiler is constructed of a number of steel tubes, which are about two and a half inches in diameter, and from three eighths to half an inch thick. These tubes are pressed into the form shown in [Fig. 13](#), the dark line in the section marked A representing the interior space. A number of tubes collapsed in this form and bent into the

shape B, are assembled as shown at C. The number of tubes depends upon the capacity of the boiler. As the tubes are very thick, they can, without any danger of bursting, be heated to so high a temperature that the water injected into them is at once turned into steam.

In [Fig. 12](#) it will be seen that the engine is located under the body of the carriage between the two axles, and that motion is imparted to the hind wheels by means of chains and sprocket wheels. The boiler is located at the back of the vehicle, the lower part projecting some distance below the rear axle. A small smoke stack at the rear of the body allows the gases of combustion to escape. Between the front wheels, a compact condenser is located, and into this the steam from the engine is exhausted. The condenser serves two purposes; it recovers a portion of the water that would otherwise escape into the air, and thus increases the distance the carriage can run without a new supply, and at the same time it lessens the noise produced by the exhaust, and also the volume of steam escaping into the atmosphere, which in cold or rainy weather becomes plainly visible.



FIG. 14. AN AMERICAN STEAM CARRIAGE OF 1900.

Although we have been rather slow in this country in taking up the automobile, inventors and manufacturers are now working at a pace that will soon make up for lost time. We already have a number of designs of steam carriages whose operation is highly creditable. [Fig. 14](#) illustrates one of these. The design of the engine, boiler and other mechanism can be well understood from [Fig. 15](#), in which a portion of the body is removed to expose the internal parts.

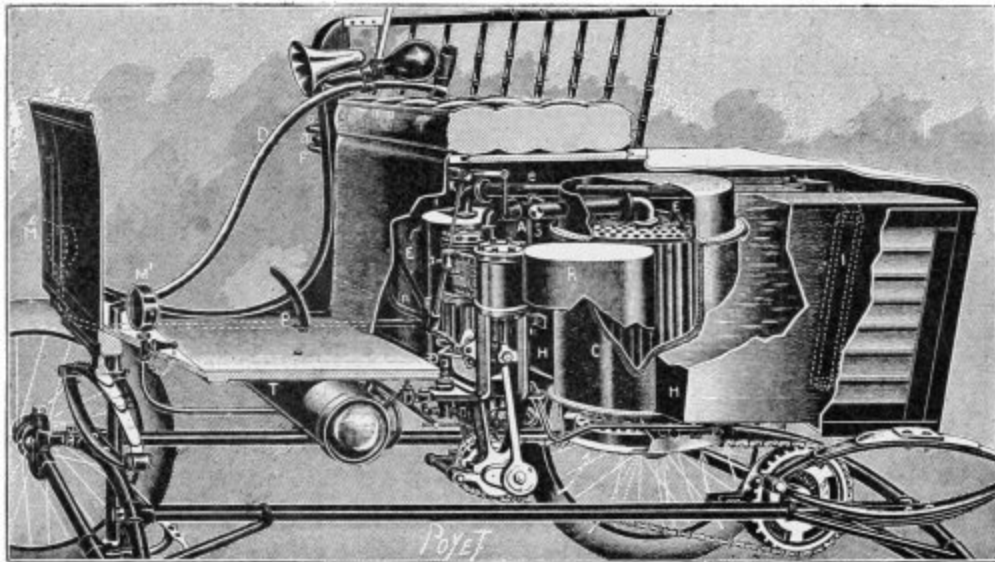
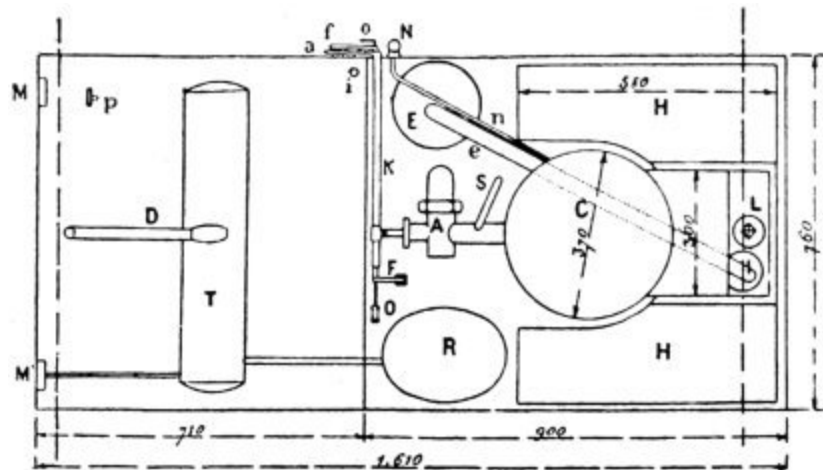


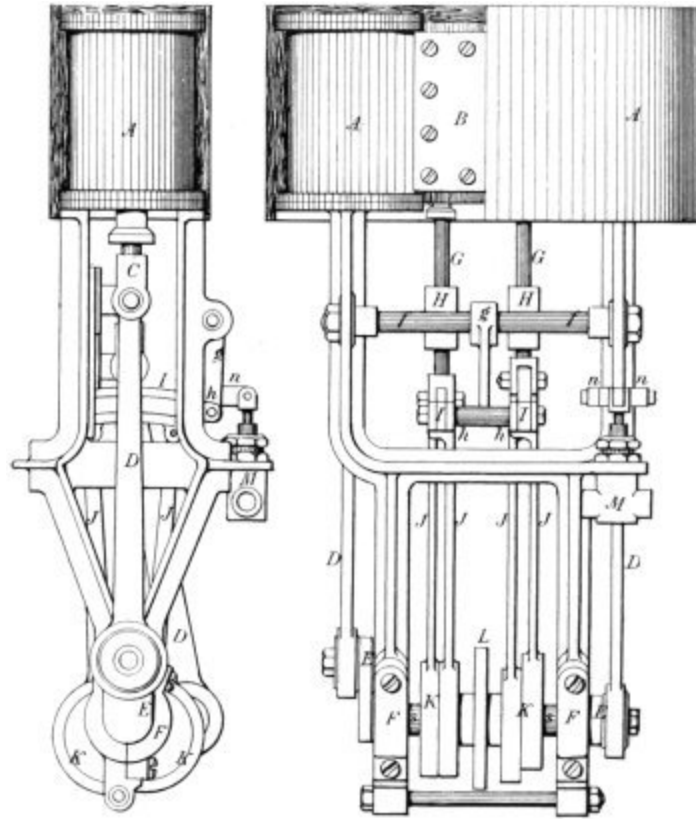
FIG. 15. SECTIONAL VIEW OF [FIG. 14](#), SHOWING LOCATION OF ENGINE, BOILER AND OTHER DETAILS.



**FIG. 15A. PLAN OF STEAM CARRIAGE SHOWN IN
[FIGS. 14](#) AND [15](#).**

The boiler is a very compact form of the upright type, such as is used in fire engines. It is about fourteen inches in diameter and twenty inches high. To increase its strength, it is surrounded with two layers of piano wire. The engine is of the locomotive type, consisting of two cylinders, the pistons of which are connected with cranks on the end of the shaft, these cranks being set at right angles, so as to prevent catching the engine on the dead center. The direction of rotation is reversed by means of the ordinary link motion. The fuel used is gasoline, which is carried in the cylindrical tank located under the front of the carriage. The gasoline is vaporized and then, mixed with a proper proportion of air, passes to a burner placed under the boiler. The amount of steam generated is regulated by the amount of gasoline supplied to the burner, and this supply in turn is regulated by the pressure of the steam, so that the action is entirely automatic. The cylinder H is a reservoir of compressed air, connected with tank I, so that the gasoline is under pressure, and therefore is forced through the pipe to the burner under the boiler. Between the burner and the tank there is a valve controlled by the steam pressure, being opened when the pressure is low and closed when it is high. When the pressure reaches a certain point the valve is closed entirely, so that even if the carriage is running very slowly, it is not possible to run the pressure above the fixed limit. The exhaust passes from the engine cylinders into a muffler, from which it escapes into the pipe K. This pipe projects downward into an opening through the center of the water

tank, and the draught produced thereby draws the gases of combustion through from the top of the boiler to the under side of the carriage body, where they escape into the atmosphere.



**FIG. 16. ENGINE OF CARRIAGE SHOWN IN
[Fig. 14](#).**

Directly in front of the exhaust muffler is seen the water gauge, which is in such a position as to be outside of the carriage body, as shown in [Fig. 14](#). A mirror is placed at the front of the vehicle, and by looking into this the water gauge can be seen. [Fig. 14](#) also shows clearly the position of the operating levers at the side of the carriage.

The actual construction of the engine is better shown in [Fig. 16](#), in which A A are the cylinders, B is the steam chest and G G are the valve rods. The piston rods connect with the crossheads C. The connecting rods D transmit motion from the latter to the cranks E, and thus rotate the shaft S.

The link motions, by means of which the direction of rotation is reversed, are at I I, and are operated by the lever G, which is mounted upon the shaft F F. This shaft is directly connected with the starting lever. The boiler feed pump is located at M. The motion of the engine is transmitted to the rear axle of the carriage by means of a chain that runs over the sprocket wheel L located between the eccentrics K K. In [Fig. 15](#), this wheel is located at D, and the chain F connects it with the axle sprocket E.

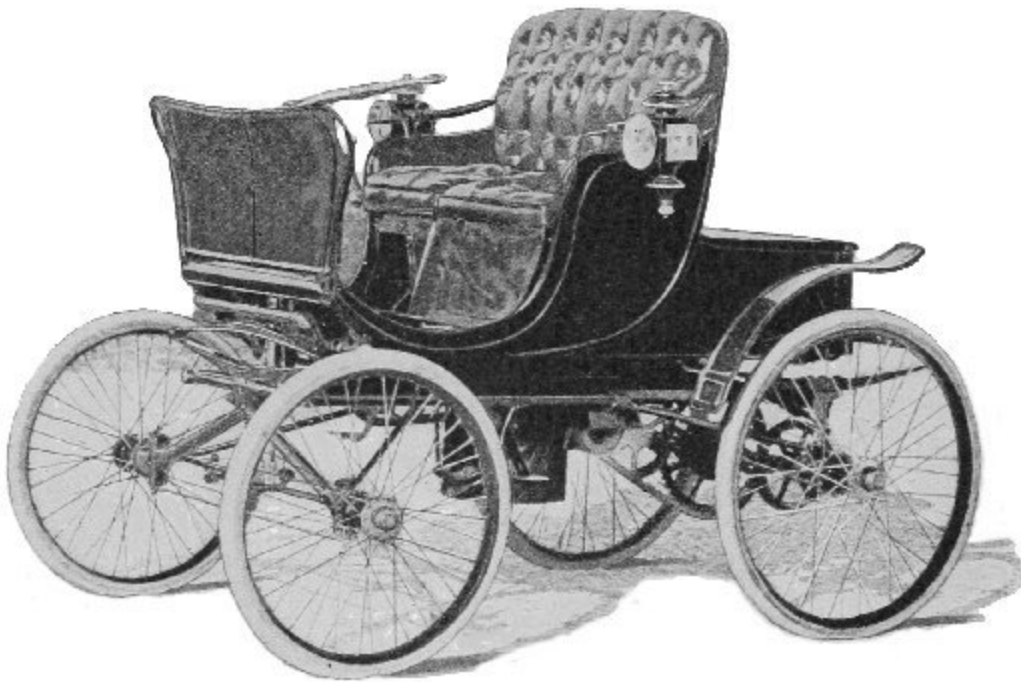


FIG. 17. AMERICAN STEAM AUTOMOBILE OF 1900.

[Fig. 17](#) shows another American steam carriage. In this vehicle the running gear is a complete truck, upon which the carriage body is supported. The appearance of the truck with the body removed is shown in [Fig. 18](#). The boiler is of the tubular type and the double cylinder engine is secured to its side. In this particular the construction differs from that of the previously described carriage, for in that the engine is attached to the cross-framing of the body of the vehicle. Although the general appearance of the mechanism of these two carriages is very similar, there are many differences in the details of their construction. In both, vertical tubular

boilers are used, and the steam is generated by the use of gasoline, which is burned in the vaporized state in specially constructed burners. The engine in both cases is of the vertical double cylinder type, and motion is transmitted to the hind axle by means of sprocket wheels and a chain; but here the similarity ends; the minor details, which it is not necessary to refer to in this connection, are with few exceptions very different.

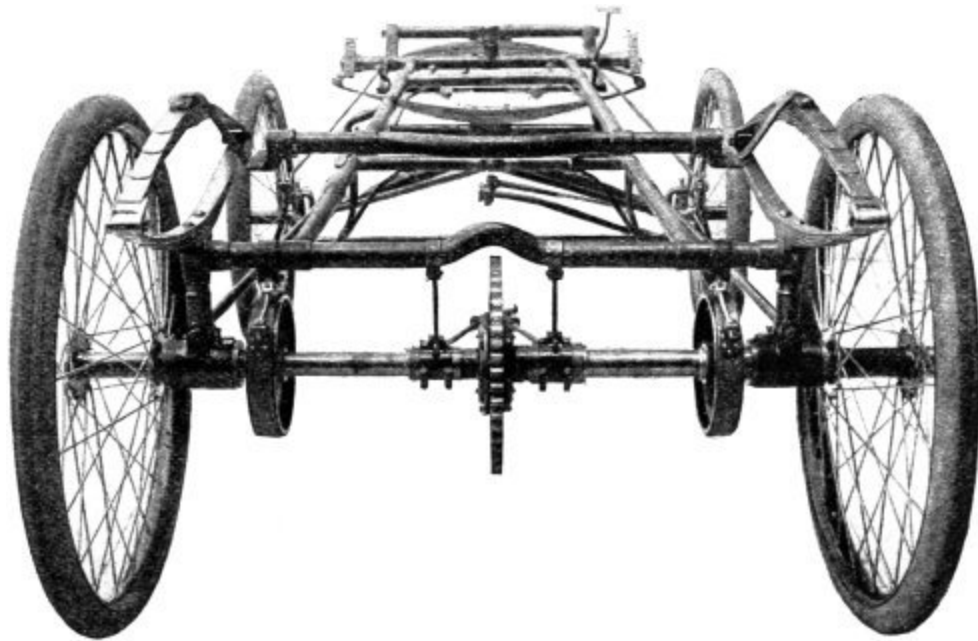


FIG. 18.

A careful examination of [Figs. 11](#), [14](#) and [17](#) will show that from an artistic point of view these examples of steam carriages are satisfactory. In regard to their operation it can be said that they have sufficient power to run up the steepest grades encountered on ordinary roads at a fair rate of speed, while on level ground their velocity is more than enough to satisfy the average rider. The danger of explosion is so remote that it need not be considered. The Serpollet boiler is practically inexplosive, while those used in the American vehicles are so constructed that they can withstand a pressure far greater than any they can be subjected to in practice. It might be expected that the motion of the machinery would produce an unpleasant vibration, but on account of the lightness of the moving parts and careful

balancing, this effect is much reduced. The use of gasoline as fuel, in connection with automatic burners, eliminates the smoke and ashes incident to the use of coal, and in addition reduces the labor of handling the vehicle, as no attention need be given to the mechanism other than to see that the water in the boiler is maintained at the proper level. In the case of the Serpollet carriages, not even this point need be looked after, as the feed of the boiler is perfectly automatic.

SCIENTIFIC RESULTS OF THE NORWEGIAN POLAR EXPEDITION, 1893–1896.^o

BY GENERAL A. W. GREELY, U. S. ARMY.

^o The Norwegian North Polar Expedition, 1893–1896. Scientific Results edited by Fridtjof Nansen. Vol. I. Longmans, Green & Co. N. Y., 1900. 1–16, 3 pl. 1–147, 3 pl. 1–26, 2 pl. 1–53 pl. 1–137, 36 pl.

FEW Arctic expeditions have done so much to increase the world's knowledge as to the physical condition of large areas of the north polar zone as has that of the *Fram*, initiated and commanded by Dr. Fridtjof Nansen.

The expedition was unique in many respects. The *Fram* was a departure from the accepted models of Arctic ships; the route followed was one unindorsed by any Arctic authority. The ship was destined to drift unprecedented distances, beset by the enormous ice-pack of the Arctic ocean. The commander himself was not only to attain the highest north, but was to make a most hazardous journey, which was to have a successful and unexpected issue partly through the aid of another polar expedition whose location and existence were unknown to the expeditionary forces of the *Fram*. Electricity made the Arctic ship a glow of light, a phonograph brought well-known voices to cheer their hours of leisure. Indeed, every device that was deemed of value was utilized.

The extent of the Arctic ocean traversed by the *Fram* is indicated by the simple fact that she passed over 120 degrees of longitude above the

eightieth parallel of north latitude, a distance of one-third around the world on that parallel.

Nansen and Johansen, in an attempt to reach the Pole, left the *Fram* March 14, 1895, in about 84° N., 100 E., but after an uneventful journey with dogs, they were obliged to turn back on April 7, 1895, in latitude 86° 14' N. They aimed to reach Spitzbergen and after months of weary effort and varying fortunes, these two hardy men landed on the east coast of the Franz Josef archipelago. Coming winter forbade further progress, so they constructed a hut and subsisted on land and sea game that was fortunately abundant. In the spring of 1896, turning southward, they attempted to reach by the kayak the east coast of Spitzbergen, hoping to be picked up by Norwegian whalers who frequent those waters. Fortunately for them, they met in April, 1896, Jackson, the commander of the Jackson-Harmsworth expedition, near Cape Flora.

Meanwhile the *Fram*, continuing its westerly drift, in which it passed the most northerly point reached by Parry in boats in 1827, emerged from the ice-floe of the Arctic ocean in the late summer of 1896 and reached Norway on August 20, about ten days later than Nansen's own arrival with the English expedition from Franz Josef Land. The *Fram* returned with its frame uninjured and its expeditionary force in health, after having covered in its voyage across the unknown Polar sea an enormous area, estimated at fifty thousand square miles.



DR. FRIDTJOF NANSEN.

The most important discovery was the oceanic depth of the Arctic Sea, where for hundreds of miles this unknown ocean disclosed a depth of over two miles. Naturally the absence of land limited the phases of the scientific work of the expeditionary force, which devoted itself to recording the phenomena of the air and the sea.

Nansen in his separate journey utilized his brief opportunities in Franz Josef Land so successfully that his contributions to the geology of that region are of no small importance.

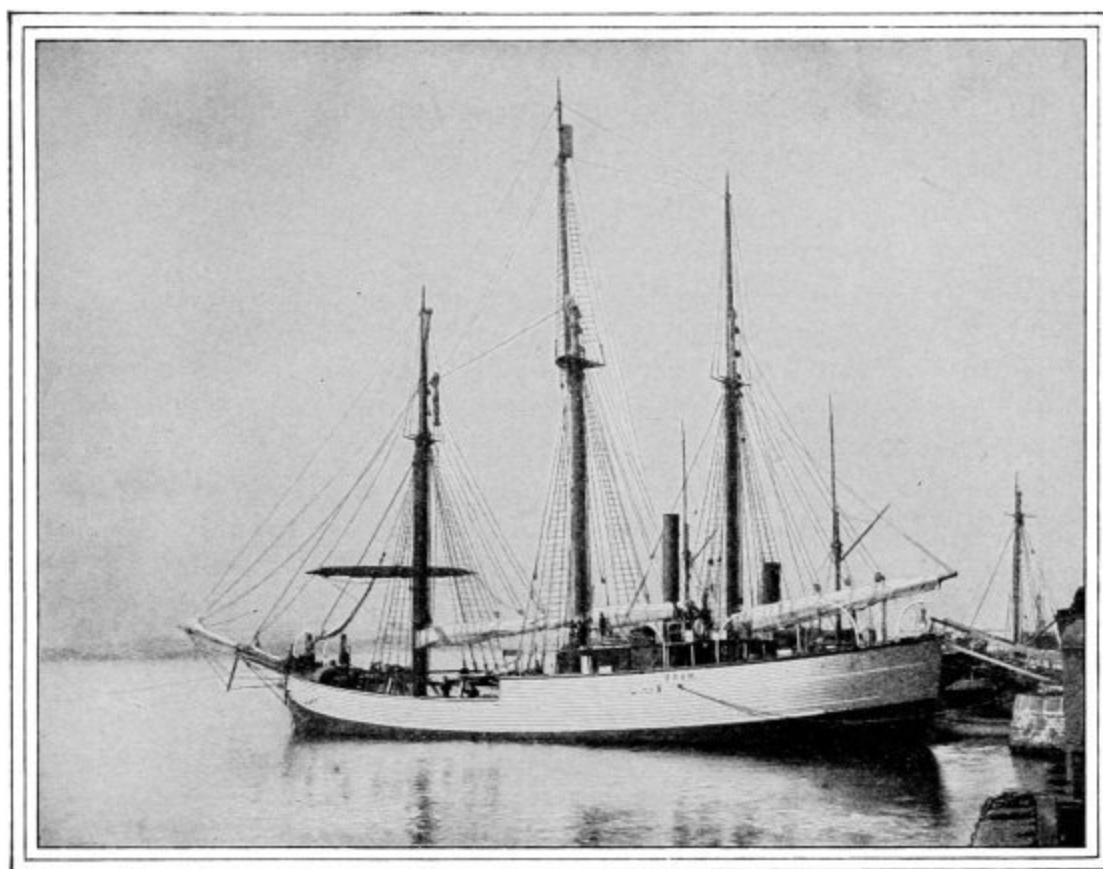
The world has looked forward with a degree of impatience to the publication of the scientific results of this expedition, and now is favored with the first volume, a beautiful quarto of some 479 pages, with 46 fine plates. It consists of a series of memoirs on the building of the ship, on the birds of the air, on the crustacean forms of sea life and a geological study of the southern part of the archipelago of Franz Josef Land. It is a striking tribute to English-speaking scientists that the work will appear in English text only. Although printed in Christiania, such has been the vigilance of the editors that typographical errors are comparatively few.

The account by Colin Archer of the construction of the *Fram* is not without interest, in view of the fact that this vessel was built on novel lines calculated to cause the ice to meet a sloping surface, so that, pressing down under the bilge, it would cause the vessel to rise and thus insure its immunity from destruction.

Archer says: "In order to utilize this principle, it was decided to depart entirely from the usual deep-bilged form of section and to adopt a shape which would afford the ice no point of attack normal to the ship's side, but would, as the horizontal pressure increased, force the attacking floes to divide under the ship's bottom, lifting her as described above.... Plane or concave surfaces were avoided as much as possible by giving her round and full lines. This, while increasing the power to resist pressure from outside, also had the advantage of making it easy for the ice to glide along the bottom in any direction."

As great length is an element of weakness, the *Fram's* length was cut down as much as possible, with a tendency to make its form circular or oval. Various expedients were adopted to reduce the dead weight of the ship by a judicious arrangement of materials. While economizing weight, the cargo-carrying capacity of the ship could not be too much reduced, and the great strength of the ship must be preserved. Inasmuch as the broadside of the ship, both structurally and from its shape, is its weakest part, it was necessary to adopt extraordinary measures to strengthen it. This was done largely by adding stays of yellow pine placed nearly at right angles to the ship's sides, and securely fastened with wooden knees. These were supplemented with upright stanchions tied by iron straps.

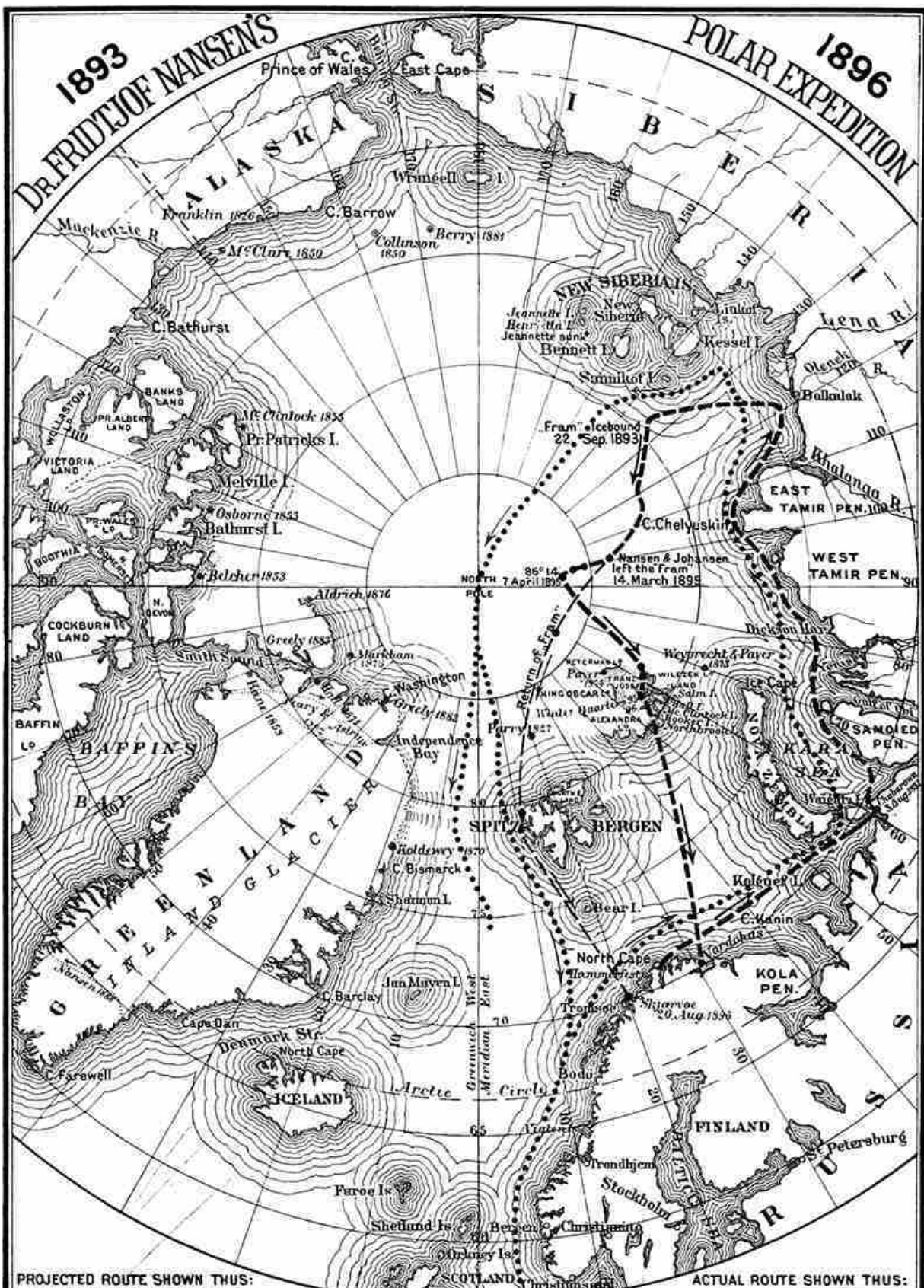
While experienced whalers strongly advocated the square rig, Archer decided to ignore their advice and rigged the *Fram* as a fore-and-aft three-masted schooner, which style of rig proved, under the circumstances, to be most suitable. The slight increase in leakage is believed by Archer to be due in part to the drawing of the oakum out of the seams and in part to the expansion and contraction of the timbers. While the *Fram* was not subjected to such tremendous ice convulsions as have been many other Arctic ships, yet her experiences were very severe and may be considered to prove that the design and system of construction adopted were the most efficient possible.



THE FRAM.

The most extensive, if not the most important, of the treatises that form this volume, relate to regions and investigations with which the voyage of

the *Fram* were only incidentally connected. Reference is had to the papers on the geological formations of Cape Flora, Franz Josef Land, by Professors Nansen, Pompeckj and Nathorst. Dr. Nansen most cordially acknowledges his great indebtedness to Mr. Jackson and Dr. Reginald Koettlitz, respectively the leader and geologist of the Jackson-Harmsworth expedition to Franz Josef Land, 1894–1896. The latter of these gentlemen, in a spirit of broad scientific generosity, accorded Dr. Nansen full and equal access to his discoveries, covering three years' work on Northbrook Island, among fossils and geological conditions of special interest.





MAP SHOWING REGIONS TRAVERSED.

(Double-tap for larger map.)

Nansen confines himself to a brief geological sketch of Cape Flora and its neighborhood; Pompeckj treats fully the Jurassic fauna, while Nathorst briefly discusses the fossil plants.

Nansen says: "Through Jackson's kindness and Koettlitz's valuable assistance, I was enabled to make a collection of fossils and rocks from the Jurassic deposits of this locality."

"(Koettlitz) took me to places where, before my arrival, he had already found fossils, or had observed anything of importance. Had it not been for him I should certainly not have been able to do what little I did during the few days at my disposal. I agree with Koettlitz on all essential points, and have nothing new of importance to add to what he has already said."

As Nansen elsewhere remarks, the memoirs of Pompeckj and Nathorst supplement the papers of Koettlitz, Newton and Teall, which appeared in the Quarterly Journal of the Geological Society, 1897, pp. 477–519, and 1898, pp. 620–651.

Pompeckj describes fully the various fossils, illustrates them with wealth of detail, discusses their stratigraphical relations, and outlines the paleographical history of Franz Josef Land.

Of the twenty-six species collected by Nansen no less than seventeen are new as compared with the Jackson-Harmsworth collection, which contains five species lacking to Nansen. There are representatives of single species only of echinoderms, vermes and gastropods, the scarcity of the last named being generally characteristic of the Jurassic fauna of the arctic regions, whether in Siberia, Greenland, or Arctic America. On the other hand, at Cape Flora the cephalopods and the lamellibranchs predominate very largely. This fact makes most notable the absence of the lamellibranch genus *Aucella*, with all other forms that are especially characteristic of the higher Jura.

The following new species have been determined by Pompeckj: *Pseudomonotis Jacksoni*, an ornamented shell of a remarkably large

Aviculid form. *Macrocephalites Koettlitz*, a shell with a very narrow umbilicus and almost completely encircling whorls. *Cadoceras Nanseni*, an ammonite showing a flat disc-like growth, with moderately thick whorls of which cross-sections are nearly elliptical. Another ammonite may possibly be a variety of *C. Nanseni*, but Pompeckj considers that it is a separate species owing to its wider umbilicus, less pronounced involution and somewhat asymmetrical lobe-line.

Pompeckj's outline of the paleontographical history of Franz Josef Land is worthy of careful consideration by all interested in this department of science, although many may differ from some of the conclusions reached by him. Commenting on the stratigraphical studies of Prof. E. T. Newton, Pompeckj states that his own investigations compel him to differ materially from the inferences drawn and theories advanced by that scientist.

Pompeckj says: "The occurrence of these three genera of Ammonites proves that the marine fauna of Cape Flora contain representatives of the Callovian. More recent marine horizons have certainly not been formed at Cape Flora, as far as I can judge from the collection of fossils before me.... The Oxfordian and all the more recent Jurassic horizons do not occur as marine deposits at Cape Flora."

He finds species pertaining to the Lower Bajocian, Lower, Middle and Upper Callovian horizons. It is most interesting to note that only one other part of the arctic regions, Prince Patrick Island, Parry Archipelago, has produced fossils, described by Haughton as Lias, that are certainly older than the Callovian. It is, however, recognized as possible that Lundgreen's fossils from East Greenland may form another exception.

Pompeckj points out that while the Bajocian fauna of Cape Flora is without analogy in the arctic regions, it nevertheless presents distinct affinities to the Central European Jura, and especially resembles the Russian Callovian.

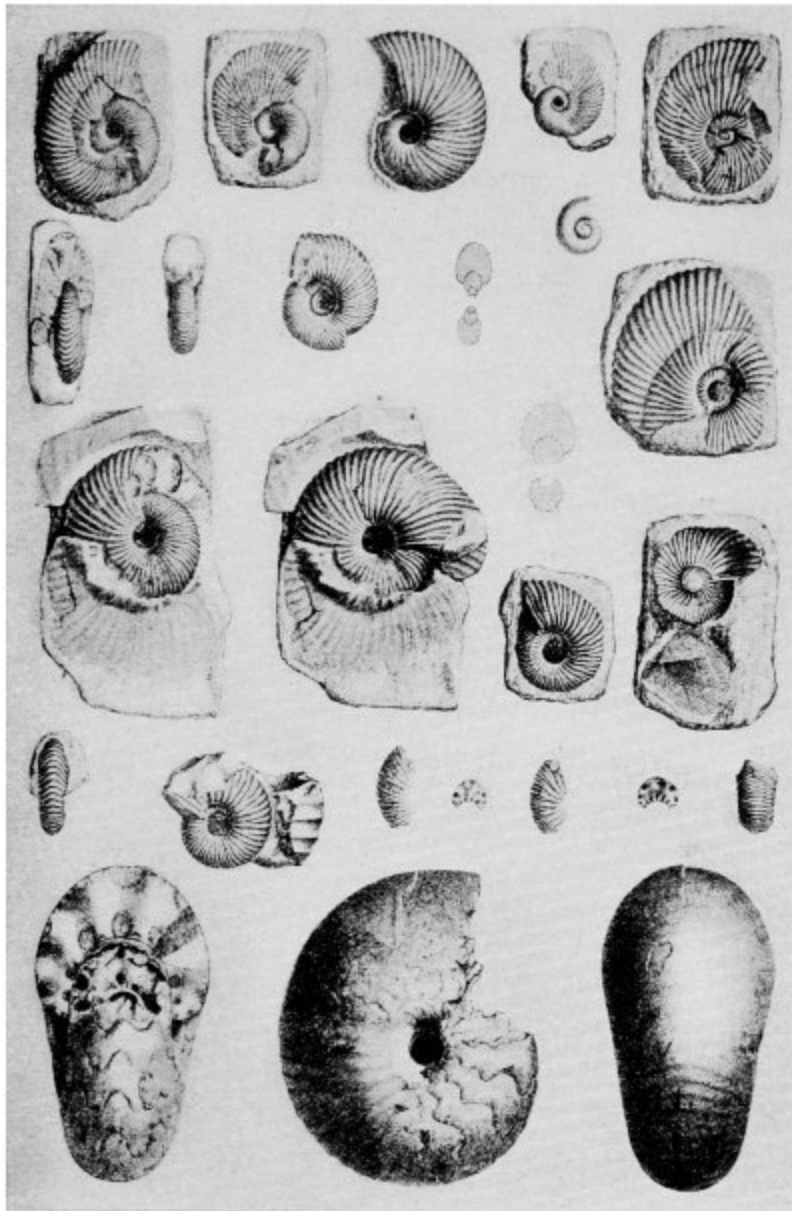
Moreover, this Jurassic collection from Cape Flora is of special importance in outlining the geographic distribution of that system. Pompeckj adds: "Hence the existence of a Bajocian sea in the north of the Eurasian Jura continent is proved beyond all doubt.... As early as the Bajocian period, there existed a Shetland Straits, which separated the

Eurasian continent, existing through the Lias period until the end of the Bathonian, from the nearctic Jura continent.”

The comments relative to the transition of Nova Zembla, Spitzbergen, Franz Josef Land, and possibly Alaska, from land to sea and sea to land, are of marked interest, indicating as they do that large areas of polar regions were exposed in the mesozoic period to repeated and very considerable oscillations of the sea level.

The more interesting of the Jurassic fossils, found at Cape Flora, are shown in the accompanying illustration. *Cadocera Nanseni* (n. sp.), 1, 2, 3, 5, 6. *Cadoceras*, sp. ex. aff. *Cad. Nanseni* (n. sp.), 4. *Cadoceras Tchefkini*, d’Orb, 7. *Cadoceras*, sp. indet., 8. *Quenstedoceras vertumnum*, Sintzow, 9. *Cadoceras Frearsi*, d’Orb, 10. *Macrocephalites*, 11. *Macrocephalites Koettlitzii*, n. sp., 12.

The collections of fossil plants, made by Nansen in Franz Josef Land through the courtesy of the Jackson-Harmsworth expedition, are of scientific value as indicating the fossil Jurassic flora of Franz Josef Land as compared with that of Spitzbergen. These collections fill in a not inconsiderable gap in the Arctic regions, and Nathorst’s investigations serve to confirm the opinions and statements made by Professor Heer, whose five volumes of *Flora Fossilis Arctica* constitute a monumental work. As is well known, research has established the fact that at one time Spitzbergen was covered with a luxuriant miocene vegetation—cypresses, birches, sequoia, oaks and planes. It moreover appears that this growth was coincident with the period when Spitzbergen, Greenland, Franz Josef Land and Nova Zembla experienced a continental climate.



JURASSIC FOSSILS FOUND AT CAPE FLORA.

As fossil collections accumulate, one appreciates more and more the masterly manner in which Heer summed up the results of polar exploration as regards Arctic vegetable paleontology. He was the first to present to the world a clear idea of the vegetation of the Cretaceous land, scarcely known to science until elucidated by him. It developed that in Heer's time, among

the fossil plants found in Spitzbergen alone were 7 ginkos, 8 pines, a short bamboo, 7 poplars, 3 maples and a fossil strawberry.

Dr. Nansen was fortunate in securing the co-operation of Prof. A. G. Nathorst in the examination of the fossil plants collected in Franz Josef Land, as he has devoted much time to the flora, present and past, of various portions of the Arctic regions, especially Spitzbergen and King Charles Land. Nathorst had the advantage of the notes of Newton, J. H. Steele and R. Curtis on the fossils of Franz Josef Land, published in the Quarterly Journal of Geological Science, London, vols. 53–54, 1897–1898.

Most unfortunately, the fossils were very fragmentary, the leaves in themselves small and often indistinguishable in color from the rock, so that their examination was made almost entirely under the magnifying lens. While the organic substance of the plants was sometimes still to be seen in a soft, brownish variety of rock, yet the harder yellowish varieties offered only impressions, or cavities, their organic substance having entirely disappeared. In cross fractures there were sometimes cavities which were complete transverse sections of coniferous leaves.

There were twenty-nine species, of which the entire number are coniferous except one fungus, one fern, two palms and one uncertain.

Nathorst says: “The plant-bearing strata of Franz Josef Land, which are yet known to us, all belong, with the exception of those from Cook’s Rock and Cape Stephen, the age of which is still uncertain, to the upper Jurassic, or the transition beds to the cretaceous, while as yet no tertiary strata have been discovered.”

In geological age, while the Franz Josef flora resembles most the previously known Jurassic floras of Siberia and Spitzbergen, yet Nathorst considers the geological age different, and naturally places it between the two, it being evidently younger than that of Siberia.

It is interesting to note that Doctor Koettlitz found in an isolated basalt nunatak (rock or hill protruding from a glacier) fossil plants similar to those found by himself and Nansen on the north side of Cape Flora. These nunatak plants, which Koettlitz believed to be *in situ*, are identified by Nathorst as Upper Jurassic, and came from an elevation variously estimated as from six hundred to seven hundred and fifty feet above the sea.

Nansen agrees with Koettlitz in believing that tree-trunks found by them, charred into charcoal or partly silicified, chiefly belonged to conifers growing on the soil over which basalt flows were discharged during the Upper Jurassic or Lower Cretaceous age, and that they have been charred by a flowing mass of lava that overwhelmed them.

These fossil plants tell the story of tremendous physical changes which have produced very important modifications in climatic conditions in the Arctic regions. The changes in the types of vegetable life are apparently as extensive in high as in low latitudes. The lower cretaceous flora is almost tropical, as is shown by the predominating forms of this vegetation. Carboniferous formations obtain extensively in the Arctic regions, as they occur in the Parry Archipelago, Spitzbergen and in Siberia. During the carboniferous age there was a great extent of land near the North Pole closely resembling that of the temperate latitude of the same period, as is shown by the small number of fossil plants that are peculiar to the Arctic regions. In the tertiary period miocene flora flourished in Spitzbergen, where even the lime, the juniper and poplars have been found near latitude 79 N. Then also thrived sequoias, which closely resemble trees growing in the southern part of the United States. The miocene flora gives evidence of a very great contrast between the climatic conditions at that epoch between Europe and the Arctic regions.

The cretaceous flora throws important light on the changes of climate in the Arctic regions, and, as has been pointed out, the tropical forms predominate in the vegetation of the Lower Cretaceous flora. Heer's prediction that the plants found on the west coast of Spitzbergen would also be found on the East Greenland coast has been fully verified. Miocene plants have been found from Spitzbergen westward through Iceland and Greenland to Banks Land and in the Parry Archipelago, and it is interesting to note that more than one fourth of the Arctic plants are common to the miocene of Europe; in Greenland and on McKenzie the percentage is nearly one half.

In all probability, the paper which is of the highest popular interest is the account of the birds by Robert Collet and Dr. Nansen. The full notes regarding Arctic birds testify fully to the fact that the observers had in view the principal points of ornithological importance. These comprise not only a

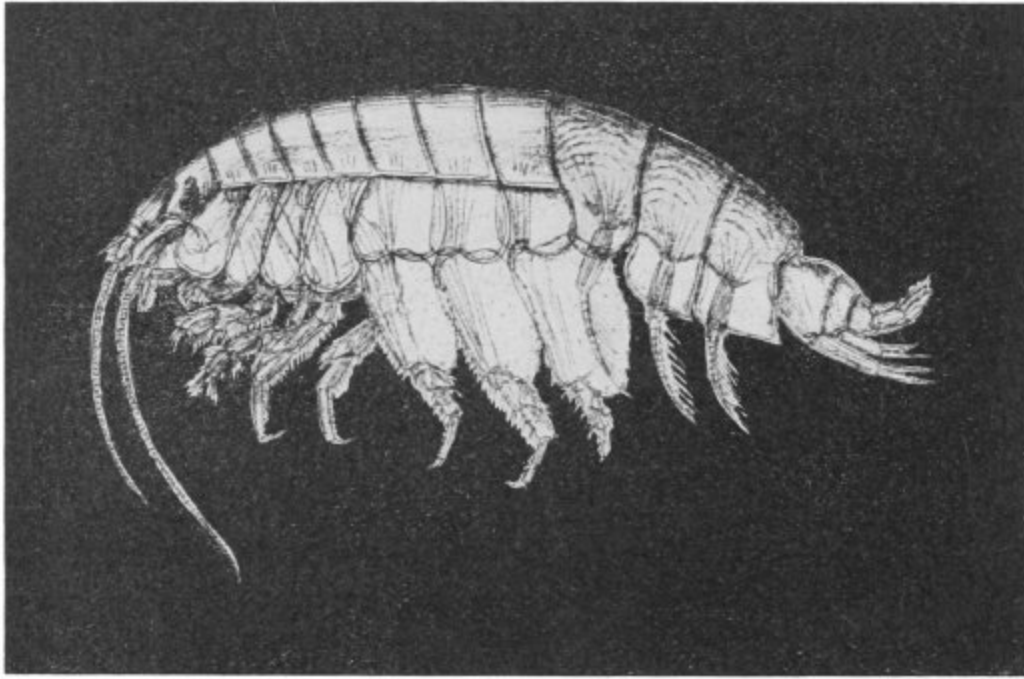
mere record of the presence or absence of certain species, but also additional observations regarding them in their Arctic habitat.

Certainly the reproach can not be brought against the expedition of the *Fram*, which has obtained in the case of many Arctic expeditions, that it has added nothing to ornithological Arctic data.

The account of the birds, prepared by Mr. Robert Collet, has been compiled from the various journals of the expeditionary force, supplemented by verbal comments of Nansen. The memoir contains such specific data as enable students to determine not only the general character of the avifauna as one moves northward in the Siberian ocean, but also the arrival and departure of the migrants and the presence of stragglers. Among the birds of special interest which were observed are the gray plover, the gray phalarope, the sabine gull and the cuneate or Ross's gull.

One of the greatest authorities on Arctic birds, Prof. Alfred Newton, of the University of Cambridge, has well said that in consideration of the avifauna of any country its peculiarities can be determined only by dismissing accidental stragglers from the discussion. In elucidating the great question of geographical distribution, one must confine himself to either the birds that breed therein, or to those species which regularly frequent it for a considerable portion of the year.

Considering the enormous area covered by the *Fram* expedition and its great diversity of physical conditions of sea and land, it was impossible to treat under a single heading the birds observed.



PSEUDALIBROTUS NANSENI, G. O. SARS.

Mr. Collet has, therefore, been wise in dividing his notes into four sections, covering the Asiatic coast, the Siberian ocean, the sledge journey to Franz Josef Land, and the Arctic Ocean to the north of Franz Josef Land and Spitzbergen. But for this division, confusion would have resulted from combining birds of regions so widely extended in longitude and latitude.

The notes show conclusively what might have been anticipated, that the avifauna of the Siberian Sea, and especially that portion of the Arctic Ocean to the north of Franz Josef Land and Spitzbergen, is strictly limited.

Including the species observed during the entire voyage, there are only thirty-three recorded. Only twenty-one species pertain to the Arctic Ocean, whether as regular migrants or stragglers, after excluding the twelve species which were observed near the Asiatic coast. The presence on the shores of the Siberian Sea of some of these twelve, however, is of ornithological interest. There may be specially mentioned the gray goose (*Anser segetum*), long-tailed duck (*Harelda glacialis*), silver gull (*Larus argentatus*), snowy owl (*Nyctea scandiaca*), gray plover (*Squatarola helvetica*) and the red-necked phalarope (*Phalaropus hyperboreus*).

Confining ourselves to birds observed to the north of 81° 30, attention is called to the abundant avifauna of the western as compared with the eastern hemisphere. In Kennedy Channel, Grinnell Land, there have been recorded no less than thirty-two species against twenty-one noted by the *Fram* in this voyage, including those seen in Franz Josef Land. This is not surprising, however, when it is considered that the drift of the *Fram* was across a deep ocean of large extent, which is covered perpetually by an unbroken ice-pack, unrelieved by any view of land until the north coast of Spitzbergen was seen.

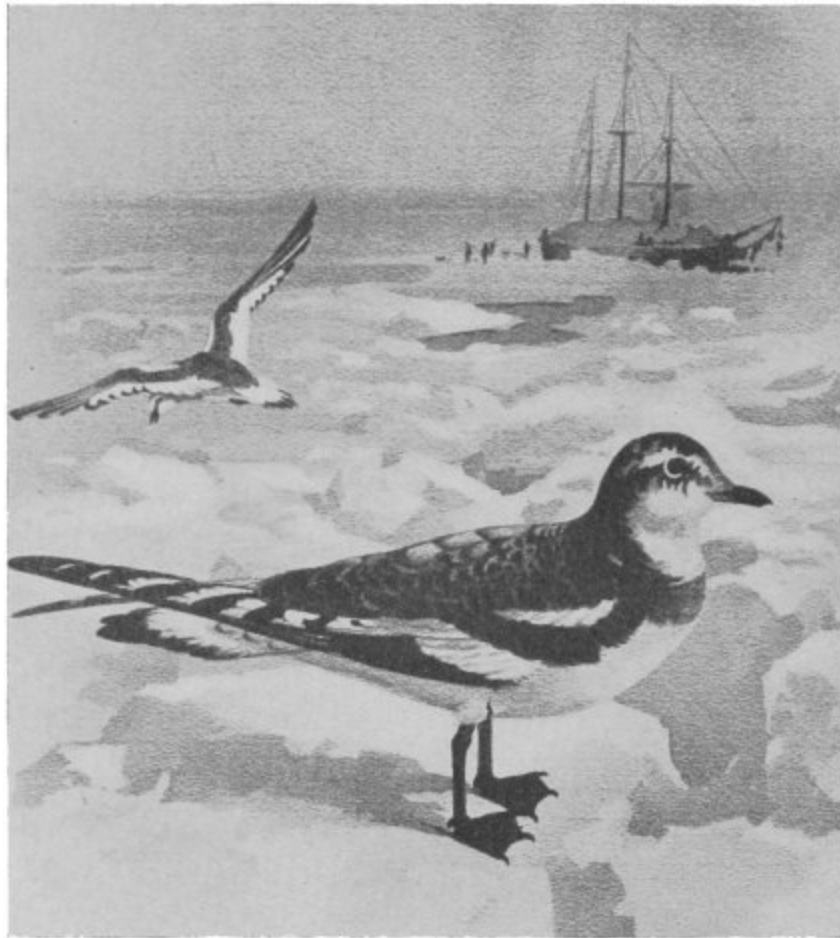
Omitting the birds observed in Franz Josef Land, the paucity of species frequenting the great western Arctic Ocean is even more apparent. The striking dissimilarity of the four regions traversed by the *Fram* is plainly evident from the bird-life recorded. While there were observed nine species in the Siberian Sea, fifteen in the Franz Josef Archipelago, eighteen in the Arctic Ocean and twenty-three on the Asiatic coast, yet only five were common to all four regions, viz.: the dovekie, the glaucous gull, the ivory gull, the kittiwake and the snow-bird.

The Siberian Sea presented a most limited avifauna, as in addition to the five common species, there were recorded in the first summer in the ice only the little auk, the fulmar, the roseate gull and a small skua. The entire absence of land or shore birds that frequent Arctic islands, omitting a single straggling snow-bird, indicates clearly that the Siberian Sea extends far northward unbroken by any land area.

The eighteen species of birds that were found in the Arctic Ocean, far to the north, naturally demand special comment. The six following species are doubtless stragglers: the ringed plover (*Aegialitis hiaticula*), 82° 59' N., the most northerly shore-bird of Spitzbergen, Nordenskiöld having observed it on Seven islands, 80° 45' N.; the eider duck (*Somateria mollissima*), 82° 55' N., near Spitzbergen; the arctic tern (*Sterna macrura*), 84° 32' N.; the puffin (*Fratercula arctica glacialis*), 83° 11' N., near Spitzbergen; the black-backed gull (*Larus marinus*), 84° 35' N. 75° E., and the Sabine gull (*Xema Sabini*), 83° N., near Spitzbergen.

Of other species, the roseate gull (*Rhodostethia rosea*), 84° 41' N., disappeared as the *Fram* drifted west from the longitude of Franz Josef Land, to be replaced as Spitzbergen was neared by a wader (*Crymophilus*

fulicarius), 83° 01' N.; forked-tailed skuas (*Stercorarius pomatorhinus*), 82° 57' N., and Bruennich's guillemot (*Uria lomvia*), 83° 11' N. The glaucous gull (*Larus glaucus*), 84° 48' N., and long-tailed skua (*Stercorarius longicaudus*), 84° 47' N., although seen both summers, were quite infrequent. These data indicate absence of land at any near distance to the north, and disclose the interesting fact that only the six following species, including the snow-bird who is more probably a straggler, can be classed as regular summer migrants to the vast ice-fields which cover the Arctic Ocean to the north of Spitzbergen and Franz Josef Land.



**RHODOSTETHIA ROSEA (MAGG), 1824. YOUNG IN
FIRST PLUMAGE.**

The little auk (*Alle alle*), 84° 48' N., was visible almost daily near the 83d parallel in great numbers during the summer season, wherever there were numerous water channels near the *Fram*. Of 40 birds killed at one time, only ten were females.

The dovekie (*Cepphus mandti*), 84° 32' N., with the little auk, was the most numerous of all birds in very high latitudes, and nearly 150 were shot for the table. Out of 40 specimens only 14 were males. The dovekie came early, May 13, 1896.

The ivory gull (*Pagophila eburnea*) is also present the entire summer. It was the first visitor in 1895, when on May 14 it was seen in 84° 38' N., and what is of special interest, was flying from the north-northeast.

The snow bunting (*Plectrophenax nivalis*), although a land-bird, was seen both summers at somewhat infrequent intervals, as far as 84° 45' N. They fed on refuse near the ships, but were also seen near water-holes, and appeared to be feeding on crustaceans. Two of three specimens were males. The first specimen in 1895 visited the *Fram* on May 22 in 84° 40' N., and then flew towards the north. In 1896 it appeared on April 25, the first bird of the year, in 84° 17' N.

The kittiwake (*Rissa tridactyla*) was much less numerous than the ivory gull. It was seen in 82° 54' N. They fed, as a rule, on crustaceans, although in one bird were found parts of a *Gadus saida* about 70 mm. in length. A *Gadus* about 120 mm. in length was observed on July 16, 1895, in 84° 42' N., the most northerly point at which any fish has been found.

The fulmar (*Fulmarus glacialis*) came early in 1895, on May 13, and in 1896 on May 22. This bold, voracious bird fed on crustaceans usually, and owing to its villainous smell was utilized principally as food for dogs. The last bird of 1895, a fulmar, was seen on September 14, when the *Fram* was in 85° 05' N., 79° E. This is the most northern latitude in which any bird has ever been observed.

The fulmars and ivory gulls were very bold and noisy, the latter being specially objectionable. Ivory gulls were seen at the winter hut in Franz Josef Land until October, when all water had long been frozen over, and appeared again as early as March 12, 1896.

The first roseate gulls were young birds observed August 3, 1894, in 81° 05' N., 120° E., about 500 kilometres from the nearest land. A long and interesting description is given of these gulls in various stages. One of the beautiful plates, which is imperfectly reproduced, shows the plumage of a very young gull about a month old. Their food consists exclusively of small fish and crustaceans, of the latter the *Hymenodora glacialis* predominating. Large numbers of these beautiful gulls were seen in 1895 to the northeast of Franz Josef Land, which points to their breeding in that locality. One was seen by Nansen on July 11, 1895, in 82° 08' N., flying from the northeast.

The very full memoir on *Crustacea* is by Dr. G. O. Sars, well known as one of the editorial committee of the scientific work of the Norwegian North Atlantic Expedition. As the greater number of marine vertebrate animals collected by the Norwegian North Polar Expedition belong to the *Crustacea*, this memoir covers the greater part of the marine collection.

The *Copepoda* are predominant, especially those belonging to the *Calanoid* group, having been taken at nearly every haul along the whole route of the *Fram*. The zoölogical equipment of the *Fram* was based unfortunately on the supposition that the Siberian basin was shallow, so that the enormous oceanic depths which were found were only inadequately explored by an extemporized sounding apparatus.

While the results of the dredging operations indicate that there was very little animal life at the bottom of the ocean, on the other hand, it appears that the entire surface of the sea, which consisted usually of small temporary openings in the ice-pack, was covered with abundant life throughout the entire year even to the most northern latitudes.

Including surface and deep-sea specimens, there were taken on October 12, 1895, no less than eleven species in latitude 85° 13' N., longitude 79° E. On June 28, 1895, in 84° 32' N., 76° E., there were taken from the surface by tow net in a large water-channel fourteen species. This indicates abundant marine life in the sea immediately near the North Pole.

The pelagic animals, therefore, were not found at the sea surface alone, but were also drawn from considerable depths. Many specimens were obtained from strata at least 250 metres below the surface, and in a number of instances from depths ranging between 500 and 1,000 metres. It is to be added that the imperfect development of the visual organs of the peculiar

amphipod, *Cyclocaris Guilelmi*, Chevreux, points to abyssal habits, as similar conditions do in the cases of other pelagic animals.

In general pelagic fauna in the Polar Sea resembles that of the northern Atlantic basin, the greater number of species being common to both. While several heretofore unknown forms collected by this expedition may be peculiar to the polar basin, yet it is not improbable that these forms also occur in the North Atlantic. This appears probable, since the western part of the *Fram*'s route lies on the border of the two basins, where the fauna does not differ essentially from that in the eastern part.

While the pelagic fauna of the Polar Sea, even in the lowest depths, resembles that of the Atlantic basin, the great salinity of its water clearly indicates that it comes from the North Atlantic, and it is therefore more than probable that the migration of pelagic animals to the North Polar Sea is also from the west.

Indeed, Doctor Sars is of the opinion that the greater part of the pelagic life of the north-polar basin comes by the underlying easterly current from the North Atlantic. On the other hand, it is evident that the westerly-flowing surface current of the Siberian Sea is of vital importance as a means of supplying nourishment to the marine animals of the western Arctic Ocean. This food supply, microscopic algæ chiefly *Diatomeae*, while very abundant on the surface of the Siberian Sea, diminishes gradually towards the west. "Indeed," says Sars, "without such a constant conveyance of nourishing matter, there could be no such rich animal life in the Polar Sea."

A very remarkable fact was the presence of certain pelagic *Copepoda*, which hitherto had only been observed in southern waters, and a *Calanoid* of the genus *Hemicalanus* Claus, previously known only from the Mediterranean and tropical parts of the Atlantic and Pacific oceans. Two species of the genus *Oncoea*, which accord perfectly with species in the Bay of Naples, were found in great abundance north of the New Siberian Islands. Another copepod, of the genus *Lubbockia* Claus, heretofore only known in the Mediterranean and tropical oceans, was found in the same locality, with which was a small perfectly hyaline copepod of the very remarkable genus *Mormonilla*, of which heretofore only two species have been recorded, both in the tropical Pacific and south of the equator.

Perhaps the most remarkable forms are those mentioned by Doctor Sars, when he says: "The very close and apparently genetic relationship between the two polar species of the amphipodous genus *Pseudalibrotos* and those occurring in the Caspian Sea, is another remarkable instance which seems fully to corroborate the correctness of the assumption of geologists as to a direct connexion in olden times between this isolated basin and the North Polar Sea."

Both species, taken near 85° N., are regarded as the primitive types from which the Caspian forms are descended. The more remarkable of the Arctic forms, *P. Nanseni*, is reproduced on page 430.

To conclude, this volume is a most valuable contribution to the scientific literature of the Arctic regions. It has but one marked objection, its publication in such beautiful form and high price as necessarily places this series beyond the means of many scientific students.

DISCUSSION AND CORRESPONDENCE.

LEGISLATION AGAINST MEDICAL DISCOVERY.^P

^P An open letter from President Eliot of Harvard University to the Chairman of the Senate Committee on the District of Columbia.

DEAR SIR: I observe that a new bill on the subject of vivisection has been introduced into the Senate, Bill No. 34. This bill is a slight improvement on its predecessor, but it is still very objectionable. I beg leave to state very briefly the objection to all such legislation.

1. To interfere with or retard the progress of medical discovery is an inhuman thing. Within fifteen years medical research has made rapid progress, almost exclusively through the use of the lower animals, and what such research has done for the diagnosis and treatment of diphtheria it can probably do in time for tuberculosis, erysipelis, cerebro-spinal meningitis and cancer, to name only four horrible scourges of mankind which are known to be of germ origin.

2. The human race makes use of animals without the smallest compunctions as articles of food and as laborers. It kills them, confines them, gelds them and interferes in all manner of ways with their natural lives. The liberty we take with the animal creation in using utterly insignificant numbers of them for scientific researches is infinitesimal compared with the other liberties we take with animals, and it is that use of animals from which the human race has most to hope.

3. The few medical investigators can not, probably, be supervised or inspected or controlled by any of the ordinary processes of Government supervision. Neither can they properly be licensed, because there is no competent supervising or licensing body. The Government may properly license a plumber, because it can provide the proper examination boards for plumbers; it can properly license young men to practice medicine, because it can provide the proper examination boards for that profession, and these boards can testify to the fitness of candidates; but the Government cannot provide any board of officials competent to testify to the fitness of the medical investigator.

4. The advocates of anti-vivisection laws consider themselves more humane and merciful than the opponents of such laws. To my thinking these unthinking advocates are really cruel to their own race. How many cats or guinea pigs would you or I sacrifice to save the life of our child or to win a chance of saving the life of our child? The diphtheria-antitoxin has already saved the lives of many thousands of human beings, yet it is produced through a moderate amount of inconvenience and suffering inflicted on horses and through the sacrifice of a moderate number of guinea pigs. Who are the merciful people—the few physicians who superintend the making of the antitoxin and make sure of its quality, or the people who cry out against the infliction of any suffering on animals on behalf of mankind?

It is, of course, possible to legislate against an improper use of vivisection. For instance, it should not be allowed in secondary schools or before college classes for purposes of demonstration only; but any attempt to interfere with the necessary processes of medical investigation is, in my judgment, in the highest degree inexpedient, and is fundamentally inhuman.

Yours very truly,
C. W. ELIOT.

HON. JAMES McMILLAN.

THE HIGHER EDUCATION FOR COLORED YOUTH.

Prof. Shaler's article in the June number of the POPULAR SCIENCE MONTHLY was in many ways sensible and timely, but it seems to the writer

that in common with many other people he is misleading in his remarks about higher education for the negro. One would think from the great outcry against the higher education for young people of the colored race, that scarcely any other kind of education was being given them. On all sides we hear the familiar refrain: "The higher education for the negro has been a failure." Now success is a relative term. If a mere handful of colored college graduates, in a few years, ought to have settled the race problem, and induced their white fellow-citizens to treat these graduates and all members of their race fairly, then it has been a failure. But if the higher education should simply give added power of mind, enlarge the mental grasp and capacity for usefulness, lift up, socially, morally, religiously and financially, not only its disciples, but also thousands who have been induced to look upward by the force of their example, then the higher education for colored youth has been a tremendous success. Is not the latter the fair test? Of course the higher education of the few has not eliminated crime. It has not done that for the white race. The writer is a colored man and a college graduate. He can not see that the higher education has any different effect on the colored youth from what it has on the white. If there be any difference it is this: It raises the colored youth from a lower social level, as a rule, and places him on a social plane, relatively, among his own people, higher than it does in the case of the white youth. The higher training, therefore, should be more valuable to the colored youth.

In a recent address before a graduating class at Howard University, the Hon. W. T. Harris, Commissioner of Education, submitted statistics which showed that the proportionate number of secondary and higher students to the whole number of children attending school in the United States had increased from 2.22 per cent in 1879 to 5.01 per cent in 1897, nearly two and a half times; while the proportion of colored students in secondary schools and colleges had increased very little indeed, from 1 per cent to only 1.16 per cent. But the story is not yet half told. According to the report of the Commissioner of Education, 1897-98, Vol. 2, page 2,097, the total number of students taking the higher education in the United States, as a whole, was 144,477, being 1,980 to each million of the total population. The same report, page 2,480, gives the total number of colored students pursuing collegiate courses in these much discussed colored colleges as 2,492. This is only 310 to the million of colored population, whereas the whole of the United States, as shown above, had 1,980 to the million, nearly

six and a half times as many in proportion to population. This does not look as if the entire colored population were rapidly stampeding to the higher education, or as if the labor supply in the Southern States were falling off from this cause.

This is an age of higher education for the masses. The increase in the number of students taking the secondary and higher education in the United States during the last ten years has been phenomenal—unprecedented. Is the person of color so much superior to the white that he does not need so much educational training? I think not. In view of the history and present condition of this race, there is an obvious necessity for a large number of educated and trained teachers, ministers, physicians, lawyers and pharmacists; and in view of the fact that this race has only one fifth of its quota pursuing studies above the elementary grades, what fair mind will not say that there is great need of more of the secondary and higher education for colored youth, instead of less of it?

According to the report above cited, 161 academies and colleges for colored youth in the United States reported. The total number enrolled was 42,328, of which 2,492 were reported in collegiate grades, 13,669 in secondary grades and 26,167 in elementary grades. Even in these colored colleges less than 6 per cent of the students are pursuing collegiate courses. Of these, perhaps not more than 2 per cent are pursuing a college course equal to that offered at Howard. Nearly two thirds of the total enrollment in these colored colleges are receiving elementary instruction in the three R's. Classified by courses of study, 1,711—217 in a million—were taking the classical course; 1,200—150 in a million—the scientific; 4,449—555 to the million—the normal course in preparation for teaching; 1,285—160 in a million—professional courses; 9,724 the English course, and 244 the business course. In each of these courses the colored race has only about one fifth or one sixth of its quota. Is there anything in these figures to alarm the nation?

About one third of the total number of students in these 161 colored schools and colleges are taking industrial training. When we consider the great demand for educated colored ministers, teachers and physicians, and the quick reward for ability in these lines, on the one hand, and the exclusiveness of some trade-unions in shutting out colored workmen, on the other, the wonder is that one third of the total number of colored youth in

these schools have chosen the industrial course. For it is by no means certain that they will be allowed to work at their trades after they have learned them.

The number of colored students who have had even a smattering of the higher education has been shown to be ridiculously small, and the total number of colored graduates with the college degree proper does not at the most liberal estimate exceed one thousand. Many of them are dead. Of the number now living, almost every one can be located in some useful and uplifting employment as ministers, teachers, physicians, lawyers, business men, or as wives presiding over happy, prosperous, cultured homes which white persons seldom enter except on business. Our critics seem to know nothing of these homes, which, as a rule, are owned by their occupants. For the most part these homes are scattered throughout the South, and are centers of culture and refinement that elevate the moral and social status of the entire community.

To deprive the youth of the colored race of the higher education is to deprive them of all the nobler incentives to study, to sacrifice, to struggle to get an education. Every thoughtful person knows that these incentives are necessary for the white race; they are equally necessary for the colored race. Neither the white youth nor the colored, in large numbers, will toil and struggle and apply himself to get an education, unless he sees that education brings power and a better living to its possessors.

The colored race, like every other part of our population, needs all kinds of education. It is a sheer fallacy and a grievous wrong to them to hold *all* of them down to the rudiments of an education, with industrial training. All can not profit by the industrial training any more than all can profit by the higher training. There is no conflict between the advocates of industrial training and the higher education. Both are right. Both are good in their respective spheres. At any rate, it is not necessary to disparage the magnificent achievements of colored persons who have received the higher training to make an argument in favor of training *all* of them in the manual trades, or to justify their elimination from politics.

ANDREW F. HILGER,

Washington, D. C.

SCIENTIFIC LITERATURE.

GEOLOGY.

In accordance with the general results of Mr. G. K. Gilbert's investigation of recent earth movements in the Great Lakes region—that the whole district is being lifted on one side or depressed on the other, so that its plane is bodily canted toward the south-southwest, and that the rate of change is such that the two ends of a line one hundred miles long, running in a south-southwest direction, are relatively displaced four tenths of a foot in one hundred years—certain general consequences ensue. The waters of each lake are gradually rising on the southern and western shores, or falling on the northern and eastern shores, or both. This change is not directly obvious, because masked by temporary changes due to inequalities of rainfall and evaporation and various other causes, but it affects the mean height of the lake surface. In Lake Ontario the water is advancing on all shores, the rate at any place being proportional to its distance from the isobase through the outlet. At Hamilton and Port Dalhousie it amounts to six inches in a century. The water also advances on all shores of Lake Erie, most rapidly at Toledo and Sandusky, where the change is eight or nine inches a century. All about Lake Huron the water is falling, most rapidly at the north and northeast; at Mackinac the rate is six inches, and at the mouth of French River ten inches a century. On Lake Superior the isobase of the outlet cuts the shore at the international boundary; the water is advancing on the American shore, and sinking on the Canadian. At Duluth the advance is six inches, and at Huron Bay the recession is five inches a century. The shores of Lake Michigan are divided by the Port Huron isobase. North of Oconto and Manistee the water is falling; south of these places it is rising, the rate at Milwaukee being five or six inches a century, and at Chicago nine or ten inches. Eventually, unless a dam is erected to prevent it, Lake Michigan will again overflow to the Illinois River, its discharge occupying

the channel carved by the outlet of a Pleistocene glacial lake. The summit in that channel is now about eight feet above the mean level of the lake, and the time before it will be overtopped may be computed. For the mean lake stage such discharge will begin in about one thousand years, and after fifteen hundred years there will be no interruption. In about two thousand years the Illinois River and the Niagara will carry equal portions of the surplus water of the Great Lakes. In twenty-five hundred years the discharge of the Niagara will be intermittent, failing at low stages of the lake, and in thirty-five hundred years there will be no Niagara. The basin of Lake Erie will then be tributary to Lake Huron, the current being reversed in the Detroit and St. Clair channels.

GEOGRAPHY.

Relating to the Royal Geographical Society the story of his exploration of the Bolivian Andes, Sir Martin Conway spoke of his journey by way of the Arequipa Railroad, Peru, to Lake Titicaca. That remarkable sheet of water is fourteen times the size of the Lake of Geneva and twelve thousand feet above the sea, and might be regarded as the remnant of a far greater inland sea, now shrunk away. Driving from Chililaya, he reached the snowy mountain called the Cordillera Real—the backbone of Bolivia—which he had come especially to visit, and in the region of which he spent four months. To the east the mountains fell very rapidly to a low hill country and the fertile valleys that send their waters to the river Beni. On the other side lay a high plateau, at a uniform altitude of from twelve thousand to thirteen thousand feet, from which the tops of low rocky hills here and there emerged. This plateau had obviously been at one time submerged; evidence was plentiful that in ancient times the glaciers enveloped a large part of the slopes that led down to it from the main Cordilleras and reached down many miles farther than now. In the immense pile of *débris* left by the glaciers deep valleys were afterward cut by the action of water, and into these valleys the glaciers of a second period of advance protruded their snouts, depositing moraines that could still be traced *in situ* as much as four or five miles below the present limit of the ice. Contrary to the apparently general impression that the peaks of the Cordilleras were volcanic, the

author had not been able to find any trace of volcanic action along the axis of the range. The Cordillera Real had been elevated by a great earth movement, and the heart of the range consisted of granites, schists and similar rocks. The whole range might be described as highly mineralized. Gold was found at several points, but the chief auriferous valleys were those on the east side of the range. Just below the snowy mass of Cacaaca on the west was a really enormous vein of tin; and antimony, cobalt and platinum have been found in different parts. The great copper deposits were not in this range, but farther west. The flora of the high regions of the Cordillera Real was apparently sparse, but is probably more abundant in the rainy season. Bird life was more prolific and birds were numerous, at suitable places, up to an altitude of seventeen thousand feet above the sea.

ZOOLOGY.

The most recent elementary text-book in zoölogy is from the press of The Macmillan Co. Professor and Mrs. Charles B. Davenport are the joint authors. It is recognized now-a-days that what the general high school or elementary student in zoölogy needs is not professional training in that subject, but rather an opportunity to view the field so that he may have as wide an acquaintance as may be of the forms of animals and of their doings. This he needs that he may have an interest in the things of nature and that he may be a more intelligent member of society in the things pertaining to his welfare as affected by animals. The book is therefore an attempt to restore the old natural history in a newer garb. The text is divided into twenty-one chapters. The first of these deals with 'The Grasshopper and its Allies,' followed by others upon the butterfly, beetle, fly, spider, etc., similarly treated. Each chapter has one or two 'keys'—that is, arrangements whereby the families of animals may be determined. The book is richly illustrated by means of half-tone and line reproduction; a number of photographs are from life, and one of these is a flash-light photograph of a slug and an earthworm crawling upon a pavement at night! Outlines for simple laboratory work and a list of books dealing with the classification and habits of American animals are to be found in an appendix. Many good things might be said of this contribution to zoölogical text-books. This

ought to be said, that it will be a book which will be of value to any person who, while upon his holiday trip, wishes to learn about the animals he may come across.

ORNITHOLOGY.

Mr. Chapman is equally at home with camera or pen. In 'Bird Studies with a Camera, with Introductory Chapters on the Outfit and Methods of the Bird Photographer,' he gives us some of his many experiences from Central Park to the swamps of Florida and the bare rocks of the Gulf of St. Lawrence. The first two chapters are devoted to a brief discussion of the outfit and methods of the bird photographer, and these any one thinking of taking up this branch of art will do well to read carefully. Mr. Chapman considers that a 4×5 plate is the size best adapted for general purposes, and notes that while a lens with short focus may serve for photographing nests and eggs, for the birds themselves a rapid lens with focus of fourteen to eighteen inches should be used. The rest of the book is for the general reader, and contains many facts of interest concerning the haunts, habits, and home life of a number of birds from the well-known sparrow to the unfamiliar pelican, the accounts of the Bird Rock and Pelican Island being the most interesting. Some of the illustrations are a little disappointing, and emphasize the difficulties of photographing wild birds, but there is ample compensation for these in the excellence of others, particularly those devoted to Percé, Bonaventure and Bird Rock. This is equally true of birds and scenery, the views of Percé Rock being the finest that have fallen under our notice. Mr. Chapman's estimate of the feathered population of Great Bird Rock, which he puts at 4,000, is by far the smallest yet made, and probably has the soundest basis, and shows a sad diminution from the hosts of fifty years ago.

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‘Bird Homes,’ by A. Radclyffe Dugmore, seems well adapted for its stated purpose of stimulating the love of birds, helping the ordinary unscientific person to get some closer glimpses of them, and aiding in the study of their wonderfully adapted nests and beautiful eggs. Furthermore, it will probably create a strong desire in the reader to become a photographer of birds and their nests. To further these aims we have a first part containing half-a-dozen chapters devoted among other things to birds’ nests and eggs, photographing nests and young birds and the approximate dates when birds begin to nest, this being adapted to the vicinity of New York.

Following this is the bulk of the volume, containing brief descriptions of the birds, their nests, nesting places and eggs, and here the author has confessedly borrowed from Bendire, Davie and other well-known authorities, although one might wish that Mr. Dugmore had introduced more of his own observations, since those given incidentally in the first part are very interesting; where he indulges in theory he is less successful. In place of the usual method of studying the nest from the bird, we have that of studying the bird from the nest, and for this purpose the nests are grouped in classes, a chapter being devoted to each class; thus we have nests open, on the ground in open fields, marshes and generally open country; open nests in trees; nests in bridges, buildings, walls, etc. By this plan any one finding a nest can, with a little care and observation, identify the bird that made it. The illustrations, largely of nests and eggs, are a noteworthy feature of the book, although the three-color process which succeeded so admirably in Dr. Holland’s *Butterfly Book*, is here as equally distinct a failure, the least bad of the colored plates being that showing the nest of the yellow-breasted chat, the worst that of the nest of the Baltimore oriole. Those in black and white, however, merit the highest praise, and this includes the smaller cuts introduced as decorative features in the first portion of the book. It would seem difficult in a half-tone to improve on the plate of young crested flycatchers for clearness of detail, while among others that deserve special mention for artistic effect is the wood thrush on nest, and the nests of the chestnut-sided, yellow, blue-winged and worm-eating warblers. The general ‘get-up’ of the book is excellent, and the

printing of the plates separately permits the use of a deadfaced paper for the text, which is pleasant to the eye.



THE PROGRESS OF SCIENCE.

We are able to publish in the present issue of the MONTHLY the address given by Mr. G. K. Gilbert as retiring president of the American Association for the Advancement of Science. The problem that he discusses is one of the most pressing for scientific workers, while at the same time it is of interest to everyone, and the address is at once an important contribution to the subject and an exposition that all can understand. The mathematical physicists find that as an abode fitted for life the earth can not be allowed a history indefinitely long—not longer perhaps than 20,000,000 years—while the geologists with equally strong arguments claim a much greater antiquity. The biologists are also concerned, owing to the time taken up by the processes of evolution, and their facts and interests range them with the geologists rather than with the physicists. The man not versed in science would also prefer to assign a long history to the earth, for while he may be ready to let the ‘dead past bury its dead,’ he looks forward even to the distant future, and the shorter the past history of the earth the less the time it will continue to be habitable. We have thus a question in the solution of which all the sciences are concerned, and one possessing a dramatic interest that appeals to everyone. The unity of science is well illustrated by such a problem. It was the subject of the address of the retiring president of the Association, a geologist; it might be taken as the subject for the address of the newly elected president, a biologist and student of the processes of evolution; and it is one to which the president of the meeting, a mathematical physicist, has given special attention.

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Dr. Robert Simpson Woodward, who presided over the New York meeting of the Association, is professor of mechanics and mathematical physics and dean of the Faculty of Pure Science in Columbia University. He was born at Rochester, Oakland County, Michigan, July 21, 1849, and spent

his early life on a farm with the exception of about two years of experience in mercantile and manufacturing pursuits. He was prepared for college at the Rochester Academy, entered the University of Michigan in 1868, and was graduated in 1872 with the degree of C. E. Twenty years later the same institution conferred upon him the degree of Ph. D. While yet an undergraduate he entered the U. S. Lake Survey, and immediately after graduation he was appointed assistant engineer in that service. He was employed in the astronomical and geodetic work of the Lake Survey until its completion in 1882. He then accepted the position of assistant astronomer to the U. S. Transit of Venus Commission and accompanied the expedition of Prof. Asaph Hall, U. S. N., to San Antonio, Tex., to observe the transit of December, 1882. He remained with the Transit of Venus Commission until 1884, when he resigned in order to take the position of astronomer in the U. S. Geological Survey. After four years of service in this bureau he resigned to accept the position of assistant in the U. S. Coast and Geodetic Survey. This he held until 1893, when he retired from the public service and accepted the call of Columbia University to the chair of mechanics. In 1895, and again in 1900, he was elected to the deanship of the graduate faculty of pure science in that institution. Professor Woodward has published many papers on subjects in astronomy, geodesy, mathematics and mechanics. He edited, and contributed several chapters to the final report of the U. S. Lake Survey, a volume of about one thousand quarto pages devoted chiefly to a discussion of the geodetic work of the Survey done during the forty years of its existence. He is the author of several of the Bulletins of the U. S. Geological Survey, and of a memoir on the Iced Bar and Long Tape Base Apparatus of the U. S. Coast and Geodetic Survey. These forms of apparatus, devised and perfected by him, involve many novel features and secure a much higher precision at a much smaller cost than apparatus previously used. He prepared for the Smithsonian Institution a volume entitled 'Geographical Tables,' being a manual for astronomers, geographers, engineers and cartographers, published in 1894. Several of his most important mathematical papers relate to geophysics, especially those bearing on the secular cooling and cubical contraction of the earth, on the form and position of the sea surface, and on the profoundly difficult problem presented by the recently discovered phenomenon of the variation of terrestrial latitudes. Although most of his publications are necessarily of a highly technical character, his semi-popular addresses and reviews have

been widely read and appreciated. Professor Woodward was an associate editor of the 'Annals of Mathematics' from 1889 to 1899 and has been an associate editor of 'Science' since 1894. He has taken an active part in the work of the scientific societies with which he is connected, and in addition to the official positions he holds in the American Association for the Advancement of Science, he has been honored by election to the presidency of the American Mathematical Society and to the presidency of the New York Academy of Sciences. Professor Woodward represents the highest type of the man of science. Eminent for his original contributions to science, a teacher of great intellectual and moral influence, an administrator with unfailing tact and unerring judgment, he confers an honor on the Association which has elected him to its highest office.

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President Low welcomed the American Association to New York and to Columbia University in an address which recounted the increased recognition given to science by the city since the Association met there thirteen years ago and the great progress of science itself. He concluded with the following words: "I am especially glad to welcome you because you are an Association for the *Advancement* of Science. That, after all, is what ought to make you feel at home in the atmosphere of this university; for a university that does not assist the advancement of science has hardly a right to call itself by that great name. I heard Phillips Brooks say, in a sermon that I heard him preach in Boston when this Association met there twenty years ago, that you can get no idea of eternity, by adding century to century or by piling æon upon æon; but that, if you will remember how little you knew when you sat at your mother's knee to learn the alphabet, and how with every acquisition of knowledge which has marked the intervening years you have come to feel, not how much more you know, but how much more there is to be known, all can get some idea of how long eternity can be, because all can understand that there never can be time enough to enable any one to learn all that there is to know. There is so much to be known, that even the great advances of the last generation do not make us feel that everything is discovered, but they appeal to new aspirations and awaken renewed energy in order to make fresh discoveries

in a region that teems with so much that is worthy of knowledge. I congratulate you upon your success, and I bid you welcome to Columbia.”

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In the course of his reply, the president of the Association, Professor Woodward, said: “But surprising and gratifying as have been the achievements of science in our day, their most important indication to us is that there is indefinite room for improvement and advancement. While we have witnessed the establishment of the two widest generalizations of science, the doctrine of energy and the doctrine of evolution, we have also witnessed the accumulation of an appalling aggregate of unrelated facts. The proper interpretation of these must lead to simplification and unification, and thence on to additional generalizations. An almost inevitable result of the rapid developments of the past three decades especially is that much that goes by the name of science is quite unscientific. The elementary teaching and the popular exposition of science have fallen, unluckily, into the keeping largely of those who can not rise above the level of a purely literary view of phenomena. Many of the bare facts of science are so far stranger than fiction that the general public has become somewhat over-credulous, and untrained minds fall an easy prey to the tricks of the magazine romancer or to the schemes of the perpetual motion promoter. Along with the growth of real science there has gone on also a growth of pseudo-science. It is so much easier to accept sensational than to interpret sound scientific literature, so much easier to acquire the form than it is to possess the substance of thought that the deluded enthusiast and the designing charlatan are not infrequently mistaken by the expectant public for true men of science. There is, therefore, plenty of work before us; and while our principal business is the direct advancement of science, an important, though less agreeable duty, at times, is the elimination of error and the exposure of fraud.”

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The meeting of the Association in New York was of more than usual importance. Not only did the nine sections of the Association hold their daily sessions, but there were also fifteen special scientific societies meeting simultaneously at Columbia University. Men of science came

together from all parts of the country to present the results of the year's research, to gain profit and pleasure from association with other workers, and to return to their homes with increased knowledge and renewed interest. It is obviously impossible to give here an account of the hundreds of scientific papers presented, or even to report upon the general proceedings of the Association. Two of the more important actions may, however, be mentioned. It was decided to send 'Science,' our weekly journal of general science, to all members of the Association without charge, and a section devoted to physiology and experimental medicine was established. It was thought that the receipt of a journal such as 'Science' would increase the membership of the Association and lead to a greater interest in its work, as even those who are unable to attend the meetings will hereafter have a definite return for membership. The Association will be greatly strengthened by giving recognition to the great group of sciences—physiology, experimental psychology, anatomy, embryology, histology, morphology, pathology, bacteriology and their applications—which have developed with such remarkable activity within the past few years.

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It is not possible to report on the scientific work of the meeting in part owing to its magnitude—the papers would fill the volumes of this journal for several years to come. It is also true that each paper taken singly is likely to be of interest only to the special student. Specialization in science is absolutely necessary for its advance, but the terminology required for exactness and economy makes the work in each department scarcely intelligible to those not immediately concerned, while the great detail necessary in careful research seems almost trivial until we realize that it is upon such special work that the general principles and the applications of science depend. We all know that our ways of thought and habits of life are chiefly based on the results of modern science. This has not been the result of a sudden revelation, but of a continual growth, scarcely perceptible until viewed from a distance. The importance of current political events is magnified by the common interest they excite, whereas in art, literature and science time is required before things can be seen in their right perspective. We can, however, take the reports of the three committees of the Association to which small grants were made for research and use these as

examples of the scientific work described at the meeting. These committees were on 'Anthropometry,' on 'The Quantitative Study of Variation' and on 'The Cave Fauna of North America.'

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The committee on anthropometry is undertaking to make measurements of the physical and mental traits of members of the Association, and to encourage such work elsewhere. At the present time there exists but little exact knowledge of how people differ from each other and of the causes and results of such differences. Much has been written regarding men of genius, criminals and other classes, but without an adequate foundation of fact. The members of a scientific society are a fairly homogeneous class, regarding whose heredity, education and achievements correct information can be secured. The measurements made at the New York meeting, determining such traits as size of head, strength, eyesight, quickness of perception, memory, etc., will supply the standard type for scientific men and their variations from this type. When other classes of the community have been measured, comparisons can be made and we shall know whether scientific men are more variable than others, have larger heads, better memory and the like. Work of this character has been carried on at Columbia University for some years. The freshmen, both the men of Columbia College and the women of Barnard College, are measured and tested with care, equal attention being paid to mental and physical traits. Then the measurements are repeated at the end of the senior year. Anthropometric work has also been done in Great Britain under the auspices of Dr. Galton and Professor Pearson, and we may perhaps hope that the time will come when we shall have as exact knowledge about human differences as we now have about different kinds of butterflies.

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Although geologists and botanists have defined hundreds of thousands of species, they have not as a matter of fact until very recently attempted to secure exact measurements of differences, and the committee of the Association on 'The Quantitative Study of Variation,' of which Prof. Chas. B. Davenport is the recorder, aims to encourage such work. It is now over forty years since the facts and arguments presented in Darwin's 'Origin of

Species' paved the way for general acceptance of the doctrine of evolution. But the objection is hardly less valid to-day than it was then that the evidence for evolution is almost wholly indirect. Over and over again naturalists have been challenged to cite one case where a species in nature has changed within historic times and repeatedly they have taken refuge in the plea that the historic period is too short for a noticeable change to have taken place. This plea can be accepted, however, only so long as we have no exact way of measuring race change. When we can express quantitatively the condition of a community to-day, we may hope to be able to say whether any change has occurred after five, ten, or a hundred years. The committee of the Association has especially concerned itself with a piece of work which may be considered typical. In the headwaters of the Tennessee River there lives a univalve mollusc which is found nowhere else in the world and which belongs to a family of molluscs that was early separated from its marine congeners as a fluviatile species. This genus, *Io*, varies greatly in different parts of the Tennessee basin. In some places it is smooth; in others, spiny; in others, long drawn out. Under a grant of the Association, Mr. C. C. Adams, of Bloomington, Ill., visited this region; travelled down one of the tributaries in a boat, collecting samples from every community of *Ios*; and went by train up a second river collecting at every stopping place. The results of this trip were, in a word, that in passing from the mouth to the headwaters of the two parallel tributaries the shells vary in parallel fashion and show a uniform, continuous change from the spiny, elongated condition characteristic of the mouths of the rivers to the smooth, more globose condition characteristic of the headwaters. The additional grant by the Association of one hundred dollars will assist Mr. Adams in making further quantitative studies on variation in the genus *Io*.

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Hardly any fact has excited more interest among evolutionists than the blindness of cave animals; and various theories have been advanced to explain the fact. It is known that the blind condition is due to a degeneration of formerly functional eyes. The difficulty has been to understand what advantage is gained by losing the eyes even in a locality where eyes are of no use. It has been affirmed that 'Nature is economical' and will not expend energy in building an unnecessary organ. Weismann has suggested that the

only reason why we have eyes at all is because Natural Selection is constantly weeding out poor eyes. Withdraw the necessity for good eyes, and poor eyes and good eyes will have an equal chance of surviving. According to a third theory, the functional activity of any organ is essential to its maintenance. Just as the unused arm withers so the unused eye degenerates. Of course all these theories assume that the ancestors of the blind species—for instance, of the blind fishes—had originally no inherent tendency to blindness or degeneration of the eyes. This assumption has, however, been recently combatted by Professor Eigenmann, who has shown that although many kinds of fish are accidentally swept into caves, only one kind has become blind; of this kind the nearest allies which live in open streams shun the light, live in crevices and under stones, and have less perfect eyes than other fishes. Some of the allies of such light-shunning fishes have made their way into caves, and have there worked out their tendency to a reduction of eyes. That has been the history of eyeless fishes. To continue the researches of Professor Eigenmann, so auspiciously begun, the Association last year granted one hundred dollars to a committee on the cave vertebrates of North America. With the aid of the grant Dr. Eigenmann has during the past year penetrated into numerous caves and obtained much additional material for his researches.

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The American Association will meet next year at Denver, beginning on August 26th. The newly elected officers are:

President.

Prof. Charles Sedgwick Minot, Harvard Medical School.

Vice-Presidents.

Mathematics and Astronomy: Prof. James McMahon, Cornell University.

Physics: Prof. D. D. Brace, University of Nebraska.

Chemistry: Prof. John H. Long, Northwestern University.

Mechanical Science and Engineering: Prof. H. S. Jacoby, Cornell University.

Geology and Geography: Prof. C. R. Van Hise, University of Wisconsin.

Zoölogy: President D. S. Jordan, Leland Stanford Jr. University.

Botany: B. T. Galloway, U. S. Department of Agriculture, Washington, D. C.

Anthropology: J. W. Fewkes, Bureau of Ethnology, Washington, D. C.

Economic Science and Statistics: John Hyde, Department of Agriculture, Washington, D. C.

Permanent Secretary.

L. O. Howard, U. S. Department of Agriculture, Washington, D. C.

General Secretary.

Prof. William Hallock, Columbia University, New York.

Secretary of the Council.

D. T. McDougal, New York Botanical Gardens.

Secretaries of the Sections.

Mathematics and Astronomy: Prof. H. C. Lord, Ohio State University.

Physics: J. O. Reed, University of Michigan.

Chemistry: Prof. W. McPherson, Ohio State University.

Mechanical Science and Engineering: William H. Jacques, Boston, Mass.

Geology and Geography: Dr. R. A. F. Penrose, Pierce, Ariz.

Zoölogy: Prof. H. B. Ward, University of Nebraska.

Botany: A. S. Hitchcock, Manhattan, Kan.

Anthropology: G. G. McCurdy, Yale University.

Economic Science and Statistics: Miss C. A. Benneson, Cambridge, Mass.

Treasurer.

Prof. R. S. Woodward, Columbia University.

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The National Educational Association, which held its annual session at Charleston during the week beginning on July 9th, is the leading representative of the many educational associations of the country. Its membership includes the ablest teachers of education in colleges and the most successful school superintendents and teachers. Its meetings give occasion for discussions of matters of educational theory and practice in many ways comparable to the discussions in scientific societies. The program of the present meeting shows that like the scientific associations, the National Educational Association has become differentiated into a number of practically isolated sections with differing interests. There are separate departments of Kindergarten Education, Manual Training, Child Study, Normal Schools, Libraries, etc. The Department of Superintendence now has a special meeting at a different time and place. There are also general sessions, and these have not become mere formal business meetings. The leading topic for discussion this year seems to have been the proposed National University at Washington. The most obviously important service which the Association has rendered to educational endeavor has been its elaboration (through efficient committees) and publication of reports on Secondary Education, Elementary Education, Rural Schools and College Entrance Requirements. These reports represent if not demonstrable facts, at least the well-considered opinion of competent judges and they have had a highly beneficial influence. Dr. J. M. Green, of Newark, will preside over next year's meeting. The decision in regard to the place has been left to the executive committee, the claims of Detroit, Cincinnati and Tacoma having been especially urged.

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The opening of a summer school at Columbia University and the attendance at Harvard University of a large proportion of all the school

teachers of Cuba are important steps towards increasing the usefulness of our institutions for higher education. The grounds, buildings and equipment of Columbia University have cost in the neighborhood of \$10,000,000, and to let these lie idle and rusting for nearly one-third of the year is evidently wasteful. But it is not only a question of the most economical administration of these trust funds that is at issue. The teachers of the country, perhaps 500,000 in number, have had just enough education to profit particularly by attendance at a university. They are engaged at their work during three-fourths of the year, but their summers can be spent in no more pleasant and useful way than by attending a university summer school. It would be good business policy for school boards to send their teachers to the summer schools, except that the benefit might not be reaped locally, as each teacher would soon deserve a better position than he now has. It is, however, not only for teachers that university sessions during the summer are needed. The long vacation is largely a tradition from the time when boys were most usefully occupied on the farm during the summer. It is doubtful whether students now come back to college in the autumn in an improved physical or moral condition. They might spend their time to advantage, but are not likely to do so at the ordinary summer resort. It is admitted by everyone that young men are too old when they leave college and the professional schools. Reforms are needed in various directions, but an obvious one is not to take four years for three years' work. Though university professors, who for the general good need freedom from routine teaching for other work, should be allowed leave of absence for a part of the year, it does not follow that they should all be away at the same time. It seems probable that the example set by the University of Chicago, which holds four sessions extending through the year, will be followed by all our universities.

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The third International Conference on a Catalogue of Scientific Literature was held in London on June 12th and 13th. It will be remembered by those who are interested in the organization of science that a conference on this subject was called by the Royal Society in 1896 at which it was proposed to undertake by international coöperation a catalogue of contributions to science. Certain details were arranged and others were

left to a committee of the Royal Society. Under the auspices of this committee schedules of classification were drawn up and estimates of the cost secured. A second conference was held in 1898, and after various changes in the plans for the catalogue it was at the recent Conference definitely decided to proceed with its publication. It is estimated that the cost will be covered by the sale of three hundred sets, and different governments or national agencies have made themselves responsible for a certain number of sets, Germany and Great Britain for example, subscribing for forty-five sets, each costing £17. The Catalogue will be published in seventeen volumes devoted to as many sciences, and will be both an author's and a subject index. The collection of material is to commence from January 1, 1901. While all scientific men welcome improvements in cataloguing scientific literature, the arrangements proposed by the Royal Society and by the different conferences have met with some criticism. The serious mistake has been made of entirely ignoring the catalogues and bibliographies already existing for most of the sciences, and it is not certain that the elaborate and expensive machinery proposed will be as useful as some plan would have been for unifying the existing agencies. Still in the end there must be some international and uniform method for cataloguing scientific literature, and it is to be hoped that our Government will do its share toward supporting the present undertaking.

Transcribers' Notes

Punctuation, hyphenation, and spelling were made consistent when a predominant preference was found in this book; otherwise they were not changed.

Simple typographical errors were corrected; occasional unbalanced quotation marks retained.

Ambiguous hyphens at the ends of lines were retained.

Page [342](#): "millenium" was printed that way.

Page [366](#): "to smear the statues of Jupiter" was misprinted as "statutes".

Page [387](#): "we have meet with a difficulty" was printed that way.

Page [387](#): "cm." originally was printed as "c. m."

*** END OF THE PROJECT GUTENBERG EBOOK THE POPULAR
SCIENCE MONTHLY, AUGUST, 1900 ***

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